

8. Medicines/therapies

Section 8 covers only the treatment recommendations for conditions covered in this manual and does not serve as a comprehensive list of indications for each medicine. Also, it does not include routine contraception, vaccines, or immunoglobulin therapy.

Information on medicines in this section is drawn from the WHO Model Formulary,¹ Pharmacological treatments of mental disorders in primary health care², manufacturers' product literature, UpToDate³, and evidence-based formularies such as the British National Formulary⁴ and Australian Medicines handbook.⁵ Other sources include the Sanford Guide to Antimicrobial Therapy.⁶

These summaries do not cover all adverse reactions and interactions. The information given should be interpreted in the light of professional knowledge and by reference to the approved product information for the individual drugs and should be supplemented as necessary by specialist advice.

Where deemed important, the principles of prescribing and medicine administration are dealt with in the relevant Sections of the manual. For example, principles of prescribing in mental health disorders are in Section 10.11.

Where possible, adverse reactions are classified according to their probable incidence: common = incidence of 1% or more; infrequent or rare = incidence of less than 1%.

Section 8 is subdivided into:

8.1 Analgesics for pain relief

8.2 Information on equivalence for interchangeability- corticosteroids

8.3 Iron content of different salts

8.4 Information on individual medicines in adolescents and adults

Section 8 should be adapted in each country to match national guidelines and essential medicines lists. Please also refer to the updated WHO essential medicines list and the *2008 WHO Model Formulary*.¹

The medicines list in 8.4 is not comprehensive. For a more complete list and for specific medicine interaction information, see the *2008 WHO Model Formulary*.¹

1 WHO Model Formulary. WHO, 2008. Available at www.who.int/selection_medicines/list/WMF2008.pdf

2 Pharmacological treatment of mental disorders in primary health care. WHO, 2009.

3 UpToDate, available at <http://www.uptodate.com/store>.

4 Joint Formulary Committee (2010). *British National Formulary*. 60th ed. London, British Medical Association and Royal Pharmaceutical Society of Great Britain. Available through HINARI (<http://extranet.who.int/hinari/en/journals.php>).

5 Rossi, S. ed, *Australian Medicines Handbook 2008*. Adelaide, Medicines Handbook Pty Ltd, 2008.

6 Gilbert DN, Ellopoulos GM, Moellering RC, Saag MS, Chambers HF, eds. *The Sanford Guide to Antimicrobial Therapy*. 40th ed. Sperryville, Antimicrobial Therapy, 2010.

For renal and hepatic impairment drug dosage adjustments, please refer to appendices 4 and 5 in *2008 WHO Model Formulary*¹ and Section 11.31 in this manual.

Some of the other common or important indications for the medicines that are not addressed in this manual are listed in the left column under “Other indications”.

Special advice for prescribing medications for the elderly

- Drugs that commonly cause problems in the elderly include hypnotics, diuretics, nonsteroidal anti-inflammatory drugs, antihypertensives, psychotropics, and digoxin.
- For some medications, start low, go slow, expect unusual side-effects and drug interactions.
- See Section 18 Geriatric care.

8.1 A guide to the use of different analgesics (see Section 20 Palliative care)

	Analgesics	Starting dose in adolescents and adults	Range	Side-effects/ cautions
STEP 1	Non-opioid			
	Paracetamol (also lowers fever)	1 gram every 4–6 hours but no more than 4 grams in 24 hours	Only 1 tablet (500 mg) may be required in the elderly or the very ill or when combined with an opioid. Mild pain might be controlled with every 6 hour dosing.	Do not exceed 4 grams in 24 hours (more can cause serious liver toxicity).
	Aspirin (acetylsalicylic acid) (also anti-inflammatory and lowers fever)	600 mg (2 tablets of 300 mg) every 4 hours		Avoid use if gastric problems. Stop if epigastric pain, indigestion, black stools, petechiae, or bleeding. Avoid in presence of any bleeding. Do not give to children under 16 years.
	Ibuprofen (also anti-inflammatory and lowers fever)	1.0 200–400 mg 3–4 times daily. Maximum 2.4 g daily.	Maximum daily dose of 2.4 g	With or after food
STEP 2	Opioid for mild to moderate pain (give in addition to aspirin or paracetamol)			
	Codeine (if not available, consider alternating aspirin and paracetamol)	Codeine phosphate 30 mg every 4 hours	Codeine phosphate 30–60 mg every 4 hours. Maximum daily dose for pain 180–240 mg – switch to morphine if pain management inadequate.	Give laxative to avoid constipation unless diarrhoea.
STEP 3	Opioid for moderate to severe pain			
	Oral morphine: 5 mg/5 ml or 50 mg/5 ml or tablets Give by mouth but, if necessary, can be given rectally. IV or IM or subcutaneously	Initially, morphine sulfate 2.5–10 mg every 4 hours, increased by 30–50% if pain persists	According to pain There is NO ceiling dose See Section 20.	Give laxative to avoid constipation unless diarrhoea. Excessive dosage can reduce respiratory rate.

8.2 Information on equivalence for interchangeability – corticosteroids

Table: Corticosteroid equivalents

Steroid	Equivalent dose	Half-life
Hydrocortisone	100 mg	8–12 hours
Prednisone/prednisolone	25 mg	12–36 hours
Methylprednisolone	20 mg	12–36 hours
Dexamethasone	4 mg	36–54 hours

Table: Classification of potencies of topical corticosteroids and interchangeability

Mild hydrocortisone 0.5%, 1% hydrocortisone acetate 0.5%, 1%	Potent betamethasone 0.05% dipropionate 0.1% betamethasone valerate 0.1% hydrocortisone butyrate 0.1% mometasone furoate 0.1% triamcinolone acetonide
Moderate betamethasone valerate 0.02%, 0.05% clobetasone butyrate 0.05% triamcinolone acetonide 0.02%	Very potent clobetasol propionate 0.05%

8.3 Iron content of different iron salts

Iron salt	Amount	Content of ferrous iron
Ferrous fumarate	200 mg	65 mg
Ferrous gluconate	300 mg	35 mg
Ferrous sulfate	300 mg	60 mg
Ferrous sulfate, dried	200 mg	65 mg

8.4 Summary of medicines/therapies in adolescents and adults

Note: These summaries do not include dosing or administration to children less than 10 years of age.

References to Sections in this manual are in parentheses.

Medicines not included in the WHO Model Formulary are in italics.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Acamprosate Maintenance of abstinence in alcohol dependence (16.5)	Tablet: 333 mg enteric coated >55 kg: 2 tablets 3 times daily. <55 kg: 2 tablets twice daily	Common: Pruritus, maculo-papular rash; diarrhoea; changes in libido Infrequent or rare: Nausea, vomiting, abdominal pain; bullous skin reactions, angioedema, anaphylactic reactions	Pregnancy/breastfeeding: Not recommended during pregnancy or breastfeeding. Use with caution: Renal or hepatic impairment (give lower doses) Does not alter CNS effects of drinking alcohol or withdrawal symptoms
Acetazolamide Pre-operative treatment of acute angle-closure glaucoma (10.12.2) (Other indications: Adjunctive treatment of chronic open-angle glaucoma)	Tablet: 250 mg Oral: Initially 250 mg then 500-750 mg per day in divided doses until patient reaches referral care.	Common: Paresthesia (of hands, face, feet, or mucocutaneous junctions), fatigue, drowsiness, depression, decreased libido, bitter or metallic taste; nausea, vomiting, abdominal cramps, diarrhoea, black faeces; polyuria, renal stones; metabolic acidosis, hypokalaemia, hyponatraemia Infrequent or rare: Transient myopia; Stevens-Johnson syndrome; aplastic anaemia (especially in the first 6 months), thrombocytopenia, agranulocytosis, neutropenia	Pregnancy/breastfeeding: Avoid in first trimester of pregnancy. Contraindications: Hypokalaemia, hyponatraemia, acidosis, severe renal impairment, severe hepatic impairment, chronic angle-closure glaucoma (may mask deterioration) Use with caution: In the elderly, diabetes mellitus, gout, history of renal stones, sulfonamide allergy Counselling: Take tablets with meals to reduce the risk of stomach upset
Acetylcysteine Paracetamol overdose (3.8.1)	Injection: 200 mg/ml (intended for antidotal, not mucolytic, use) Oral: 100 mg/ml IV: Initially 150 mg/kg over 15 minutes; then 50 mg/kg over 4 hours; then 100 mg/kg over 16 hours Oral: Loading dose of 140 mg/kg; THEN 4 hours after loading dose, initiate maintenance dose of 70 mg/kg administered at 4-hour intervals for 17 doses. Continue until 72 hours post-ingestion (longer if LFTs abnormal).	Common: Flushing, urticaria, itch Infrequent or rare: Anaphylactoid or hypersensitivity-like reactions including angioedema, bronchospasm/ respiratory distress, hypotension, and, rarely, tachycardia or hypertension. Usually occur 15–60 minutes after start of infusion. (Manage by reducing infusion rate or suspending infusion until reaction has settled; stop infusion if severe anaphylaxis occurs; rash – use an antihistamine; acute asthma – use short-acting beta2 agonist such as salbutamol.) Once an anaphylactoid reaction is under control, infusion can normally be restarted at the lowest infusion rate (100 mg/kg in 1 litre over 16 hours).	Pregnancy/breast feeding: Clinical experience indicates that use of acetylcysteine for treatment of paracetamol overdose is effective and benefits outweigh risks. Use with caution: In patients with asthma or history of bronchospasm, but do not delay acetylcysteine treatment Administration: IV: Dilute requisite dose in glucose intravenous infusion 5% as follows: initially, 200 ml given over 15 minutes, then 500 ml over 4 hours, then 1 litre over 16 hours. Oral: Dilute to a 5% solution in soda pop, juice, or water prior to oral or nasogastric administration. Storage: A change in the colour of the solution to light purple has sometimes been noted; this colour change is not thought to indicate significant impairment of safety or efficacy. Store below 25 °C.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Acetylsalicylic acid (aspirin) Acute coronary syndrome/myocardial infarction (Quick Check page 20) Mild to moderate pain (20.2, 2.4); fever (10.1.2) (Other indications: acute ischaemic stroke; prophylaxis of cerebrovascular disease or myocardial infarction; acute migraine attack; inflammatory arthritis)	Tablet: 100 mg, 300 mg Oral: 300 mg (preferably chewed or dispersed in water) given immediately as a single dose 300 mg–600 mg every 4 hours as needed; higher doses (900mg) may be useful for analgesia in some patients; maximum 4 grams in 24 hours	Common: Gastrointestinal irritation with slight blood loss, tinnitus; deafness (large doses), nausea, dyspepsia, vomiting, increased bleeding time, headache, dizziness Infrequent or rare: Steven Johnson syndrome, iron deficiency anaemia, renal impairment, oesophageal ulceration, major GI bleeding, blood dyscrasias, Reyes syndrome. Allergy, bronchospasm angioedema, urticaria or rhinitis in hypersensitive patients (particularly in people with asthma) * If overdose, see Section 3.8.	Pregnancy/breastfeeding: Precautions in the first and second trimesters – low doses probably not harmful. Consider alternative for analgesia. Use not recommended in third trimester due to impaired platelet function and risk of haemorrhage. Delayed onset and increased duration of labour with increased blood loss; avoid analgesic doses if possible in last few weeks (low doses probably not harmful); with high doses, closure of fetal ductus arteriosus <i>in utero</i> and possibly persistent pulmonary hypertension of newborn; kernicterus in jaundiced neonates. Breastfeeding: Short course safe in breastfeeding at usual dosage (monitor infant, regular use of high doses could produce hypoprothrombinaemia in infant; if neonatal vitamin K stores low; possible risk of Reye syndrome; avoid breastfeeding 1–2 hours after dose to minimise infant exposure. Contraindications: In hypersensitivity (including asthma, angioedema, urticaria or rhinitis) to acetylsalicylic acid or any other NSAID, active peptic ulceration, bleeding disorders, and in children and adolescents under 16 years (Reye syndrome) Not for treatment of gout Use with caution: in asthma, heart failure, uncontrolled hypertension, allergic disease, previous peptic ulceration, renal impairment, hepatic impairment, G6PD deficiency, dehydration, and in the elderly Administration: Give with food, large quantities (~240 ml) of water (unless fluid restricted) or milk to minimize gastric irritation. Counselling: If you develop swollen ankles, difficulty in breathing, black stools or vomit that looks like coffee grounds stop taking the medicine.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Aciclovir Herpes simplex virus (HSV); genital herpes; initial or recurrent episode (11.15) Prophylaxis for recurrent herpes simplex (chronic suppression)(11.15) HSV encephalitis or hepatitis (11.15) HSV oesophagitis (11.15, 10.7b.3): • If NOT immuno-compromised • If immuno-compromised • HSV oesophagitis suppression Varicella (chickenpox) (11.45.1) Herpes zoster (shingles) (11.45.2) Herpes keratitis (10.12.2)	Tablet: 200 mg, 400 mg, 800 mg Oral suspension: 200 mg/5 ml Infusion: 250 mg vial Eye ointment: 3% W/W Oral: 400 mg 3 times daily OR 200 mg 5 times daily for 7–10 days (longer if new lesions appear or incomplete healing) Oral: 200 mg 3–5 times daily OR 400 mg twice daily interrupt every 6–12 months for reassessment. IV: 10 mg/kg 3 times daily for 14–21 days Oral: 200 mg 5 times daily OR 400 mg 3 times daily for 7–10 days Oral: 400 mg orally 5 times daily for 14–21 days Oral: 400 mg twice daily AND ART See dosing in 11.45.1 Oral: 800 mg 5 times daily for 7 days (begin within 72 hours of appearance of rash to be effective) AND 3% eye ointment every 4 hours if ophthalmic involvement Apply 1 cm ointment 3% directly to eye 5 times daily; continue for at least 3 days after healing is complete. OR oral aciclovir 400 mg 5 times daily until healing is complete	Common: Nausea, vomiting, diarrhoea, hallucinations (with high dose), headaches, lethargy, confusion, seizures Infrequent or rare: Hypersensitivity reactions; agitation, vertigo, dizziness, weakness; oedema, renal impairment; arthralgia; sore throat; abdominal pain, constipation; hepatitis, jaundice; blood disorders (anaemia, thrombocytopenia, leukopenia). Stevens-Johnson syndrome, toxic epidermal necrolysis, anaphylaxis Severe local inflammation on intravenous infusion	Pregnancy: Precautions in the first trimester Breastfeeding: Considered safe Use with caution: In renal impairment, with concurrent administration of other nephrotoxic drugs, use of IV form in patients with underlying neurological abnormalities, in elderly patients Administration: Maintain adequate hydration with IV use (balance with risk of cerebral oedema in patients with encephalitis). Infuse IV dose over 1 hour to prevent renal damage. Ensure adequate fluid intake (1.5–2 litres/day), which should also be maintained with oral administration to prevent crystallization in the renal tubules. Changes in renal function during treatment usually respond to rehydration and/or dosage reduction. Counselling: Drink plenty of fluids (at least 1.5–2 litres/daily). Tablet can be dissolved in water if desired. If taking 5 times daily, take every 4 hours during waking hours.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Albendazole Strongyloidiasis (if ivermectin not available) (11.36) Cysticercosis, as alternative to praziquantel in uncomplicated cases (11.7) Filariasis (11.12) Ascariis (10.7), hookworm (10.18) • Treatment • Prophylaxis	Tablet: 400 mg (chewable) Oral: 400 mg twice daily for 3 days (consider longer course in immunocompro-mised patients). THEN maintenance at 400 mg monthly. Oral: 15 mg/kg daily for 8 days Oral: 400 mg as a single dose + diethyl-carbamazine OR albendazole 400 mg/twice daily for 2 weeks + ivermectin as a single dose Oral: 400 mg as a single dose Oral: 400 mg as a single dose every 6 months	Common: During treatment of neurocysticercosis, neurological symptoms (fever, headache, worsening of disease, probably due to CNS reaction to dying parasites) Infrequent or rare: GI intolerance, increase in liver enzymes, reversible alopecia, rash, reversible leukopenia, bone marrow suppression (pancytopenia, aplastic anaemia, agranulocytosis), Stevens-Johnson syndrome	Pregnancy/breastfeeding: Recommended for use only during the second and third trimesters of pregnancy and only when there are no alternatives and benefit outweighs risk. Administration: Check liver function tests and blood counts before longer-term treatment and twice during each cycle (every 2 weeks). Cease therapy if enzymes greater than twice normal limit. Counselling: Chew and take on an empty stomach. Contraindications: Do not co-administer ivermectin if onchocerciasis or loaisis are co-endemic with filariasis.
Amiloride Oedema (10.4.4) (Other indications: Liver cirrhosis)	Tablet: 5 mg Oral: 10 mg daily in 1–2 divided doses; give in combination with furosemide	Common: Hyperkalaemia, hyponatraemia, hypochloreaemia (especially when combined with thiazide diuretics), weakness, headache, nausea/vomiting, constipation, impotence, dizziness, muscle cramps Infrequent or rare: Diarrhoea, anorexia, dry mouth, abdominal pain, flatulence, polyuria, rash, pruritus, visual disturbances, mild psychiatric disturbances, palpitations	Pregnancy/breastfeeding: Not recommended in pregnancy or breastfeeding; consider alternatives. Contraindications: Hyperkalaemia, renal failure Use with caution: In the elderly, debilitated patients with cardiopulmonary disease or uncontrolled type 1 diabetes; in patients with cirrhosis, may precipitate renal failure, hyperchloraemic metabolic acidosis, hepatic encephalopathy; with medicines that can increase potassium concentration (ACE inhibitors) Counselling: Take in the morning. Dizziness with standing may occur.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Amitriptyline Depression (10.11.6); neuropathic pain (10.10a); herpetic neuralgia (11.45)	Tablet: 25 mg, 50 mg Oral: Start 50 mg at bedtime AND Increase gradually as necessary by 25–50 mg every 1–2 weeks to 100–150 mg daily. Maximum dose of 200 mg daily for depression; up to 300 mg for neuropathic pain (single dose, preferably at bedtime or in divided doses). Note: delay in onset of effect For elderly and medically ill: Start at 25 mg at night, can be increased gradually to maximum tolerated dose of 100 mg per day	Common: Orthostatic hypotension (fall risk), dizziness, sedation, dry mouth, constipation, nausea, difficulty urinating, blurred vision, headache, confusion, disorientation, increased liver enzymes, worsening depression, anxiety, insomnia, ejaculatory problems, impotence, appetite and weight changes Serious side-effects: Cardiac arrhythmias, heart attack, stroke, seizures, hyperthermia, heat stroke, mania/hypomania If overdose, see Section 3.8.	Pregnancy: Manufacturer advises against use unless essential, particularly during first and third trimesters. Breastfeeding: Use with caution if indicated and if the drug of choice; reversible withdrawal symptoms and adverse effects possible (monitor infant for drowsiness). Preferably, give as a single nightly dose after breastfeeding. Contraindications: Patient has taken an MAO-I within 2 weeks; recent myocardial infarction, arrhythmias (especially heart block), bipolar disorder, severe liver disease. Do not use in adolescents. Use with caution: In elderly, patients with cardiac history; epilepsy, hepatic impairment, thyroid disease, pheochromocytoma, history of mania, psychoses, angle- closure glaucoma, history of urinary retention, concurrent electroconvulsive therapy, anaesthesia. May increase the risk of suicidal thinking and behaviour. Prone to multiple significant drug interactions. Counselling: Blurred vision and dry mouth may be troublesome but may lessen or disappear after about 7 days. Try to take it at night to reduce daytime drowsiness. You may feel dizzy on standing when taking this medicine; get up gradually from sitting or lying to minimize this effect. Avoid driving and operating machinery until you know how you react to this medicine. Do not stop taking the medicine suddenly. If you suddenly stop taking amitriptyline, you may experience withdrawal symptoms such as nausea, headache, and lack of energy. Your clinician will probably decrease your dose gradually. May increase the effects of alcohol.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Amoxicillin	Tablet/capsule: 250 mg, 500 mg Powder for oral liquid: 125 mg/5 ml		
Non-severe pneumonia (10.6.3)	Oral: 500–1000 mg 3 times daily for 5–7 days	Common: Nausea, vomiting, diarrhoea; hypersensitivity reaction (discontinue if severe rash, urticaria, wheezing); superinfection including candidiasis	Pregnancy: Not known to be harmful Breastfeeding: Considered safe
Sinusitis (11.35)	Oral: 1 gram 3 times daily for 7 days	Infrequent or rare: Fever, erythema, exfoliative dermatitis, angioedema, <i>C. difficile</i> colitis, anaphylaxis, bronchospasm, interstitial nephritis, serum sickness-like syndrome, haemolytic anaemia, toxic epidermal necrolysis, erythematous rashes (common in glandular fever, lymphocytic leukaemia, cytomegalovirus infection, Epstein-Barr virus infection, possibly HIV)	Contraindications: Known hypersensitivity/ anaphylaxis to penicillins or other beta-lactams Use with caution: If history of allergy or renal impairment. Maintain adequate hydration with high doses (risk of crystalluria). Reduce dose if severe renal failure.
Non-severe cutaneous anthrax (10.2.10)	Oral: 500 mg every 8 hours for 7–10 days		Counselling: Swallow the capsule whole with a glass of water at the start of a meal or slightly before.
Refractory gastritis/ PUD—eradication regimen for documented <i>H. pylori</i> infection (10.7a.2)	Oral: 1 g twice daily AND clarithromycin AND omeprazole for 1 week		
Dental abscess (10.17.5) and peritonsillar abscess (10.17.9)	Oral: 500 mg to 1 g 3 times daily for 5–7 days AND metronidazole		
(Other indications: Bronchitis; otitis media; osteomyelitis; endocarditis prophylaxis; post-splenectomy prophylaxis)			

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Amoxicillin with clavulanic acid (co-amoxiclav) Doses are based on amoxicillin component. Septic abortion (10.15.6) Lower urinary tract infection (11.44) Dental abscess (10.17.5); peritonsillar abscess (10.17.9) Cholecystitis (10.7a) Sinusitis (11.35) (Other indications: Otitis media; cellulitis)	Tableted/capsule: amoxicillin 500 mg + clavulanic acid 125 mg; amoxicillin 875 mg + 125 mg clavulanic acid Oral: 500 mg 3 times daily for 5 days (double in severe infections) Oral: 500 mg 3 times daily for 3–7 days Oral: 500 mg 3 times daily or 875 mg every 12 hours for 14 days. Oral: 1 g 3 times daily or, if severe, 4 times daily 875 mg twice daily for 7 days	Common: Transient increase in liver enzymes and bilirubin See amoxicillin. Infrequent or rare: Acute generalised exanthematous pustulosis; very rarely hepatic events have been reported, predominantly in males and elderly patients; these may be associated with prolonged treatment. In some cases may not become apparent until several weeks after treatment has ceased. These are usually reversible.	Pregnancy/breastfeeding: Not known to be harmful. Breastfeeding: Considered safe. Contraindications: Known severe hypersensitivity to penicillins or other beta-lactams Use with caution: In history of mild hypersensitivity to beta-lactams, renal impairment, hepatic impairment Administration: Renal dose adjustment required (do not use high doses with CrCl $\leq$ 30 ml/hour). 875 mg orally every 12 hours can be substituted for 500 mg orally every 8 hours. Counselling: To minimize stomach upset, take at the start of a meal.
Amphotericin B (conventional) Cryptococcosis (11.5) Histoplasmosis (moderate–severe) (11.16), penicilliosis (moderate–severe) (11.29) Visceral and cutaneous leishmaniasis (11.20) (Other indications: disseminated deep fungal infections)	Infusion, intravenous: 50 mg vial (dissolve 50 mg in 10 ml sterile water and make up to 500 ml with 5% glucose to give 100 mcg/ml) See Section 11.5 for dosing and administration. IV: 0.7 mg/kg until clinical improvement (usually 14 days) THEN itraconazole maintenance therapy See Section 11.20	Common: Acute infusion reactions (fever, chills, headache, hypotension) – these become less frequent over time; thrombophlebitis, anaemia, nephrotoxicity (major dose-limiting toxicity); hypokalaemia, hypomagnesaemia Infrequent or rare: Anaphylaxis occurs rarely with any IV amphotericin product. Headache, anorexia, weight loss, nausea, vomiting, diarrhoea; muscle and joint pain; cardiovascular toxicity (arrhythmias and cardiac arrest, due to rapid infusion and electrolyte disturbances); neurological disorders, abnormal liver function (discontinue treatment)	Pregnancy/breastfeeding: Use with caution in pregnancy or breastfeeding, if clinically indicated. Available safety data are limited. Use with caution: In renal impairment use liposomal amphotericin if possible (see Section 11.5). Administration: Follow instructions in Section 11.5. Give via D5W IV infusion (incompatible with NS, 1/2 NS, other saline-containing solutions, or preservatives) over 2 to 6 hours. Infusion time may be reduced to approximately 1 hour in patients who tolerate treatment well. If the patient experiences discomfort during infusion, the duration of infusion may be increased. Existing IV line should be flushed with D5W prior to infusion (if not feasible, administer through a separate line).

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Amphotericin B (liposomal) Visceral and cutaneous leishmaniasis (11.20) Cryptococcal meningitis in patients with renal impairment (11.5)	<i>Injection, powder for reconstitution: 50 mg (contains soy, sucrose 900 mg)</i> See Section 11.20 for dosing. See Section 11.5 for dosing.	See conventional amphotericin B. Incidence of decreased renal function and infusion-related events are lower than with conventional amphotericin B. For a patient who experiences chills, fever, hypotension, nausea, or other nonaphylactic infusion-related reactions, premedicate with the following drugs 30–60 minutes prior to drug administration: paracetamol OR hydrocortisone 50–100 mg.	If available, liposomal amphotericin B should be used instead of conventional amphotericin B in patients with renal failure; or switch from amphotericin B if renal impairment develops. Available at decreased cost through WHO-Gilead partnership for leishmaniasis (see http://www.gilead.com/visceral_leishmaniasis). Pregnancy/breastfeeding: Use with caution in pregnancy or breastfeeding, if clinically indicated. Available safety data are limited. Use with caution: In renal impairment; continue to monitor electrolytes and renal function tests despite lower incidence of adverse renal effects than with conventional amphotericin. Administration: See conventional amphotericin B.
Ampicillin Severe infections (septic abortion (10.15.6), septic shock (3.1.5), upper urinary tract infection (pyelonephritis) (11.44)) Empirical therapy for meningitis, if ceftriaxone not available (10.10b.3) Initial empirical antibiotics for emergency management (QC p. 19) Cholangitis (10.7a.2), peritonitis (10.7a.2) Listerial meningitis (10.10b.3) Leptospirosis (11.22) (Other indications: Mastoiditis; osteomyelitis)	<i>Injection: 500 mg vial, 1 g vial</i> IM/IV: 250 mg/kg daily, divided every 4 hours (range 6–12 g/day) AND gentamicin for 10–14 days AND, for septic abortion, clindamycin IM/IV: 250 mg/kg daily divided every 4 hours (range 6–12 g/day) AND gentamicin + cotrimoxazole for 10–14 days IM/IV: 2 g in a single dose AND gentamicin 240 mg IV: 2 g every 4 hours AND gentamicin + metronidazole for 10–14 days IV: 2 g IV every 4 hours for at least 21 days PLUS gentamicin IV: 500 mg to 1 g every 6 hours for 7 days	Common: Nausea, vomiting, diarrhoea Infrequent or rare: Haemolytic anaemia, interstitial nephritis, blood disorders, <i>C. difficile</i> colitis Hypersensitivity reaction (discontinue if severe), anaphylaxis	Pregnancy: Not known to be harmful. Breastfeeding: Considered safe Contraindications: Severe hypersensitivity to penicillins and other beta-lactams Use with caution: In mild hypersensitivity to beta-lactams with renal impairment Erythematous rashes common in glandular fever, lymphocytic leukaemia, Epstein-Barr virus, cytomegalovirus infection. Administration: Injection contains 2.7 mmol (62 mg) sodium/gram. Reduce dose in severe renal failure.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
ANTIRETROVIRALS for HIV infection Note: antiretroviral therapy must include at least 3 ARVs. See Sections 13 and 14.			
Abacavir (ABC)	Tablet: 300 mg Oral liquid: 100 mg/ml	Common: Nausea, vomiting, diarrhoea Infrequent or rare: Life-threatening hypersensitivity reactions, blood disorders, lipodystrophy, lactic acidosis	Pregnancy/breastfeeding: Limited safety data available, but continuation of antiretroviral therapy throughout pregnancy is recommended. Potential alternate if AZT and TDF are not tolerated during pregnancy. Use with caution: In hepatic disease. Can lead to potential life-threatening lactic acidosis. Safe to use after lactic acidosis. Do not use after ABC hypersensitivity reactions.
Antiretroviral (13, 14)	Oral: 300 mg twice daily or 600 mg once daily If hepatic impairment (mild), 200 mg twice daily (maximum)		
Atazanavir + ritonavir (ATV/r)	Capsule 300 mg atazanavir; capsule ritonavir 100 mg	Common: Diarrhoea, indirect hyperbilirubinaemia, clinical jaundice, hyperglycaemia, fat maldistribution, nephrolithiasis; prolonged PR interval – first-degree symptomatic AV block in some patients	Pregnancy/breastfeeding: Generally safe Use with caution: In haemophilia: possible increased bleeding episodes Administration: Take both at same time with food
Antiretroviral (13, 14)	Oral: 300 mg ATV and 100 mg ritonavir once daily		
Didanosine (ddI)	Capsule: 400 mg	Common: Nausea, vomiting, diarrhoea, Infrequent or rare: Acute pancreatitis, lactic acidosis, peripheral neuropathy, lipodystrophy, hyperuricaemia	Pregnancy/breastfeeding: Lactic acidosis with hepatic steatosis may be more frequent in pregnant women. Should be used during pregnancy only if there is NO alternative. Contraindications: Do not use with stavudine (d4T). Use with caution: In renal or hepatic impairment; see dose adjustment in Section 11.31.
Antiretroviral (13, 14)	Oral: 250 mg once daily if <60 kg; 400 mg once daily if >60 kg Ensure sufficient antacid from buffered tablets at least 1 hour before food or on empty stomach		

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Efavirenz (EFV) Antiretroviral (13, 14)	Capsule: 100 mg, 200 mg Tablet: 600 mg Oral solution: 150 mg/5 ml Oral: 600 mg once daily See Section 14 for use of EFV as ARV prophylaxis against MTCT of HIV.	Common: Hypersensitivity reaction: Rash – generally mild, often resolves within 3–5 days without need to change ART (but discontinue if severe) Often self-limiting CNS toxicities: Insomnia, abnormal dreams; less commonly, persistent and severe CNS toxicity (depression, confusion) Hepatic toxicity: Elevated liver enzymes (if seropositive for hepatitis B or C) Hyperlipidaemia; male gynaecomastia; potential teratogenicity (first trimester of pregnancy) Infrequent or rare: Stevens-Johnsons syndrome	Pregnancy/breastfeeding: Potential risk of teratogenicity in first trimester. Safe after first trimester. Provide effective contraceptives after delivery. Discuss risk and benefit of using EFV with women who are planning to become pregnant or who may become pregnant. If a woman is diagnosed as pregnant before 28 days of gestation or plans to become pregnant, EFV should be stopped and NVP or a PI substituted. There is no indication for termination of pregnancy in women exposed to EFV in the first trimester. Provide effective contraceptive for women of reproductive age who are taking EFV and choose to avoid conception. Use with caution: In hepatic impairment (avoid if severe), severe renal impairment, elderly, history of mental illness or substance abuse. Rash usually resolves within 3-5 days, but discontinue ART and monitor AST or ALT if rash severe or if accompanied by blisters, desquamation, involvement of mucous membranes, or fever; seek urgent medical care. See Section 13. Seek urgent medical care also in severe CNS toxicities or if depression, confusion occurs. Contraindications: History of severe psychiatric illness Administration: Take at bedtime on an empty stomach. Suggested substitution if not tolerated: NVP, or bPI if neither NNRTI is tolerated; or triple NRTI if no other options
Emtricitabine (FTC) Antiretroviral (13)	Capsule: 200 mg Oral liquid: 10 mg/ml Oral: 200 mg once daily	Common: Nausea/vomiting, abdominal pain, diarrhoea, headache, peripheral neuropathy	Pregnancy/breastfeeding: There is limited experience with use during pregnancy. Use with caution: In renal impairment, hepatic disease Active against HBV (see Section 11.14)

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Lamivudine (3TC) Antiretroviral (13, 14)	Tablet: 150 mg Oral solution: 50 mg/5 ml Oral: 150 mg twice daily OR 300 mg once daily See Section 14 for use as ARV prophylaxis against MTCT of HIV.	Common: Well tolerated; occasional nausea and diarrhoea, pancreatitis	Pregnancy/breastfeeding: Favourable safety profile Active against HBV (see Section 11.14)
Nevirapine (NVP) Antiretroviral (13, 14)	Tablet: 200 mg Oral: 200 mg once daily for first 14 days; THEN (if no rash present) 200 mg twice daily See Section 14 for use as ARV prophylaxis against MTCT of HIV.	Common: Nausea and vomiting, rash (including Stevens-Johnson syndrome) Infrequent or rare: Toxic epidermal necrolysis, hepatotoxicity jaundice, abdominal pain, diarrhoea, hypersensitivity reactions	Pregnancy/breastfeeding: Avoid using in pregnant women with CD4 count >350. In women with CD4 count of 250–350 increased risks of maternal hepatotoxicity; use, with close monitoring, as benefit exceeds risk in those who require ART. Contraindications: Severe hepatic failure, pregnancy with CD4 >350 Use with caution: In hepatic impairment or history of chronic hepatitis; monitor liver function. Counselling: Advise patients about the signs or symptoms of hypersensitivity reactions. Seek immediate medical attention if such symptoms develop.
Lopinavir + ritonavir (LPV/r) Antiretroviral (13, 14)	Tablet: Fixed dose combination of (LPV 200 mg + RTV 50 mg) and (LPV 100 mg + RTV 25 mg) Oral: LPV 200 mg + RTV 50 mg (2 tablets) twice daily In patients taking rifampicin, use ritonavir super-boosting (LPV 400 mg + RTV 400 mg OR LPV 800 mg + RTV 200 mg) twice daily.	Common: GI intolerance, nausea, vomiting, diarrhoea, headache Infrequent or rare: Hyperlipidaemia (especially hypertriglyceridaemia), elevated transaminases, hyperglycaemia, fat maldistribution, PR interval prolongation, QT interval prolongation, torsade de pointes, lipodystrophy	Pregnancy/breastfeeding: Continuation of antiretroviral therapy throughout pregnancy is recommended; close monitoring of blood glucose recommended, as risk of pregnancy-related hyperglycaemia may be increased. Use with caution: Close clinical and hepatic enzyme monitoring needed in patients taking ritonavir super-boosting. Administration: Possible increased bleeding episodes in patients with haemophilia Suggested substitute: ATV/r

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Saquinavir (SQV) Antiretroviral (13, 14)	Capsule: 200 mg Oral: 1 g twice daily (with 100 mg ritonavir booster)	Common: Diarrhoea, headache, increased transaminases, dyslipidaemia, hyperglycaemia, buccal and mucosal ulceration, unconjugated hyperbilirubinaemia	Pregnancy/breastfeeding: Continuation of antiretroviral therapy throughout pregnancy is recommended; close monitoring of blood glucose recommended, as risk of pregnancy-related hyperglycaemia may be increased. Use with caution: In severe hepatic impairment Administration: Take with a fatty meal or up to 2 hours after meal. Avoid garlic capsules, which reduce plasma saquinavir concentration)
Tenofovir (TDF) Antiretroviral (13, 14)	Tablet: 300 mg Oral: 300 mg tablet once daily	Common: Nausea, vomiting, diarrhoea, abdominal pain, headache, dizziness Infrequent or rare: Renal insufficiency, Fanconi syndrome, hypertriglyceridaemia, osteomalacia. Severe acute exacerbation of hepatitis may occur in HBV co-infected patients who discontinue TDF.	Pregnancy/breastfeeding: Concern of HBV flare if HBV–HIV co-infected mother stops the medication postpartum. Limited data available on potential maternal and infant bone toxicity. Contraindication: Renal impairment Use with caution: In underlying renal disease, age >40 years, BMI <18.5 (or body weight <50 kg), diabetes mellitus, hypertension, concomitant use of a bP1 or nephrotoxic drug Administration: Do not use with ddl (levels increased). Active against HBV (see Section 11.14)
Zidovudine (ZDV, AZT) Antiretroviral (13, 14)	Capsule: 250 mg Oral: 250–300 mg twice daily See Section 14 for use as ARV prophylaxis against MTCT of HIV.	Common: Nausea, vomiting, diarrhoea, headache, fatigue, myalgia Infrequent or rare: Bone marrow suppression (macrocytic anaemia, neutropenia); pancreatitis, lactic acidosis with hepatic steatosis, CNS problems, insomnia, pigmentation of nails, skin, oral mucosa	Pregnancy/breastfeeding: Well tolerated, risk of anaemia Contraindications: Abnormally low neutrophil counts or severe anaemia (Hb <7.0 g/dl and/or ANC <750 cells/mm ³); high risk for anaemia and neutropenia if CD4 count <200, BMI <18.5 (or body weight <50 kg), anaemia at baseline. Use with caution: In vitamin B12 deficiency, renal or hepatic impairment, elderly patients

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Artemether Severe malaria (note: IV artesunate is preferred) (QC, p.20 and 11.25)	Injection: 80 mg/ml in 1 ml ampoule Loading dose: IM (in anterior thigh): 3.2 mg/kg then 1.6 mg/kg daily until patient can take oral. Start appropriate oral treatment as soon as tolerated and give full course (see Section 11.25).	Common: Refer to artesunate. Neurotoxicity has been reported in animal studies, particularly with very high doses of intramuscular artemether, but has not been substantiated in humans.	Pregnancy: Both artesunate and quinine may be considered as options for the treatment of severe malaria during pregnancy. See note under artesunate Use with caution: In all patients, artemether IM should only be used if parenteral formulations of artesunate or quinine are not available, as its absorption may be erratic.
Artemether + lumefantrine Uncomplicated <i>P. falciparum</i> malaria, first-line (11.25)	Co-formulated tablets of 20 mg artemether + 120 mg lumefantrine Oral: 80 mg artemether and 480 mg lumefantrine twice daily for 3 days	Common: Abdominal pain, anorexia, diarrhoea, nausea/vomiting, palpitation, cough, headache, dizziness, sleep disturbances, asthenia, arthralgia, myalgia, pruritus, rash, QT prolongation Infrequent or rare: Paraesthesia, ataxia, hypoaesthesia, increased liver transaminases	Pregnancy/breastfeeding: Not recommended in first-trimester pregnancy unless no other treatment immediately available. Contraindications: History of arrhythmias, clinically relevant bradycardia, or CHF with reduced LV ejection fraction; family history of sudden death or prolonged QT interval. Administration: Lumefantrine absorption is enhanced by co-administration with fat. Counselling: Take this ACT immediately after a meal or drink containing at least 1.2 g fat. Dizziness may impair ability to perform skilled tasks such as operating machinery and driving.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Artesunate Severe malaria (QC p. 20, 11.25.5)	<i>Injection: 60 mg ampoule with separate ampoule of 5% sodium bicarbonate</i> IM/IV: 2.4 mg/kg on admission AND repeat at 12 hours, 24 hours; then once daily. Start appropriate oral treatment as soon as tolerated, and give full course. See Section 11.25.5.	Generally well tolerated Infrequent or rare: Mild gastrointestinal disturbances; dizziness, tinnitus; reticulocytopenia, neutropenia, elevated liver enzyme values, ECG abnormalities (bradycardia, prolongation of the QT interval). Type 1 hypersensitivity reactions in approximately 1 in every 3000 patients. Post-treatment hemolysis has been observed with high cumulative doses. (Patients should be monitored for signs of hemolysis after parasitologic cure.)	Pregnancy/breastfeeding: Both artesunate and quinine may be considered as options for the treatment of severe malaria during pregnancy. Treatment must not be delayed; so, if only one of the drugs artesunate, artemether, or quinine is available, then that should be started immediately. Give quinine if possible during first trimester; for second and third trimesters, artesunate is preferred, as it is for all adolescents and adults. Use with caution: In patients who must perform skilled tasks, such as operating machinery or driving, that would be impaired by dizziness. Administration: The solution should be used immediately after the powder is dissolved. It should not be used for intravenous infusion if the solution appears cloudy or if sediment is present.
Artesunate + amodiaquine Uncomplicated <i>P. falciparum</i> malaria, first-line (11.25)	<i>Co-formulated tablets of artesunate + amodiaquine, 25/67.5 mg, 50/135 mg, or 100/270 mg. Blister packs of separate scored tablets also exist.</i> Oral: 200 mg artesunate and 540 mg amodiaquine fixed-dose combination once daily for 3 days	Common: Anorexia, abdominal pain, nausea; somnolence, insomnia, cough. Infrequent or rare: Weakness, anaemia, vertigo. Amodiaquine (at higher doses and/or during prolonged treatment) may lead to leukopenia and neutropenia, agranulocytosis; nervous system disorders. Rare: Neuromyopathy, transient accommodation disorders, corneal opacification (regresses once treatment stops) but, very rarely, irreversible retinopathy; hepatobiliary disorders; severe, sometimes fatal hepatitis; slate-gray pigmentation of the skin, notably affecting the fingers and mucous membranes.	Pregnancy: Not recommended in first-trimester pregnancy unless no other treatment immediately available Breastfeeding: No data available on the excretion of artesunate/amodiaquine fixed-dose combination in breast milk. Continuation can be considered while taking into account safety profile of artesunate/amodiaquine fixed-dose combination tablets. Contraindications: Previous hypersensitivity to amodiaquine or artesunate, history of liver toxicity or neutropenia during treatment with amodiaquine, retinopathy (in cases of frequent treatment) Use with caution: Avoid if possible in PLHIV on zidovudine or efavirenz. Counselling: Somnolence, dizziness, or weakness may occur. May impair ability to perform skilled tasks such as operating machinery and driving.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Artesunate + sulfadoxine-pyrimethamine Uncomplicated <i>P. falciparum</i> malaria, first-line (11.25)	<u>Co-blistered, scored tablets of 50 mg artesunate and tablets of 500 mg sulfadoxine + 25 mg pyrimethamine</u> Oral: 200 mg artesunate once daily for 3 days AND 1500 mg/75 mg sulfadoxine- pyrimethamine as a single dose on day 1	Common: Nausea/vomiting, anorexia, diarrhoea Infrequent or rare: See cotrimoxazole.	Pregnancy: Not recommended in the first trimester unless no other treatment immediately available. Breastfeeding: Safe to use Use with caution: Do not choose this option if patient is taking cotrimoxazole prophylaxis. See cotrimoxazole.
Artesunate + mefloquine Uncomplicated <i>P. falciparum</i> malaria, first-line (11.25)	<u>Co-blistered, scored tablets of 50 mg artesunate and 250 mg base of mefloquine</u> Oral: 200 mg artesunate once daily for 3 days AND 1500 mg mefloquine, usually split on days 2 and 3 (e.g. 4 tablets on day 2 and 2 on day 3)	Common: Nausea/ vomiting, diarrhoea, abdominal pain, anorexia, headache, dizziness, loss of balance, sleep disorders (abnormal dreams) Infrequent or rare: Neurological and psychiatric disturbances (convulsions, depression, hallucinations, panic attacks, emotional instability, aggression, suicidal ideation), cardiac conduction problems, muscle weakness/rash, disturbances in liver function tests	Pregnancy: Not recommended during the first trimester unless no other treatment immediately available. Breastfeeding: Safe to use Contraindications: Do not give mefloquine within 60 days of prior administration. History of neuropsychiatric disorders, epilepsy, hypersensitivity to quinine Use with caution: Avoid in severe hepatic impairment, cardiac conduction disorders. Counselling: May impair ability to perform skilled tasks such as operating machinery and driving. These effects may continue up to 3 weeks after the last dose.
Artesunate + clindamycin Uncomplicated <i>P. falciparum</i> malaria, first-line (11.25)	<u>See clindamycin and artesunate</u> Oral: Artesunate 2 mg/kg once daily AND clindamycin 10 mg/kg twice daily for 7 days	See artesunate and clindamycin.	Pregnancy: Not recommended during the first trimester unless no other treatment available. Breastfeeding: See clindamycin.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Atropine Organophosphate poisoning (3.8.1) Beta-blocker overdose with hypotension (3.8.1) Beta-blocker overdose: prophylaxis before inserting NG tube (3.8.1) Block muscarinic effects of neostigmine (3.9.2) (Other indications: Bradycardia: heart block with hypotension)	<i>Injection: 1 mg (sulfate) in 1 ml ampoule</i> IM/IV: Give bolus 1–3 mg (aim for clear lungs, stable blood pressure and dry mucous membranes) THEN double initial dose if no improvement at 5 minutes. See Section 3.8.1 for further treatment and monitoring. IV: 0.5–1 mg. Repeat every 3–5 minutes to a total dose of 0.04 mg/kg IV: 0.6 mg IV: 25–30 mcg/kg given 30–60 seconds before neostigmine administration	Common: Dry mouth, tachycardia, blurred vision, photophobia, constipation, urinary retention, flushing, delirium, fever Infrequent or rare: Vomiting, headache, paralytic ileus, rash, acute angle-closure glaucoma, seizures	Pregnancy: May be used at recommended doses. Can affect fetal heart rate. Breastfeeding: Use with caution in breastfeeding; monitor infant (e.g. drying of secretions, temperature rise). May suppress milk production. Contraindications: Angle-closure glaucoma. Use with caution: In Down syndrome, myasthenia gravis, pyloric stenosis, ileus, prostatic enlargement, cardiac disorders, hypoxia, and in the elderly. Patients with pyrexia and in warm environments: Monitor temperature and keep patients cool. Use may precipitate acute attack of angle-closure glaucoma, particularly in the elderly or long-sighted.
Atropine eye drops Iritis, uveitis; preretinal treatment for bacterial corneal infection (10.12.2) (Other indications: Cycloplegic refraction procedures)	<i>Solution (eye drops): 0.5%, 1% (sulfate)</i> 1 drop (0.5% or 1% solution) up to 4 times daily	Common: Transient stinging, raised intraocular pressure, local irritation, hyperaemia, oedema, contact dermatitis, systemic toxicity (in very young and elderly)	Precautions: May cause sensitivity to light and blurred vision. Contraindications: Angle-closure glaucoma Counselling: Avoid skilled tasks, such as operating machinery or driving, until your vision is clear.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Azithromycin Uncomplicated genital chlamydial infections (10.15.4) Trachoma (10.12.5) Early syphilis (11.37) Mycobacterium avium complex (MAC) (11.27) Yaws (10.2.5)	Capsule: 250 mg, 500 mg, 600 mg Oral: 1 g as a single dose (>45 kg) OR 20 mg/kg as a single dose (<45 kg) Oral: 20 mg/kg up to 1 g once annually AND tetracycline ointment Oral: 2 g once (do not use in PLHIV) Oral: 600 mg daily AND ethambutol for 6 months Oral: 30 mg/kg (maximum 2000 mg) single dose	Common: Nausea/ vomiting, diarrhoea, abdominal pain and cramps, headache, dizziness, drowsiness, candida infections, taste disturbances Infrequent or rare: Constipation, hepatitis, hepatic failure, syncope, insomnia, agitation, anxiety, asthenia, paraesthesia, hyperactivity, thrombocytopenia, haemolytic anaemia, interstitial nephritis, acute renal failure, photosensitivity, tooth and tongue discoloration, <i>C. difficile</i> colitis; hypersensitivity reactions including anaphylaxis	Pregnancy/breastfeeding: Use, with caution, in pregnancy or breastfeeding, if clinically indicated. Available data on safety are limited. Contraindications: Hypersensitivity to macrolides Use with caution: In renal impairment, hepatic impairment; in combination with other medications that can prolong QT interval; myasthenia gravis Administration: Capsule should be taken on an empty stomach. Oral suspension can be taken with food. Counselling: Swallow whole at least 1 hour before or 2 hours after food. Do not take with aluminium or indigestion remedies containing magnesium.
Beclometasone inhaler Chronic asthma, COPD (10.6.4, 10.6.5)	Inhalation: aerosol 50 mg per dose (dipropionate); 250 mg (dipropionate) per dose See Section 10.6 for dosing.	Common: Oropharyngeal candidiasis: cough, dysphonia; bruising, facial skin irritation following nebulisation Infrequent or rare: Adrenal suppression, growth retardation in adolescents; impaired bone metabolism; cataract, glaucoma with high doses or when combined with oral steroids; paradoxical bronchospasm; urticaria, rash, angioedema, sleep disorders, anxiety, behavioural changes	Pregnancy: Benefit of treatment greater than risk. Not known to be harmful. Breastfeeding: Considered safe Use with caution: If active or quiescent TB possible.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Benzathine benzylpenicillin Streptococcal pharyngitis (10.17.9) Secondary prophylaxis of rheumatic fever (11.32) Early syphilis (11.37) Late syphilis (11.37) Yaws (10.2.5)	Injection: benzathine benzylpenicillin 1.8 g (equal to 2.4 million IU) in 5 ml vial IM: 900 mg (1.2 million IU) as a single dose IM: 900 mg (1.2 million IU) every 3–4 weeks IM: 1.8 g (2.4 million IU), divided between 2 sites, as a single dose IM: 1.8 g (2.4 million IU), divided between 2 sites, once weekly for 3 consecutive weeks IM: 900 mg (1.2 million IU) as a single dose	Common: Pain/inflammation at injection site Infrequent or rare: Hypersensitivity reactions, including anaphylaxis; haemolytic anaemia, interstitial nephritis, blood disorders, CNS toxicity, Jarisch-Herxheimer reaction	Pregnancy: Not known to be harmful. Breastfeeding: Considered safe (monitor infant) Contraindications: Severe hypersensitivity to penicillins or other beta-lactams Use with caution: With history of allergy, renal failure, heart failure. Administration: Given by deep IM injection only. Give doses of more than 900 mg as two injections at separate sites. Do not give IV. Avoid intrathecal injection.
Benznidazole Chagas disease (11.42)	Tablet: 100 mg Oral: 5 mg/kg daily, divided in 2 or 3 daily doses, over 60 consecutive days (maximum 300 mg/day)	Common: Dermatitis with cutaneous eruptions (rash and rash erythematous); generalized oedema; fever, myalgias, arthralgias; gastrointestinal disorders; depression of bone marrow; polyneuropathy, paresthesia, peripheral neuropathy, among others	Pregnancy/breastfeeding: Not recommended in first trimester; use, with caution, in second and third trimester or breastfeeding if clinically indicated. Available data on safety are limited. Contraindications: Renal impairment, hepatic impairment Administration: Preferably after meals

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Benzyl benzoate Scabies (10.2.3); empirical treatment of itching papular lesions (10.2.3) Pediculosis (10.2.8)	Lotion: 25% Topical: Follow instructions in 10.2.3. Apply to affected area THEN wash off 24 hours later (further applications may be needed after 7 and 14 days).	Infrequent or rare: Local irritation, burning sensation, itch dermatitis, CNS stimulation (e.g. seizures with excessive use)	Pregnancy: May be used in pregnancy at the recommended dose, but permethrin is the preferred treatment during pregnancy. Breastfeeding: May be used at the recommended dose if it is the treatment of choice; systemic absorption likely to be minimal with topical use, but excess lotion should be wiped from nipple areas, and infant skin contact, minimized. Use with caution: Do not use on inflamed or broken skin. Avoid contact with eyes and mucous membranes. Administration: See Section 10.2.3. Do not bathe before application. Application to the face and genitals can cause irritation; an alternative agent such as permethrin is preferable. Counselling: Remember to apply also between fingers and toes, under nails, in skin folds, navel, between the buttocks, and on groin area. If you wash your hands or any other parts of the body during the treatment period, you should reapply the lotion to the washed areas.
Benzoyl peroxide Acne (10.2.3)	Cream or lotion: 2.5%, 5% Initially apply directly to clean skin on alternate days, increasing frequency to 1–2 times daily as tolerance to irritant effect develops. Continue until 2 weeks after lesions disappear.	Common: Initial irritation: skin dryness or peeling, feeling of warmth, mild stinging, erythema, but subsiding with continued use (in some cases may need to reduce frequency of application or temporarily suspend use) Infrequent or rare: Contact sensitivity (occasionally, even one application can cause severe irritation).	Pregnancy/breastfeeding: Safe to use Use with caution: Avoid contact with eyes, mouth, and mucous membranes; avoid use of occlusive dressings. Counselling: Before applying, wash affected area with mild soap or soap substitute and warm water. Gently pat dry. Then apply a thin layer to the affected area and rub in gently. May bleach fabrics, hair, and skin. Avoid excessive exposure to sunlight.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Benzylpenicillin (penicillin G) Streptococcal endocarditis (11.10) Neurosyphilis (11.37) Severe anthrax (10.2.10, 10.6.2) Leptospirosis (11.22) (Other indications: Otitis media; gas gangrene; actinomycosis; osteomyelitis; brain abscess)	Intravenous: 600 mg (1 million IU); 3 g (5 million IU) (sodium or potassium salt) in vial IV: 7.2–10.8 g (12–18 million IU) daily in 4 or 6 equally divided doses for 4 weeks IV: 1.8–2.4 g (3–4 million IU) every 4 hours for 2 weeks IV: 2.4–3.6 g (4–6 million IU) every 6 hours for 7–10 days IV: 900 mg (1.5 million IU) every 6 hours for 7 days	See benzathine benzy/penicillin.	See benzathine benzy/penicillin.
Betametasone (topical) Severe inflammatory skin conditions including papular lesions; eosinophilic folliculitis; pityrosporum folliculitis; papular urticaria; eczema; contact atopic dermatitis; numular eczema; seborrhoeic dermatitis; psoriasis; itchy papular lesions (10.2) (Other indications: Lichen planus)	Cream, ointment: 0.1% Apply sparingly 1 or 2 times daily.	Common: Folliculitis, steroid rosacea, perioral dermatitis, atrophy-thinning of the skin, striae, depigmentation, dilated vessels, acne at site of application, exacerbation of local infection or worsening of the condition; local trophic changes (particularly on the face and in skin folds) Infrequent or rare: Contact dermatitis, hyperaesthesia, subcutaneous tissue atrophy, hypertrichosis	Pregnancy/breastfeeding: May be used in pregnancy and breastfeeding; systemic absorption likely to be minimal with topical use. Contraindications: Untreated skin infections, broken skin, rosacea, acne, perioral dermatitis Use with caution: In psoriasis (may precipitate severe pustular psoriasis on withdrawal; avoid in widespread plaque psoriasis). If used on a large area of the body or for a long time, risk for adrenal suppression (particularly with an occlusive dressing). Avoid use on the face for more than 7 days. Secondary infection requires treatment with an appropriate antimicrobial. Counselling: Apply a thin layer by smoothing gently into skin, preferably after bathing.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Biperiden Acute dystonic reaction (QC p. 29) Antipsychotics causing extrapyramidal side-effects such as parkinsonism or dystonia (10.11.4)	Injection: 5 mg IM/IV Tablet: 2 mg IM: 5 mg (give IV if condition life-threatening); maximum 20 mg in 24 hours Oral/IV: Start with 1 mg twice daily. Increase to 2 mg 3 times daily to a target dose of 3–12 mg daily.	Common: Nausea, vomiting, dry mouth, constipation, dyspepsia, blurred vision, mydriasis, dry eyes; urinary retention, tachycardia; sedation, confusion, memory disturbance especially in elderly Infrequent or rare: arrhythmia, dizziness, drowsiness, headache, hallucinations, fever, anaphylaxis, acute angle-closure glaucoma, myasthenia gravis, gastrointestinal obstruction	Pregnancy/breastfeeding: Appears to be safe
Buprenorphine Opioid withdrawal (3.6.2) Opioid substitution treatment (17.4)	Tablet given sublingually: 2 mg, 8 mg SL: 2–16 mg/day for 3–14 days SL: Start at 2–8 mg and increase by up to 8 mg daily as needed up to a maximum of 32 mg. Average dose is usually 12–16 mg daily.	Opioid agonist effects Common: Euphoria; constipation, anorexia, nausea, vomiting, sweating, headache, dizziness, vasodilatation; dry mouth, fatigue, sedation, anxiety, postural hypotension, miosis, decreased libido Infrequent or rare: Hallucinations, confusion, spasm of urinary or biliary tract, hypotension, vertigo, bradycardia, tachycardia, palpitations, hypothermia, rash, facial flushing, urticaria Opioid antagonist effects (which can occur if used soon after a full opioid agonist such as methadone) mimic those of naltrexone in opioid dependence. Common: Vomiting, nausea, diarrhoea, anxiety, agitation, restlessness, insomnia, muscle aches, sweating, dilated pupils, tachycardia, hypertension. Infrequent or rare: Delirium, involuntary ejaculation	Pregnancy: Previously commenced therapy can be maintained. Breastfeeding: Caution in breastfeeding; poor oral bioavailability, but monitor infant for opiate side-effects. Use with caution: Must not be given while the person has any signs of opioid toxicity due to risk of precipitating withdrawal syndrome. Buprenorphine has both opioid agonist and antagonist properties and may precipitate withdrawal symptoms in people with high degree of tolerance to opioids, including those prescribed opioids for pain. When injected, buprenorphine has a similar abuse potential to injected heroin and may lead to dependence. Unlike most opioid analgesics, the effects of buprenorphine are only partially reversed by naloxone. Counselling: Place the tablet under the tongue and keep in place until dissolved. Do not chew or swallow the tablet.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Calamine lotion Mild pruritus; mild drug reactions (10.2)	Lotion: <i>calamine 15%, zinc oxide 5%, bentonite 3%, sodium citrate 0.5%, liquid phenol 0.5%, glycerine 5% + water to 100 ml</i> IV: 100 mg/ml in 10 ml ampoule (10%)		
Calcium gluconate Hypotension with calcium channel blocker overdose (3.8.1) Antidote for magnesium sulfate toxicity (QC p. 28) Hyperkalemia with ECG changes or K >6.5 mmol/l (5.2.2) (Other indication: Hypocalcaemic tetany)	IV: 0.6 ml/kg (600 mg/kg) to a maximum of 30 ml over 5 minutes; can be repeated every 10–20 minutes, up to 4 doses IV: 1 g (10 ml of 10% solution) over 10 minutes IV: 10 ml of calcium gluconate 10% slowly, given over 2–5 minutes; may repeat after 5 minutes titrated and adjusted to ECG improvement	Common: Nausea, vomiting, constipation; injection site reactions; fall in blood pressure Infrequent or rare: Bradycardia/arrhythmia, peripheral vasodilation, renal calculi, severe tissue damage with extravasation	Pregnancy: Use only when indicated (no controlled trials or animal studies). Breastfeeding: Calcium in breastmilk is normal nutritional component. Contraindications: Conditions associated with hypercalcaemia and hypercalciuria (for example, some forms of malignant disease) Use with caution: In renal impairment, sarcoidosis, history of nephrolithiasis Administration: Monitor ECG (and plasma calcium if feasible) with IV administration. Do not give by subcutaneous or IM route, as it will cause tissue necrosis. For infusion, dilute 100 ml of calcium gluconate 10% in 1 litre of glucose 5% or sodium chloride 0.9%, and give at an initial rate of 50 ml/hour, adjusted according to response.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Carbamazepine Generalized tonic-clonic seizures, partial seizures (3.5, 10.10c)	Oral liquid: 100 mg/5 ml Tablet (chewable): 100 mg, 200 mg Tablet (scored): 100 mg, 200 mg Oral: initially 100–200 mg 1–2 times daily (tablets) OR 50–100 mg every 6 hours as oral suspension THEN increase gradually by ~200 mg every week as needed (usual maintenance dose is 400–1,400 mg daily in divided doses) Oral: initially 200 mg daily at bedtime THEN gradually increase to 400–600 mg daily (divided doses); in severe cases may need 1000 mg (In elderly or medically ill: see Section 10.11.5)	Common: Drowsiness, dizziness, ataxia, headache, diplopia (may be associated with high plasma levels); dry mouth; mild transient generalized erythematous rash (withdraw if worsens or other symptoms); diarrhoea or constipation; leukopenia, thrombocytopenia; increased liver enzymes (usually not clinically significant) Infrequent or rare: Antibody deficiency, exfoliative dermatitis, Stevens-Johnson syndrome, systemic lupus erythematosus, agranulocytosis; aplastic anaemia, multiorgan hypersensitivity syndrome (including fever, severe skin disease, lymphadenopathy, haematologic abnormalities, hepatitis); psychiatric disorders, or facial dyskinesia, jaundice/hepatitis, acute renal failure, cardiovascular problems (arrhythmias, heart block heart failure), neuropsychiatric problems, impotence/male infertility, photosensitivity, pulmonary hypersensitivity, confusion/agitation (elderly), SIADH	Pregnancy/breastfeeding: Avoid in pregnancy. May be used in breastfeeding (monitor infant for drowsiness). Contraindications: Atrioventricular conduction abnormalities, history of bone marrow depression, porphyria; history of sensitivity to tricyclic antidepressants; oxcarbazepine, or other structurally related drugs; use of MAO-Is within past 2 weeks; on antiretrovirals for HIV (lowers levels) Use with caution: In hepatic impairment, renal impairment, cardiac disease, skin reactions, history of blood disorders, glaucoma, elderly. Monitor blood counts before and during treatment. Avoid sudden withdrawal. Prone to multiple drug interactions. Monitor for drug interactions when starting any new medication. Counseling: Take with food to help prevent stomach upset. This medicine may cause drowsiness, dizziness, or blurred vision, especially at the start of treatment or when the dose is increased. If affected, do not drive or operate machinery. May increase the effects of alcohol. Tell your health care provider that you are taking carbamazepine before starting any new medicine including herbal and OTC products.
Cefixime Gonorrhoea, uncomplicated (10.15.4) Gonococcal dermatitis-arthritis syndrome (10.13.2) Gonococcal septic arthritis (10.13.2)	Capsule: 400 mg Oral: 400 mg as a single dose Oral: 400 mg twice daily to complete total 7–10 days, after ceftriaxone for 3 days. Oral: 400 mg twice daily to complete 14–21 days, after ceftriaxone for 3 days.	Common: Nausea, vomiting, diarrhoea, abdominal discomfort, headache Infrequent or rare: Allergic reactions (including anaphylaxis), erythema multiforme; transient hepatitis, jaundice; leukopenia, thrombocytopenia, agranulocytosis, aplastic anaemia, haemolytic anaemia, interstitial nephritis	Pregnancy/breastfeeding: Safe to use Contraindications: Cephalosporin hypersensitivity Use with caution: In sensitivities to beta-lactam antibacterials (avoid if history of immediate hypersensitivity reaction); renal impairment Note: Can cause false positive urinary glucose (if tested for reducing substances) and false positive Coombs' test.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Ceftriaxone First-line empirical antibiotic coverage for emergency management (QC p. 19); septic shock (3.1.5); severe pneumonia (3.2.3); acute pyelonephritis (11.44); septic abortion (10.156) Endocarditis - viridans streptococci (11.10) Meningitis (10.10b.3) Cholangitis (10.7a.2) Spontaneous bacterial peritonitis (10.9.2) Gonorrhoea, uncomplicated (10.15.4, 11.13) Prophylaxis of contacts of meningococcal meningitis (10.10b.2) Gonorrhoea, arthritis (10.13.5) Gonorrhoea, conjunctivitis (10.12.2)	Injection: 250 mg, 1 g (as sodium salt) in vial IV: 1 g daily for 7–14 days for severe infection IV: 2 g daily for 4 weeks IV: 2 g twice daily for 5–14 days IV: 1 g daily AND metronidazole for 10–14 days IV: 2 g daily for 5–10 days IM: 250 mg as a single dose IM: 250 mg as a single dose IV: 1 g daily, continuing 1–2 days after improvement; THEN cefixime to complete 14–21 days treatment IM: 1 g as single dose pre-referral	Common: Diarrhoea, nausea; rash, electrolyte disturbances, pain and inflammation at injection site Infrequent or rare: Antibiotic-associated colitis (particularly with higher doses), hypersensitivity reactions (including anaphylaxis), erythema multiforme, transient hepatitis/jaundice, blood disorders (leukopenia, thrombocytopenia, agranulocytosis, aplastic anaemia, haemolytic anaemia), interstitial nephritis, pancreatitis, cholecystitis, pseudolithiasis (dose-dependent, asymptomatic, and reversible biliary sludge formation due to calcium-ceftriaxone complex, which usually resolves after treatment stopped), nephrolithiasis (formation of calcium-ceftriaxone renal stones, sometimes requiring treatment, usually reversible) pain, tenderness at IM injection site (can reconstitute with 1% lidocaine for patient comfort)	Pregnancy/breastfeeding: Considered safe in both pregnancy and breastfeeding (monitor infant for side-effects) Contraindications: Cephalosporin hypersensitivity (anaphylaxis, hives), porphyria, hypoalbuminaemia or impaired bilirubin binding Use with caution: In history of allergy to beta-lactams (minor rash), pre-existing gallbladder disease Administration: Incompatible with calcium; do not give via calcium-containing solutions. Divide IM dose over 1 g between 2 sites.

PID (10.15.5)	IM: 250 mg as a single dose AND doxycycline + metronidazole IV: 1 g daily for 7–10 days IV: 1 g daily for 7 days IV/IM: 1–2 g daily for 10–14 days		
Severe cellulitis (10.2.2)			
Leptospirosis- moderate to severe disease (11.22)			
Typhoid fever (11.43)			

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Charcoal, activated Antidote for some poisons (3.8)	Powder Oral (adults and adolescents >13 years of age): 50–100 g	Common: Nausea/vomiting, constipation, black stools, colicky abdominal pain Infrequent or rare: Diarrhoea, aspiration pneumonia, dehydration and electrolyte imbalances, gastrointestinal obstruction/faecal impaction in dehydrated patients	Pregnancy/breastfeeding: Considered safe in both pregnancy and breastfeeding Contraindications: Poisoning by hydrocarbons, with high potential for harm if aspirated; poisoning by corrosive substances — may prevent visualization of lesions caused by poison Use with caution: In drowsy or unconscious patients — risk of aspiration (intubate via nasogastric or gastric tube before administration) Not effective for poisoning with alkalis, acids, heavy metals, iron, lithium, toxic alcohols, glycols, or hydrocarbons such as kerosene Administration: Improve palatability by chilling. It may be easier for some patients to take it in a covered container with a large straw or with eyes shut.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Chloramphenicol Severe infections such as sensitive H. influenzae meningitis (10.10b.3); H. influenzae epiglottitis (3.2.2) Empirical treatment of bacterial meningitis if anaphylaxis to penicillin (10.10b.3) Epidemics of meningococcal meningitis (10.10b) Typhoid fever if known antibiotic sensitivity (11.43) Rickettsial diseases in pregnant women (11.33) Tetanus (11.39) (Other indications: Cerebral abscess; mastoiditis; relapsing fever; plague; psittacosis; tularemia; Whipple disease)	Capsule: 250 mg Injection: 1 g (sodium succinate) vial Oral injection: 500 mg/ml, 2 ml ampoule (for IM use only) IV: 1 gram every 6 hours IV: 1 gram every 6 hours AND cotrimoxazole IM (oily): 100 mg/kg (maximum 3 g) as a single dose; repeat after 24–48 hours if necessary Oral: 2 to 3 g in 4 divided doses for 14 days Oral: 500 mg 4 times daily for 5–7 days IV: 1 gram every 6 hours	Common: Nausea/vomiting, headache, reversible bone marrow suppression Infrequent or rare: Diarrhoea, stomatitis/glossitis, depression, hypersensitivity reactions (including anaphylaxis), aplastic anaemia, peripheral/optic neuritis, minor disulfiram-like reactions, grey baby syndrome (premature and newborn infants), <i>C. difficile</i> colitis	Pregnancy: Use with caution in pregnancy, risk of neonatal "grey syndrome" with high doses close to term. Breastfeeding: Not recommended with systemic use Use with caution: In porphyria reduce high doses as soon as clinically indicated; avoid repeated courses and prolonged use. Reduce dose in hepatic and severe renal impairment. Blood counts required before and during treatment. Plasma concentration monitoring required in the elderly and in hepatic or renal impairment; renal and hepatic dose adjustment required. Counselling: Tell your doctor if you get pale skin, sore throat, fever, tiredness or weakness, or unusual bleeding or bruising in the months after you stop taking the medicine.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Chloramphenicol – eye drops and ointment Superficial bacterial infections of the eye; prophylaxis against bacterial infection following minor ocular trauma (10.12.2) (Other indications: blepharitis)	Drops: 0.5% Ointment: 1% Eye drops: 1 or 2 drops, every 2 hours for the first 24 hours; THEN decrease to every 6 hours until 48 hours after resolution Ointment: May be used at bedtime (if eye drops used during day) or 3–4 times daily (if eye ointment used alone).	Common: Transient stinging Infrequent or rare: Unpleasant taste, hypersensitivity reactions (local allergy, angioedema, anaphylaxis), dermatitis. (Large, population-based studies have found no association between use of chloramphenicol eye drops and aplastic anaemia.)	Pregnancy/breastfeeding: Safe to use Contraindications: Chloramphenicol hypersensitivity Maximum duration of use: 7 days; prolonged use may lead to overgrowth of non-susceptible organisms
Chlorhexidine Multiple skin infections, itching (10.2.8) (Other indications: Disinfection of clean instruments; pre-operative skin disinfection and hand washing)	Solution: concentrate for solution, 5% 0.05% aqueous solution applied to affected areas	Infrequent or rare: Skin irritation/local contact dermatitis	Pregnancy/breastfeeding: May be used in pregnancy and breastfeeding. Systemic absorption likely to be minimal with topical use. Use with caution: Not for use in body cavities. Avoid contact with middle ear, eyes, brain and meninges.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Chloroquine Malaria caused by <i>P. malariae</i> , <i>P. vivax</i> , or <i>P. ovale</i> (11.25.3) (Other indications: Malaria prophylaxis; rheumatoid arthritis)	Tablet: chloroquine base (as phosphate or sulfate) 100 mg, 150 mg Oral: Initially 10 mg/kg (600 mg) as a single dose THEN 5 mg/kg (300 mg) 6–8 hours later THEN 5 mg/kg (300 mg) daily on next 2 days (24 and 48 hours after initial dose) THEN full anti-relapse treatment of 14 days of primaquine in <i>P. vivax</i> and <i>P. ovale</i> infections, except during pregnancy and breastfeeding	Common: Nausea, vomiting, diarrhoea, abdominal pain or cramps; headache; rash, pruritus Infrequent or rare (with prolonged treatment): Psychotic episodes, anxiety, personality changes, visual disturbances (dose-related retinopathy), hair loss, blue-black pigmentation of mucous membranes and skin, photosensitivity, tinnitus, hearing loss, bone marrow suppression, hypersensitivity reactions, atrioventricular block, porphyria, exacerbation of psoriasis, neuromyopathy	Pregnancy: Benefit exceeds risk when indicated for treatment of acute malaria. Breastfeeding: Use with caution. Use with caution: In hepatic and renal impairment, neurological disorders (avoid for prophylaxis if epilepsy history), severe gastrointestinal disorders, G6PD deficiency, pre-existing auditory damage, elderly. May exacerbate psoriasis or myasthenia gravis. If patient continues to deteriorate after chloroquine, administer quinine intravenously (suspect resistance). Counselling: Take with food to minimize nausea and vomiting. If part or all of a dose is vomited, the same amount should be taken again immediately.
Chlorphenamine (chlorpheniramine) Itching (10.2.8) Allergic reactions, anaphylaxis (adjunct) (3.1.3)	Tablet: 4 mg Injection: 10 mg/ml, 1 ml ampoule; (if necessary, injection solution can be diluted with sodium chloride 0.9% injection) Oral: 4 mg every 4–6 hours (maximum 24 mg daily) SC/IM/IV: 10–20 mg (maximum 40 mg in 24 hours); give IV over 1 minute	Common: Drowsiness, headache, psychomotor impairment, antimuscarinic effects such as urinary retention, dry mouth, blurred vision, and gastro-intestinal disturbances Infrequent or rare: Hypotension, palpitations, arrhythmias, extrapyramidal effects, dizziness, confusion, depression, sleep disturbances, tremor, convulsions, hypersensitivity reactions (including bronchospasm, angioedema, and anaphylaxis, rashes, and photosensitivity reactions), blood disorders, liver dysfunction, angle-closure glaucoma.	Pregnancy/breastfeeding: May be used at recommended doses in pregnancy and breastfeeding (monitor infant for drowsiness). Use with caution: In the elderly, may cause a paradoxical stimulation. Also, use with caution in prostate enlargement, urinary retention, ileus or pyloroduodenal obstruction, glaucoma, renal impairment, hepatic impairment, epilepsy. Counselling: Drowsiness may impair ability to perform skilled tasks such as operating machinery or driving. Drowsiness may diminish after a few days of treatment.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Chlorpromazine	Tablet: 100 mg Oral liquid: 25 mg/5 ml Injection: 25 mg/ml, 2 ml ampoule	Common: Extrapyramidal side-effects: Acute dystonic reaction or severe muscle spasm, stiffness, tremor, motor restlessness, agitation, Parkinson's syndrome; with prolonged administration, potentially irreversible involuntary movements (tardive dyskinesia) Autonomic side-effects: Drowsiness; orthostatic hypotension, dizziness; tachycardia; dry mouth; blurred vision; constipation; urinary retention Other: Nausea, anorexia, dyspepsia; headache; apathy, confusion, depression, nightmares, insomnia; weight gain; photosensitivity; rash; galactorrhoea, gynaecomastia, sexual dysfunction, impotence, menstrual irregularities; jaundice, altered liver enzymes Infrequent or rare, serious: Neuroleptic malignant syndrome, hyperthermia, hyperpyrexia, heat stroke; blood disorders, including leukopenia, thrombocytopenia, aplastic anaemia, lupus erythematosus-like syndrome; exfoliative dermatitis; seizures; cholestatic jaundice; arrhythmias; respiratory depression at high doses Other infrequent or rare: ECG changes; hypothermia; corneal and lens opacities; purplish pigmentation of the skin, cornea and retina (with prolonged high doses)	Pregnancy: Use, with caution, in pregnancy, if benefit is greater than risk. Breastfeeding: Use caution (monitor infant for drowsiness). Contraindications: CNS depression/impaired consciousness; bone marrow depression; phaeochromocytoma, porphyria, basal ganglia disease, parkinsonism Use with caution: In cardiovascular and cerebrovascular disorders, dementia, respiratory disease, epilepsy, acute infections, renal and hepatic impairment (avoid if severe), history of jaundice, leukopenia (blood count required if unexplained fever or infection), hypothyroidism, myasthenia gravis, prostatic hypertrophy, angle-closure glaucoma, organophosphate poisoning, subarachnoid haemorrhage, metabolic disturbances (hypokalaemia, hypocalcaemia, hypomagnesaemia), seizure disorders; in patients on other QT-prolonging medications Elderly/debilitated (including HIV stage 3 or 4): Haloperidol preferred; alternative is chlorpromazine at one-third to half adult dose. Avoid abrupt withdrawal. Check blood counts if unexplained fever or infection. Administration: Avoid skin contact with injection solution or oral liquid, as there is a risk of contact dermatitis. Oral hygiene is very important with regular use of oral liquid. Counselling: Warn patients that this medication may impair ability to perform skilled tasks such as operating machinery or driving. If taking liquid form, daily oral hygiene is very important.
Psychosis (including schizophrenia) (10.11.4)	Oral: Initially 75 mg at night Typical effective dose is 75–300 mg daily, but up to 1000 mg may be necessary in severe cases.		
Vomiting (10.7c.3, 14.1.11)	Oral/IM/IV: 25–50 mg 4 times daily until vomiting stops		
Tetanus spasms (11.39)	IM: 50–150 mg every 4–8 hours		

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Ciprofloxacin	Tablet: 250 mg Solution for IV infusion: 2 mg/ml	Common: Nausea, vomiting, diarrhoea, dyspepsia, abdominal pain Infrequent or rare: Hepatitis, jaundice, pancreatitis; dizziness, sleep disorders; convulsions, paraesthesia, movement disorders (discontinue use); hypersensitivity reactions (if severe rash, discontinue), petechiae, haemorrhagic bullae, erythema nodosum; photosensitivity reaction; psychiatric symptoms (depression, confusion, hallucinations – discontinue if occurs); haemolytic anaemia; C. difficile colitis. Tendon damage (including rupture) has been reported rarely in patients receiving quinolones. Tendon rupture may occur within 48 hours of starting treatment; cases have also been reported several months after stopping a quinolone. If tendinitis is suspected, the quinolone should be discontinued immediately.	Pregnancy/breastfeeding: Contraindicated in pregnancy and breastfeeding due to theoretical risk of injury to developing cartilage; consider alternatives where possible; use only if no available alternatives and benefit is greater than risk. Contraindications: History of quinolone-associated tendon disorder Use with caution: In elderly patients and those on corticosteroids (higher risk of tendonitis/rupture); in history of epilepsy or seizure, renal impairment, G6PD deficiency, myasthenia gravis (risk of exacerbation). Prone to multiple drug interactions. If TB infection is suspected, limit use if there are alternate antibiotics available. Administration: Give IV infusion over at least 60 minutes. Counselling: Oral: Take 1 hour before or 2 hours after meals. Drink plenty of fluids while taking it. Dairy products, antacids; and iron, zinc, or calcium supplements may reduce absorption; do not take within 2 hours of a ciprofloxacin dose. This medication may impair ability to perform skilled tasks such as operating machinery or driving. Avoid extended exposure to sunlight (discontinue if photosensitivity occurs; report to clinician).
Severe infections: acute pyelonephritis (11.44); typhoid fever (11.43)	Oral: 500 mg twice daily OR IV: 400 mg twice daily for 10–14 days		
Cholecystitis (10.7a.2)	Oral: 500 mg twice daily for 5–7 days AND metronidazole		
Cholangitis (10.7a.2)	IV: 400 mg twice daily AND metronidazole for 10–14 days		
Isosporiasis with intolerance to cotrimoxazole or resistance to ampicillin and cotrimoxazole (11.18)	Oral: 500 mg twice daily for 7 days Prophylaxis: Oral: 500 mg 3 times a week		
PID (10.15.5)	See table in Section 10.15.5.		
Uncomplicated gonorrhoea if susceptible (10.15.4); gonococcal conjunctivitis (10.12.2)	Oral: 500 mg as a single dose		
Anthrax: cutaneous (10.2.10)	Oral: 500 mg twice daily for 7–10 days		
Anthrax: severe (10.2.10)	IV: 400 mg twice daily for 7–10 days		
Prophylaxis, meningococcal meningitis in non-epidemic situations (10.10b.3)	Oral: 500 mg as a single dose		
Cholera (10.7d.2)	Oral: 1 g as a single dose		

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Clarithromycin	Tablet: 250 mg, 500 mg Oral suspension: 125 mg/5 ml, 250 mg/5 ml	Common: Dyspepsia. Infrequent or rare: Tooth and tongue discoloration; smell and taste disturbances; stomatitis, glossitis; headache, arthralgia, myalgia, hepatitis, pancreatitis; tinnitus, dizziness, insomnia, nightmares, anxiety, confusion, psychosis, paraesthesia, convulsions; hypoglycaemia, interstitial nephritis, renal failure; leukopenia, thrombocytopenia; pulmonary infiltration with eosinophilia; prolonged QT interval, torsade de pointes; on IV infusion, local tenderness, phlebitis. See also erythromycin adverse effects.	Pregnancy: Use alternative macrolides in pregnancy where possible. Breastfeeding: Caution in breastfeeding (monitor infant for side-effects) Contraindications: Sensitivity to macrolide antibiotics. Known interactions with many drugs. Use with caution: In renal impairment Counselling: Before starting or stopping any other medicines, tell your doctor that you are taking this medication.
MAC treatment in PLHIV (11.27)	Oral: 500 mg twice daily AND ethambutol 15 mg/kg daily for 6 months		
Helicobacter pylori infection eradication (10.7a.2)	Oral: 500 mg twice daily AND amoxicillin + omeprazole for 7 days		
Buruli ulcer (10.2.10)	Oral: 7.5 mg/kg twice daily (not to exceed 500 mg twice daily) AND rifampicin for 8 weeks		

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Clinداميسين Cellulitis (10.2.2) Complicated soft tissue infection including necrotizing fasciitis (10.2.2) Septic abortion (10.15.6) PID (10.15.5) Uncomplicated <i>P. falciparum</i> malaria in first trimester pregnant woman (11.25) (Other indications: Staphylococcal bone and joint infections; endocarditis prophylaxis)	Capsule: 150 mg Injection: 150 mg/ml, 2 ml ampoule Oral: 300-450 mg every 8 hours (maximum 450 mg every 6 hours) IM/IV: 900 mg every 8 hours AND ampicillin or cloxacillin IV: 900 mg every 8 hours AND ampicillin/penicillin + gentamicin for 14 days See table in Section 10.15.5. Oral: 600 mg twice daily AND quinine (600 mg every 8 hours) for 7 days	Common: Nausea, vomiting, diarrhoea, abdominal pain or cramps; rash, contact dermatitis Infrequent or rare: Unpleasant taste in the mouth; oesophagitis; altered liver enzymes, jaundice; blood disorders (leukopenia, granulocytosis, eosinophilia, thrombocytopenia); pain/induration/abscess after intramuscular injection, thrombophlebitis after intravenous injection, erythema multiforme, polyarthritis; C. difficile colitis (see Section 10.7d)	Pregnancy: Not known to be harmful Breastfeeding: Unknown safety; amount in milk probably too small to be harmful. Discontinue immediately if diarrhoea or colitis develops. Contraindications: Diarrhoeal states, porphyria Use with caution: In hepatic impairment. Monitor liver function during prolonged therapy. Administration: Dilute in glucose 5% or normal saline to concentration not more than 12 mg/ml and infuse slowly (not more than 30 mg/minute) to reduce risk of adverse cardiac effects. Single doses over 600 mg by intravenous infusion only and should not exceed 1.2 g over 1 hour. Counselling: Take with a full glass of water. Stop taking this medication and tell your doctor immediately if you develop diarrhoea.
Clinداميسين, topical Moderate acne (10.2.3)	Gel/lotion: clindاميسين 1% (as phosphate) Gel: Apply to lesions once daily. Lotion: Apply to the lesions twice daily. Continue until 2 weeks after lesions disappear.	Common: Dry, scaly, or peeling skin Infrequent or rare: Contact dermatitis, irritation, burning sensation, itch	Pregnancy/breastfeeding: Safe to use Counselling: Noticeable improvement usually seen in 6 weeks; however, 8-12 weeks of treatment may be required before maximum benefit is seen. Before applying, wash affected area with a mild soap and warm water, rinse thoroughly, and pat dry. Avoid contact with eyes, lips, and inside of your nose or mouth.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Clofazimine Leprosy (multibacillary) (11.21) Type 2 lepra reaction if inadequate response to prednisolone or if steroids contraindicated (11.21)	Capsule: 50 mg, 100 mg See Section 11.21 for dosing of clofazimine as part of multidrug treatment for multibacillary leprosy 200-300 mg daily in 2-3 divided doses for maximum 3 months	Common: Nausea/ vomiting (hospitalize if persistent), abdominal pain, headache, tiredness, brownish-black discoloration of lesions, pink to brownish-black discoloration of skin including areas exposed to light; reversible hair discoloration; dry skin; red discoloration of faeces, urine, and other body fluids; rash, pruritus. Infrequent or rare: Photosensitivity, acne-like eruptions, anorexia, eosinophilic enteropathy, bowel obstruction, dry eyes, dimmed vision, macular and subepithelial corneal pigmentation; elevation of blood sugar, weight loss, splenic infarction, lymphadenopathy, GI bleeding, constipation, taste disorder, dizziness, drowsiness	Pregnancy: No information Breastfeeding: May cause reversible skin discoloration in nursing infants Use with caution: In liver and renal impairment and pre existing GI symptoms Counselling: This medication is absorbed best if taken with food. Your skin may become pink to brownish-black while you are taking this medicine, but this is reversible. Avoid exposure to sunlight. Tears and urine may turn reddish-brown.
Clomipramine Obsessive-compulsive disorders; phobic states, panic attacks (10.11.7)	Capsule: 10 mg, 25 mg Oral: see instructions in Section 10.11.7.	See amitriptyline.	Pregnancy: Use, with caution, in pregnancy if clinically indicated and drug of choice. Breastfeeding: Caution in breastfeeding (monitor infant for drowsiness) Contraindications: See amitriptyline.
Cloxacillin Cellulitis (10.2.2) S. aureus endocarditis (11.10) Septic arthritis due to S. aureus (10.13.2)	Capsule: 500 mg, 1 g Powder for injection: 500 mg in vial Oral: 500 mg every 6 hours IV: 1-2 g every 6 hours IV: 2 g every 4 hours or 3 g every 6 hours for 6 weeks IV: 2 g every 6 hours for 2-4 weeks	See benzylpenicillin. Common: Nausea/vomiting, transient increases in liver enzymes and bilirubin Infrequent or rare: Cholestatic hepatitis	Pregnancy/breastfeeding: Considered safe in both pregnancy and breastfeeding (monitor infant for side-effects) Contraindications: Known severe hypersensitivity to penicillins or other beta-lactams Use with caution: In patients with history of mild hypersensitivity to beta-lactams, renal/hepatic impairment Counselling: This medicine is absorbed best if taken on an empty stomach at least half an hour before food or 2 hours after food.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Coal tar Chronic psoriasis, either alone or in combination with exposure to ultraviolet light (10.2.7)	Solution: 5%. Topical: apply directly to the affected area 1–3 times daily, preferably starting with lower strength preparation or add 100 ml to bath of tepid water and soak affected area for 10–20 minutes, once daily to once every 3 days for at least 10 baths.	Common: irritation, photosensitivity reactions; skin, hair, and fabrics discoloured. Infrequent or rare: hypersensitivity	Contraindications: inflamed, broken or infected skin. May be used with salicylic acid in psoriasis Administration: Skin protection possibly required to reduce photosensitivity reactions. Bathing can be alternated with exposure to ultraviolet (UVB) rays, allowing at least 24 hours between exposure and treatment with coal tar.
Codeine Acute or chronic pain (20.2, 20.4)	Tablet: 30 mg (phosphate) Oral: 30 mg every 4 hours; maximum dose for pain 240 mg; consider switch to morphine when a dose of 180 mg is reached.	Common: Nausea, vomiting, constipation; dizziness, headache, miosis, difficulty with micturition, urinary retention, dry mouth, dyspepsia Infrequent or rare: Dependence, biliary spasm, palpitations, bradycardia, tachycardia, hypothermia, hallucinations/mood changes, rash, in large doses, convulsions, respiratory depression, hypotension	Pregnancy/breastfeeding: Considered safe at recommended doses in both pregnancy and breastfeeding Contraindications: Conditions where inhibition of peristalsis should be avoided; abdominal distension; acute diarrhoeal conditions such as ulcerative colitis or antibiotic-associated colitis; acute respiratory depression Use with caution: With prolonged use, tolerance or dependence may occur in elderly and debilitated patients and patients with hepatic or renal impairment. 6–10% of Caucasians and 1–2% of Asians lack the enzyme CYP2D6 (necessary to metabolise codeine to morphine) and are, therefore, unlikely to obtain analgesia with codeine.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Cotrimoxazole- trimethoprim with sulfamethoxazole (TMP-SMX) (doses based on the trimethoprim component) <i>P. jirovecii</i> (<i>P. carinii</i>) pneumonia (PCP) (10.6.3) Primary prophylaxis for PLHIV (see 13.3 for indications for primary prophylaxis); secondary prophylaxis after PCP pneumonia (10.6.3), isoporiasis (11.18), toxoplasmosis (11.40) Acute bacterial meningitis- empirical treatment in absence of ceftriaxone (10.10b.3) Acute bacterial meningitis with anaphylaxis to penicillin (empirical or for confirmed <i>N.</i> <i>meningitidis</i>, <i>H. influenzae</i>, <i>S. pneumoniae</i>) (10.10b.3)	Tablet: sulfamethoxazole- 100 mg + trimethoprim 20 mg; sulfamethoxazole 400 mg + trimethoprim 80 mg (single- strength-SS); sulfamethoxazole 800 mg + trimethoprim 160 mg (double-strength=DS); Injection: sulfamethoxazole 80 mg + trimethoprim 16 mg/ml, 5 ml and 10 ml ampoules IV: 5 mg/kg (based on the trimethoprim component) 4 times daily for 21 days 1 DS tablet or 2 SS tablets once daily. Total daily dose is 960 mg (800 mg SMX + 160 mg TMP) IV: 10–20 mg/kg (based on the trimethoprim component) daily divided into 2–4 doses AND ampicillin IV: 10–20 mg/kg (based on the trimethoprim component) daily divided into 2–4 doses AND chloramphenicol	Common: Nausea, vomiting, diarrhoea; headache; hyperkalaemia; rash; anorexia, sore mouth, fever Infrequent or rare: Stevens-Johnson syndrome, toxic epidermal necrolysis, photosensitivity – discontinue immediately; drowsiness, liver damage (including jaundice and hepatic necrosis), pancreatitis, <i>C. difficile</i> colitis, myocarditis, cough and shortness of breath, pulmonary infiltrates, aseptic meningitis, depression, convulsions, peripheral neuropathy, ataxia, tinnitus, vertigo, hallucinations, hypoglycaemia, blood disorders (including leukopenia, thrombocytopenia, megaloblastic anaemia, eosinophilia), hyponatraemia, renal disorders (including interstitial nephritis), arthralgia, myalgia, vasculitis, systemic lupus erythematosus; rhabdomyolysis reported in HIV- infected patients; crystalluria	Pregnancy: Avoid in first trimester. However, if a pregnant woman with HIV requires cotrimoxazole prophylaxis, it should be started regardless of the stage of pregnancy. Pregnant women in malarial areas who are taking cotrimoxazole should not be given sulfadoxine- pyrimethamine-based intermittent preventive therapy for malaria. See Sections 13 and 14 for PMTCT and management of cotrimoxazole side-effects. Breastfeeding women living with HIV should continue to receive cotrimoxazole prophylaxis. Contraindications: Severe renal or hepatic impairment; previous severe reactions to sulfa-containing drugs. Use with caution: In predisposition to folate deficiency, patients with a sulfa allergy, elderly, and G6PD deficiency. See Section 13.3 for response to rash. Discontinue immediately if anaemia or new jaundice appears. Renal dose adjustment required. Avoid in blood disorders except with specialist supervision. Administration: IV: Dilute each 5 ml in 100–125 ml fluid, preferably glucose 5%, and infuse over 60–90 minutes. Maintain adequate fluid intake. Counselling: Take this medicine with food to reduce stomach upset. Tell your doctor if you get a sore throat, fever, troublesome rash, cough, difficulty breathing, joint pain, dark urine, or pale stools. If taking prolonged high dose treatment: Drink a lot of fluid – at least 2–3 litres daily.

Listeria meningitis with anaphylaxis to penicillin (10.10b.3) Severe pneumonia, suspect community- associated MRSA where cotrimoxazole has activity (3.2.3) Septic arthritis where MRSA suspected Epididymitis with suspected coliforms (10.16.4); donovoniasis (granuloma inguinale) (10.14.3) Empirical treatment for prostatitis (10.16.5) Persistent diarrhoea in immunocompromised patients (10.17d.2), isosporiasis (11.18) Toxoplasmosis (11.40) Uncomplicated cellulitis, furuncle, carbuncle, abscess (10.2.2), sinusitis (11.35)	IV: 10–20 mg/kg (based on the trimethoprim component) daily divided into 2–4 doses Add IV cotrimoxazole to regimen IV: 15–20 mg/kg (based on the trimethoprim component) daily divided into 2–4 doses for 2–4 weeks Oral: 2 SS or 1 DS twice daily for 14 days. Oral: 2 SS or 1 DS twice daily for 21 days Oral: 2 DS tablets twice daily for 14 days THEN 1 DS tablet twice daily for 3 weeks Oral: 2 DS tablets 3 times daily for 6 weeks Oral: 1 to 2 DS tablets twice daily for 7–10 days Oral: 1 DS tablet twice daily for 7 days	
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Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Dapsone Leprosy, paucibacillary (11.21) Leprosy, multibacillary (11.21) Alternative to cotrimoxazole prophylaxis for PLHIV allergic to sulfa-based medicines (13.3)	Tablet: 25 mg, 50 mg, 100 mg Oral: 100 mg daily for 6 months AND rifampicin on day 1 of each month Oral: 100 mg daily for 12 months AND rifampicin + clofazimine (see Section 11.21) Oral: 100 mg daily	Common: Nausea/vomiting, diarrhoea, abdominal pain or cramps; hepatitis, rash, fever, jaundice, headache, nervousness, blurred vision Infrequent or rare (dose-related and uncommon at doses used for leprosy): haemolysis, methaemoglobinemia, allergic dermatitis including Stevens-Johnson syndrome	Pregnancy: Third trimester use can result in neonatal haemolysis and methaemoglobinemia; folic acid 5 mg daily should be given to mother. Breastfeeding: Continue breastfeeding, monitor infant for jaundice. Use with caution: In anaemia, susceptibility to haemolysis Administration: Obtain full blood count before starting treatment; then again each week for the first month, and then each month during treatment. Counselling: Take dapsone with food to reduce stomach upsets. Stop medication and inform your doctor if troublesome rash occurs.
Deferoxamine (desferrioxamine mesilate) Acute iron poisoning (3.8.1) (Other indications: Chronic iron overload including hemoglobinopathies; aluminium overload in end-stage renal disease; diagnosis of iron or aluminium overload)	Powder for injection: 500 mg (mesilate) in vial IV (slow): Initially 15 mg/kg per hour THEN reduce after 4–6 hours so that total dose does not exceed 80 mg/kg in 24 hours.	Common: Nausea, vomiting, diarrhoea, abdominal cramps or pain; injection-site reactions (including redness, pain, swelling rashes and itch); hypotension (especially when given too rapidly by intravenous injection); asthma, fever, headache, arthralgia and myalgia; growth retardation; bone deformities Infrequent or rare: Disturbances of hearing and vision (including lens opacity and retinopathy); acute respiratory distress syndrome; neurological disturbances (including dizziness, neuropathy and paraesthesia), Yersinia and mucormycosis infections, rash, renal impairment, blood dyscrasias, anaphylaxis	Pregnancy: Use only if benefit greater than risk (risk of teratogenicity). Breastfeeding: Caution in breastfeeding; not recommended, but low oral bioavailability so unlikely to cause adverse effects Use with caution: In renal impairment, aluminium encephalopathy (may exacerbate neurological dysfunction) Administration: Perform eye and ear examinations before treatment and at 3-month intervals during treatment.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Dexamethasone Also, see steroid equivalents table in Section 8.2. Addison's disease (adrenal insufficiency) (3.4.3) Cerebral oedema associated with malignancy (20.3) Bacterial meningitis (consider before antibiotics but do not delay antibiotics) (10.10b.3) (Other indications: Some malignant neoplasms, to help prevent chemotherapy-induced emesis)	Tablet 500 mg, 4 mg Injection: 4 mg/ml, 1 ml ampoule IV: 8 mg; then repeat every 8 hours IV/IM: Initially 24 mg daily; THEN reduce by 2 mg/day to lowest effective maintenance dose. IV: 10 mg every 6 hours for 4 days	See prednisolone. In addition: Burning and tingling in perineal area (high dose IV treatment) Dexamethasone has a long duration of action and very high glucocorticoid activity in conjunction with insignificant mineralocorticoid activity. This makes it particularly suitable for high-dose therapy in conditions where fluid retention would be a disadvantage.	Pregnancy/breastfeeding: Use in pregnancy only if drug of choice and benefit greater than risk (risk of intrauterine growth retardation); unlikely to be harmful at low doses (<40 mg daily prednisolone equivalent) Breastfeeding: Caution in breastfeeding; prednisolone preferred Contraindications: Systemic infection (unless life-threatening or specific antimicrobial therapy given); avoid live virus vaccines in those receiving immunosuppressive doses. Use with caution: In patients with infections (symptoms may be masked until advanced stage; clinical presentation may be atypical); can activate or exacerbate TB, amoebiasis, strongyloidiasis; in the elderly and adolescents, hypertension, recent myocardial infarction, congestive heart failure, liver failure, renal impairment, diabetes mellitus (including family history), osteoporosis, glaucoma (including family history), severe affective disorder (particularly if history of steroid-induced psychosis), epilepsy, psoriasis, peptic ulcer, hypothyroidism, history of steroid myopathy. Adrenal suppression can occur during prolonged treatment and persist for years after stopping treatment. Administration: Monitor weight, blood pressure, fluid and electrolyte balance, and blood glucose throughout prolonged treatment.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Diazepam Anxiety (10.11.7); adjuvant analgesia for muscle spasms and anxiety-related pain (20.3) Convulsions (QC pages 6 and 19, 3.5); organophosphate or chloroquine poisoning (3.8) Acute alcohol withdrawal (3.7.1) Amphetamine or cocaine acute intoxication with severe agitation or anxiety (3.6) Insomnia	Tablet: 2 mg, 5 mg Injection: 5 mg/ml rectal solution: 10 mg/2 ml Oral: see Section 10.11.7 10 mg IV slowly, or rectally if no IV access; can be repeated after 10 minutes if convulsion does not stop. DO NOT give IM. Up to 20 mg, according to the severity of alcohol withdrawal, every 1-2 hours until patient is calm and mildly sedated (See Section 3.7.1 for how to titrate diazepam dose.) See Section 3.6.3. See Section 20.7.	Common: Sedation and other side-effects that appear similar to alcohol intoxication Infrequent or rare: Nausea/vomiting, diarrhoea, abdominal pain or cramps, respiratory depression (usually due to an excessive dose), withdrawal syndrome, hypotension, bradycardia, dependence and abuse, drowsiness and light-headedness the next day, confusion and ataxia (especially in elderly), amnesia, dependence, muscle weakness, paradoxical increase in aggression, visual disturbances, changes in libido, incontinence or urinary retention, blood disorders, raised liver enzymes/ jaundice, hypotension with rapid IV administration	Pregnancy: Avoid regular use (use only if necessary, e.g. seizure control). Use minimum effective dose for shortest duration; associated with increased risk of cleft palate; withdrawal symptoms in newborns have been reported Breastfeeding: Use with caution; short-acting benzodiazepines preferred if needed. Adverse effects possible (monitor infant for drowsiness and poor feeding) Contraindications: Respiratory disease, acute pulmonary insufficiency, sleep apnoea, severe hepatic impairment, myasthenia gravis, acute angle closure glaucoma Use with caution: In muscle weakness, history of alcohol or drug abuse, marked personality disorder, elderly or debilitated, porphyria, hepatic impairment (avoid if severe), renal impairment. Repeated use even in therapeutic doses can lead to a dependence syndrome. This is unlikely when it is prescribed in standard doses for up to 4 weeks, but beyond this the risk increases in proportion to the duration of treatment and daily dose. Avoid abrupt withdrawal. Counselling: This medication may impair ability to perform skilled tasks such as operating machinery or driving.
Diethylcarbamazine (DEC) Filariasis (11.12)	Tablet: 50 mg, 100 mg Oral: 6 mg/kg in 3 divided doses for 12 days OR 6 mg/kg as a single dose AND albendazole 400 mg in a single dose. Repeat every 6 months if microfilaraemic or symptomatic.	Common: Nausea, vomiting, headache, dizziness Infrequent or rare: immunological reaction	Pregnancy/breastfeeding: Not for use in pregnancy Contraindications: DEC should not be used in areas where onchocerciasis or loiasis is co-endemic, due to possible severe adverse reactions. Patients should be examined for co-infection before using DEC. Use with caution: In renal impairment, cardiac disorders

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Dihydro-artemisinin + piperazine A first-line antimalarial regimen for symptomatic, uncomplicated P. falciparum malaria (11.25.3)	Co-formulated tablet of 40 mg dihydroartemisinin and 320 mg piperazine See dosing in table in 11.25.3.	Common: Headache, eosinophilia, cough, potential for QTc prolongation	Pregnancy: Not recommended in the first trimester of pregnancy Administration: Take with water on an empty stomach. If a patient vomits within 30 minutes of taking drug, the whole dose should be re-administered; if a patient vomits within 30-60 minutes, half the dose should be re-administered.
Diloxanide Luminal amoebicide (11.1.1)	Tablet: 500 mg (furoate) Oral: 500 mg 3 times daily for 5 days	Common: Flatulence Infrequent or rare: Vomiting, pruritus, urticaria	Pregnancy: Defer treatment until after the first trimester. Breastfeeding: Manufacturer advises to avoid.
Dithranol Moderate to severe psoriasis (10.2.7) Note: Stability is a problem with dithranol preparations, especially those of low strength. Addition of salicylic acid, ascorbic acid or oxalic acid as an antioxidant stabilises dithranol products and prevents discolouration and inactivation. Stability appears to be best in soft paraffin and least in cream bases.	Ointment: 0.1-2% Topical: start with 0.1%, carefully apply directly to lesions only, leave in contact for 30 minutes, then wash off thoroughly, repeat application daily, gradually increasing strength to 2% and contact time to 60 minutes at weekly intervals; some 0.1–0.5% strength preparations are suitable for overnight use	Common: Local irritation; excessive erythema or spread of lesions (discontinue use); conjunctivitis following contact with eyes; staining of skin, hair, and fabrics.	Precautions: irritant (avoid contact with eyes, mucous membranes and healthy skin). Contraindications: hypersensitivity; avoid use on face, groin, acute eruptions, excessively inflamed areas. Administration: Localize application to plaques by application in Lassar's paste (zinc paste 98%, salicylic acid 2%, liquid paraffin 2) or white soft paraffin to surrounding skin. Dithranol must be protected from light and should be supplied in appropriate light-occlusive containers. Counselling: Wash hands thoroughly after use

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Dopamine Shock not responding to fluid boluses (3.1)	Infusion, intravenous: 200 mg in 5 ml ampoule IV: See vasopressor dosing table and instructions in Section 3.1.	Common: Nausea/vomiting, tachycardia, ectopic beats, palpitations, anginal pain; hypotension with dizziness, hypertension (in overdosage), headache, dyspnoea Infrequent or rare: Allergic reaction including anaphylaxis (due to sodium metabisulphite in products), abnormal ventricular conduction, bradycardia, piloerection, uraemia, mydriasis, peripheral vasoconstriction, asthma exacerbation, necrosis/gangrene at injection site	Pregnancy: Use only if benefit is greater than risk and no alternatives are available. Breastfeeding: May be used at recommended doses if benefit is greater than risk; short half-life and rapidly destroyed in the GI tract. Contraindications: Tachyarrhythmia, ventricular fibrillation, ischaemic heart disease, phaeochromocytoma, hyperthyroidism. Correct hypovolaemia before administration; maintain blood volume during treatment. Correct hypoxia before or at same time as starting treatment. See Section 3.1. Use with caution: If history of peripheral vascular disease Administration: Dilute before use in glucose 5% or normal saline and infuse into a large vein. Do not add to sodium bicarbonate or other strongly alkaline solutions.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Doxycycline Non-severe pneumonia (10.6.3) In combination with quinine or artesunate to complete course of treatment for severe malaria; as second-line <i>P. falciparum</i> antimalarial treatment; or for travellers (11.25)	Capsule: 100 mg Oral: 100 mg twice daily for 5-7 days See Section 11.25.	Common: Nausea/vomiting, diarrhoea, anorexia, epigastric burning, staining of growing teeth and occasional dental hypoplasia, photosensitivity Infrequent or rare: Tinnitus, hypersensitivity reactions, visual disturbances, hepatotoxicity, blood disorders, <i>C. difficile</i> colitis, oesophagitis and oesophageal ulceration	Pregnancy/breastfeeding: Not indicated after week 8 due to effects on fetal bone growth and dental discolouration; short courses can be used if alternative not appropriate; implicated in causing maternal hepatotoxicity, especially in third trimester (dose-related). May be used for a single short course of 7-10 days (monitor infant for side-effects). Use with caution: In hepatic impairment, porphyria, systemic lupus erythematosus
Syphilis, early latent- only for non-pregnant and penicillin allergic (11.37)	Oral: 100 mg twice daily for 14 days		Counselling: Advise patients to take capsules whole, with plenty of fluid, while sitting or standing to prevent oesophageal irritation; may give with milk/food to counter gastric irritation. Avoid exposure to sunlight or sunlamps.
Syphilis, late latent or undetermined duration (11.37)	Oral: 100 mg twice daily for 30 days		
Uncomplicated genital chlamydia (10.15.4); non-gonococcal urethritis; rickettsial diseases (11.33); leptospirosis (11.22)	Oral: 100 mg twice daily for 7 days		
Lymphogranuloma venereum (10.14.3)	Oral: 100 mg twice daily for 14 days		
Granuloma inguinale (donovaniasis) (10.14.3)	Oral: 100 mg twice daily until healed		
PID (10.15.5)	Oral/IV: See table in Section 10.15.5.		
Trench fever (11.2.3)	Oral: 100 mg twice daily for 4-6 weeks		

Bacillary angiomatosis (11.2.4)	Oral: 100 mg twice daily for 3 weeks		
Cat scratch fever (11.2.2)	Oral: 100 mg twice daily for 10-14 days AND rifampicin		
Cholera (10.7d.2)	Oral: 300 mg as a single dose		
Severe acne (10.2.3)	Oral: 50 mg daily for 3-6 months		
Anthrax: cutaneous (10.2.10); uncomplicated cellulitis, furuncle, carbuncle, abscess (10.2.2)	Oral: 100 mg twice daily for 7-10 days		
Leptospirosis (11.22)	Oral: 100 mg twice daily for 7 days		
Eosinophilic folliculitis (10.2.3)	Oral: 100 mg twice daily for 8-12 weeks		

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Effornithine Human African trypanosomiasis - <i>T.b. gambiense</i> meningo-encephalitic stage (11.41)	Injection: 200 mg/ml in 100 ml bottle IV (slow infusion over 2 hours): 200 mg/kg every 12 hours for 7 days AND oral nifurtimox (if nifurtimox is not available, give effornithine 100 mg/kg IV every 6 hours for 14 days.)	Common: Anaemia, leukopenia, thrombocytopenia, convulsions, impaired hearing, vomiting, abdominal pain, headache, facial oedema	Pregnancy/breastfeeding: Contraindicated in pregnancy and breastfeeding Use with caution: In renal impairment hospitalize and supervise closely during treatment. Administration: Monitor blood count for bone marrow suppression.
Enalapril HIV-associated nephropathy (11.31.5) (Other indications: Hypertension; heart failure)	Tablet 2.5 mg Oral: 2.5 to 40 mg daily. See Section 11.31.5 for dose adjustment.	Common: Hypotension if diuretics co-prescribed, cough, hyperkalaemia, headache, dizziness, fatigue, nausea, renal impairment, stomatitis, glossitis Infrequent or rare: See current formulary such as BNF	Pregnancy: Avoid in pregnancy. Breastfeeding: Amount excreted probably too small to be harmful. Contraindications: Hypersensitivity to ACE inhibitors (including angioedema) Use with caution: In renal impairment, peripheral vascular disease, aortic stenosis, hepatic impairment Administration: When starting, stop diuretics for 24 hours and start with a low dose at bedtime. Check renal function and electrolytes before starting. Counselling: While taking this medicine, you may feel dizzy on standing. Get up gradually from sitting or lying to minimize this; sit or lie down if you feel dizzy. Do not take potassium supplements while you are taking this medicine unless prescribed by your clinician.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Epinephrine (adrenaline) Anaphylaxis (QC p. 11, 3.1.3) Septic shock not responding to fluid boluses (3.1.5); cardiogenic shock (3.1.4) Severe bronchospasm if no inhaled salbutamol available (3.2.4)	Injection: 1 mg in 1 ml ampoule (1:1000) IM (1:1000): 0.5 ml in a single dose if 50 kg (0.3 ml if 30 kg, 0.4 ml if 40 kg) THEN may repeat at 5-minute intervals IV infusion: See vasopressor dosing table and instructions in Section 3.1 IM: 0.5 mg (0.5 ml of 1:1000)	Common: anxiety, headache, fear, palpitations, tachycardia, tremor, dizziness, sweating, pallor, nausea, vomiting, hyperglycaemia. Infrequent or rare: hypertension, arrhythmias, angina, pulmonary oedema (a sign of excessive dosage or extreme sensitivity), tissue necrosis (if injected IM or SC)	Pregnancy: May be used at the recommended doses Breastfeeding: Caution in breastfeeding. Monitor infant for side-effects such as irritability, restlessness, tremor. Use with caution: In hyperthyroidism, hypertension, diabetes mellitus, heart disease, arrhythmias, cerebrovascular disease, second stage of labour, and the elderly Administration: Intravenous epinephrine should be given only by those experienced in its use and in a setting where patients can be carefully monitored.
Ergometrine Haemorrhage in early pregnancy (QC p. 24) Postpartum haemorrhage: heavy bleeding continues with soft uterus after oxytocin given and placenta delivered or after manual removal of placenta (QC p. 25, 27)	Injection: 200 mcg (0.2 mg) (hydrogen maleate) in 1 ml ampoule IM: 0.2 mg in a single dose AND repeat IM or IV if bleeding continues IV (slow) or IM: 0.2 mg	Common: Nausea, vomiting Infrequent or rare: Headache, dizziness, tinnitus, abdominal pain, chest pain/palpitations, dyspnoea, bradycardia, transient hypertension, stroke, myocardial infarction, pulmonary oedema	Pregnancy: Contraindicated at induction of labour or first and second stages of labour as can cause premature uterine contractions and decrease uterine blood flow (possible fetal hypoxia and death) Breastfeeding: Single dose post-partum compatible with breastfeeding; avoid prolonged or multiple dosing Contraindications: Vascular disease, severe cardiac disease (angina pectoris), severe hypertension, severe renal or hepatic impairment, sepsis, pre-eclampsia or eclampsia Use with caution: In cardiac disease, hypertension, renal impairment, multiple pregnancy, porphyria

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Erythromycin	Tablet: 250 mg Capsule: 250 mg Infusion: 500 mg vial		
Early syphilis (11.37)	Oral: 500 mg 4 times daily for 14 days	Common: Nausea/vomiting, diarrhoea, abdominal pain or cramps	Pregnancy/breastfeeding: Not known to be harmful in pregnancy or breastfeeding
Late latent syphilis (11.37)	Oral: 500 mg 4 times daily for 30 days	Infrequent or rare: Allergic reactions, reversible hearing loss, jaundice, cardiac effects, myasthenia-like syndrome, erythema multiforme (Stevens-Johnson syndrome)/toxic epidermal necrolysis, QT prolongation, <i>C. difficile</i> colitis	Contraindications: Hypersensitivity to erythromycin or other macrolides, porphyria
Uncomplicated genital Chlamydia; non-gonococcal urethritis (10.15.4)	Oral: 500 mg twice daily for 7 days		Use with caution: In hepatic impairment, predisposition to QT interval prolongation (including electrolyte disturbances and concomitant use of drugs that prolong the QT interval). Prone to multiple drug interactions.
Lymphogranuloma venereum (10.14.3)	Oral: 500 mg 4 times daily for 14 days		Administration: Reconstitute injection with water for injection only. Dilute further for administration.
Granuloma inguinale (donovaniasis) (10.14.3)	Oral: 500 mg 4 times daily until healed		Parenteral erythromycin is an irritant and may cause thrombophlebitis.
Chancroid (10.1.3)	Oral: 500 mg 4 times daily for 7 days		Infuse at a rate of 1–5 mg/ml over 60 minutes or slower, via a central vein where possible. Avoid extravasation.
Severe pneumonia (3.2.3) or non-severe pneumonia not responding to oral therapy after 3 days (10.6.3)	IV/oral: 500 mg 4 times daily AND ceftriaxone OR ampicillin + gentamicin for 10–14 days		
Non-severe pneumonia (10.6)	Oral: 500 mg 4 times daily for 5–7 days		
Rheumatic fever, primary treatment if allergic to penicillin (11.32)	Oral: 250 mg 4 times daily for 10 days		

Rheumatic fever, secondary prophylaxis (11.32)	Oral: 250 mg twice daily		
Streptococcal pharyngitis (10.17.9)	Oral: 250 mg 4 times daily for 10 days		
Severe acne (10.2.3)	Oral: 500 mg twice daily		
Cholera in areas with tetracycline resistance (10.7d.2)	Oral: 500 mg 4 times daily for 3 days		

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Erythromycin, topical Mild to moderate acne (10.2.3)	Gel/lotion: erythromycin 2% Gel: Apply once a day in a thin film to affected area. Lotion: Apply twice daily in a thin film to affected area. See Section 10.2.3	Common: Dry skin, itch, stinging, burning feeling Infrequent or rare: Desquamation, erythema If irritation occurs, apply less frequently; if it persists, stop treatment.	Pregnancy/breastfeeding: Safe to use Counseling: Before applying, wash affected area with a mild soap and warm water, rinse thoroughly, and pat dry. Avoid contact with eyes, lips, and inside of your nose or mouth. Noticeable improvement may be seen in 3–4 weeks. However, 6–12 weeks of treatment may be required before maximum benefit is seen.
Ethambutol Tuberculosis, initial phase of combination therapy (15.3) MAC (in HIV-infected patients) (11.27)	Tablets: 100 mg, 400 mg Oral: 15 mg/kg daily or 30 mg/kg 3 times a week 15 mg/kg/day AND clarithromycin or azithromycin for 6 months	Common: Optic neuritis Infrequent or rare: peripheral neuritis (especially in legs), rash, pruritus, urticaria, thrombocytopenia, gout, jaundice	Pregnancy/breastfeeding: Considered safe in both pregnancy and breastfeeding (monitor infant for side-effects, including jaundice) Contraindications: History of optic neuritis and severe renal impairment Use with caution: In elderly with ocular defects, renal impairment Administration: Ocular examination recommended before and during treatment; patients should report visual disturbances immediately and discontinue treatment. Renal dose adjustment required. Counseling: If you see less clearly or colour vision is affected, stop the medication and tell your clinician.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Ethanol Ethylene glycol or methanol poisoning (3.8.1)	Oral liquid: 40–43% alcohol (vodka or whisky) Oral (or by NG): Loading: 1.8 mL/kg over 15–30 minutes (diluted). Maintenance dose: 0.2 mL/kg/hour (non-drinker) or 0.46 mL/kg/hour (heavy alcohol user)	Common: Signs of intoxication, confusion, drowsiness, coma, respiratory depression, hypoglycaemia	Pregnancy/breastfeeding: Judgement needed; however, fetus is more likely to be at risk from metabolic derangements from ethylene glycol/methanol poisoning than from ethanol as antidote. Avoid breastfeeding during ethanol treatment since ethanol passes into breast milk. Contraindications: Hypersensitivity to ethanol. Use with caution: Causes CNS depression, and effects are additive with other CNS depressants, e.g. benzodiazepines. Increased risk of hypoglycaemia in those with alcohol dependence. In patients taking drugs that inhibit aldehyde dehydrogenase, e.g. disulfiram, metronidazole, griseofulvin, use can result in acetaldehyde syndrome (nausea, flushing, autonomic instability).
Ferrous sulfate See elemental iron equivalents in Section 8.3. Iron-deficiency anaemia (10.18.3) Preventive iron supplementation in pregnant women without anaemia (14.1.1)	Tablet: equivalent to 60 mg iron Elemental iron, 60 mg for mild anaemia, 120 mg (plus 400 mcg folic acid) for moderate to severe anaemia daily Elemental iron 100 mg AND 400 mcg folic acid daily	Common: Constipation (particularly in older patients; occasionally leads to faecal impaction), diarrhoea, dark stools, nausea, epigastric pain, gastrointestinal irritation Infrequent or rare: Haemosiderosis	Pregnancy/breastfeeding: Safe to use Contraindications: Haemosiderosis, haemochromatosis; any form of anaemia not caused by iron deficiency; patients receiving repeated blood transfusions; parenteral iron therapy Administration: If side-effects occur, the dose may be reduced; alternatively, another iron salt may be used, but an improvement in tolerance may simply be a result of a lower content of elemental iron. See Section 10.18.3 for monitoring, duration of treatment. Counselling: Although iron preparations are best absorbed on an empty stomach, they may be taken after food to reduce gastrointestinal adverse effects. Keep out of children's reach.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Fluconazole	Capsule: 50 mg Infusion, intravenous: 2 mg/ml Oral liquid: 50 mg/5 ml	Infrequent or rare: Nausea/vomiting, abdominal pain, diarrhoea, headache, hepatic disorders, dizziness, seizures, alopecia, rash (withdraw treatment), hypersensitivity reactions, blood disorders, hypokalaemia	Pregnancy: Single dose unlikely to pose a risk to fetus, but avoid high dose or prolonged treatment. Breastfeeding: Safe in usual dosage for short-term treatment (monitor infant for side-effects) Use with caution: In renal impairment, sensitivity to other azoles. Monitor liver function (discontinue if signs or symptoms of hepatic disease). Prone to multiple drug interactions. Renal dose adjustment required. Counselling: Tell your doctor if you feel unusually tired, nauseous, or are not eating, or if you notice dark urine, pale faeces, or yellowing of the white of your eyes or skin.
Vaginal candidiasis (10.15.4, 11.4)	Oral: 150 mg as a single dose		
Recurrent oral candidiasis (10.17.3, 11.4)	Oral: 100-200 mg daily for 7-14 days		
Oesophageal candidiasis (10.7b.3, 11.4)	IV/oral: 100-200 mg daily for 14-21 days		
Invasive candida disease and candidemia (11.4)	IV/oral: 400 mg daily, continued for 14 days after last fever		
Cryptococcal meningitis, with or without amphotericin B (11.5)	See options in Section 11.5		
Cryptococcal meningitis (secondary prophylaxis) (11.5)	Oral: 200 mg daily until the patient is on successful ART and CD4 count is maintained above 200 for 6 months		
Pityriasis versicolor (10.2.8)	Oral: 400 mg as a single dose		
Dermatophytosis (10.2.7)	Oral: 150-300 mg weekly until cure (6-12 months)		
Leishmaniasis, cutaneous (11.20.1)	Oral: 200 mg daily for 6 weeks		

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Flucytosine (5-FC) Cryptococcal meningitis (as an adjunct to amphotericin) (11.5)	Capsule: 250 mg Infusion: 2.5 g in 250 ml IV/oral: 100 mg/kg daily in 4 divided doses for 14 days	Common: Nausea/vomiting, rash, diarrhoea Infrequent or rare: Cardiotoxicity, confusion, hallucinations, convulsions, headache, sedation, vertigo, alterations in liver function tests (hepatitis and hepatic necrosis reported), toxic epidermal necrolysis; blood disorders including thrombocytopenia, leukopenia, aplastic anaemia	Pregnancy: Avoid in pregnancy (teratogenic in animal studies); consider alternatives; use only if benefit greater than risk. Breastfeeding: Not recommended until more information available; use only if benefit greater than risk; consider alternatives. Use with caution: In elderly, renal impairment, pre-existing bone marrow suppression Administration: Monitor liver function, kidney function, and blood counts when use with amphotericin B (check weekly in renal impairment or in blood disorders). Renal dose adjustment required. Storage: Keep at 15–25 °C (forms fluorouracil above 25 °C and can precipitate below 15 °C). Counselling: Take the capsule with food.
Fluoxetine Moderate or severe depression (10.11.6); chronic anxiety disorders (10.11.7)	Capsule/tablet: 20 mg Oral: Initiate treatment with 20 mg daily (to reduce risk of side-effects that undermine adherence, may start at 10 mg (e.g. half a tablet) once daily and increase to 20 mg if the medication is tolerated), THEN if no response in 4–6 weeks or partial response in 6 weeks, increase dose by 20 mg (maximum dose 60 mg) according to tolerability and symptom response. For elderly and medically ill, see Section 10.11.6	Common: Restlessness, nervousness, insomnia, anorexia and other gastrointestinal disturbances, headache, sweating, decreased libido Infrequent or rare: Marked akathisia (inner restlessness), bleeding abnormalities in patients taking aspirin or non-steroidal anti-inflammatory drugs	Pregnancy: Use with caution; self-limiting withdrawal symptoms have been reported in newborns (e.g. distress, poor feeding, sleep disturbances). Breastfeeding: Not recommended; long half-life, may accumulate in breast milk; if required, use lowest effective dose. Contraindications: In combination with MAO-Is Use with caution: Renal or hepatic failure (consider dose reduction), diabetes mellitus. Prone to multiple drug interactions. Administration: Although symptomatic relief may be apparent within the first 1–3 weeks, optimum antidepressant effect usually requires at least 4 weeks or more of therapy. Watch for agitation and suicidal ideation and behaviour (see Section 10.11). If history of mania or bipolar disorder, use a mood stabilizer first (see Section 10.11.5 on bipolar disorder for details).

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Fluphenazine Antipsychotic: maintenance treatment of schizophrenia and other psychosis (10.11.4)	Injection: 25 mg, 1 ml ampoule Deep IM injection: Initially 12.5 mg in gluteal region THEN repeat every 2–4 weeks. Typical effective dose is 12.5–100 mg IM every 2–5 weeks. In elderly or medically ill patients: IM: Initially 6.25 mg deep injection in gluteal region AND repeat IM injections every 2–5 weeks using lowest effective dose.	See chlorpromazine. Additionally for fluphenazine: pain at injection site With the exception of tardive dyskinesia, fluphenazine has more prominent extrapyramidal side effects but fewer autonomic side-effects, than chlorpromazine.	Pregnancy: Use with caution; monitor infant for reversible extrapyramidal side-effects. Breastfeeding: Use, with caution, if drug of choice (monitor infant for side-effects such as sedation). Contraindications: See chlorpromazine. Use with caution: See chlorpromazine. Elderly/debilitated (including HIV stage 3 or 4): See chlorpromazine. Counselling: See chlorpromazine. Warn patients that this medication may impair ability to perform skilled tasks such as operating machinery or driving.
Folic acid Folate-deficiency megaloblastic anaemia (10.18.3) Sickle-cell disease prophylaxis (10.18.3) (Other indications: Prevention of neural tube defect in pregnancy; prevention of recurrence of neural tube defect)	Tablet: 1 mg, 5 mg Oral: 5 mg daily for 4 months (in pregnancy, continue to term); up to 15 mg daily for malabsorption states if needed AND vitamin B12 Oral: 1 mg daily	Infrequent or rare: Nausea/vomiting, diarrhoea, abdominal pain or cramping	Pregnancy/breastfeeding: May be used at recommended dose in pregnancy and breastfeeding Contraindications: Folate-dependent malignant disease Women receiving antiepileptic therapy need counselling before starting folic acid. Should never be given without vitamin B12 in undiagnosed megaloblastic anaemia or other vitamin B12 deficiency states due to risk of precipitating subacute combined degeneration of the spinal cord.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Folinic acid (calcium folinate, calcium leucovorin) Methanol poisoning (3.8.1) Toxoplasmosis (11.40) (Other indications: Given with methotrexate and 5 FU chemotherapy)	Injection, powder for reconstitution, folinic acid (as calcium salt) 3 mg/ml in 10 ml ampoule Tablet: 15 mg IV: 50 mg every 4 hours for 6 doses Oral: 10-25 mg with each dose of pyrimethamine	Common: Hypersensitivity reactions Infrequent or rare: Pyrexia after parenteral use	Pregnancy/breastfeeding: Not recommended during pregnancy or breastfeeding. Manufacturer advises use only if potential benefit outweighs risk. Use with caution and at lower doses if renal or hepatic impairment. Does not alter the CNS effects of drinking alcohol or withdrawal symptoms
Furosemide Oedema (not lymphoedema) (10.4.3) Acute pulmonary oedema (3.2.5) Oliguria in acute kidney injury (11.31.3)	Tablet: 40 mg Injection: 10 mg/ml Oral: Initially 20–80 mg daily in morning (higher starting doses recommended when patient not naive to drug or renal impairment) Maintenance dose: 20–40 mg daily (may be increased to 80 mg daily or more in resistant oedema) IV (slow): 20–50 mg (if necessary, increase by 20 mg steps every 2 hours) (see flowchart in Section 3.2.5) IV: Initially give 20 mg; monitor urinary response. (see dosing in Section 11.31.3)	Common: Electrolyte imbalance (hypokalaemia, hyponatraemia, hypomagnesaemia), hyperuricaemia and gout, orthostatic hypotension, hypovolaemia, syncope, dizziness Infrequent or rare: Hypochloraemic alkalosis, hypocalcaemia, hyperglycaemia (less than with thiazide diuretics), paraesthesia, blood disorders (including thrombocytopenia, leukopenia, agranulocytosis, aplastic anaemia, haemolytic anaemia), bone marrow depression (withdrawal treatment); deafness (with rapid administration of large parenteral doses and in renal impairment), hypotension, hypersensitivity reaction (including anaphylaxis), temporary increase in plasma cholesterol and triglyceride concentration	Pregnancy: Consider alternatives; loop diuretics not recommended unless absolutely necessary (e.g. cardiac failure). Neonatal ototoxicity has been reported. Breastfeeding: Limited data. Not recommended; theoretically, may suppress lactation due to diuresis Contraindications: Anuria due to renal failure, precomatose states associated with liver cirrhosis, electrolyte depletion Use with caution: In elderly (reduce dose), hypotensive patients, renal impairment, hepatic impairment, prostatic enlargement Administration: Dose to be diluted in suitable amount of infusion fluid (saline or LR; glucose solutions are unsuitable), depending on hydration of patient, at rate not to exceed 4 mg/minute. Monitor electrolytes, particularly potassium and sodium. Correct hypovolaemia before using in oliguria. IV to PO conversion is 1:2 (e.g. 20 mg IV is equal to 40 mg PO)

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Gabapentin Neuropathic pain (if poor response to amitriptyline in HIV patient on ART) (10..10a.6)	Tablet: 300 mg Oral: Initial dose of 300 mg daily; THEN increase to 300 mg twice daily; THEN 300 mg 3 times daily as needed THEN titrate with 100 mg increments every 3 days to a maximum of 3.6 g daily given as 1200 mg 3 times daily or 900 mg 4 times daily	Common: Nausea, vomiting, diarrhoea, dry mouth, dyspepsia, constipation, abdominal pain, flatulence; appetite changes, gingivitis, weight gain; hypertension, vasodilation, oedema; dyspnoea, cough, rhinitis Infrequent or rare: Confusion, depression, hostility, sleep disturbances, headache, dizziness, anxiety, amnesia, ataxia, dysarthria, myasthenia, tremor, asthenia, paraesthesia, hyperkinesia; influenza-like symptoms; impotence, urinary incontinence; leukopenia; myalgia, arthralgia; diplopia, amblyopia; rash, purpura, pruritus, acne; rarely, pancreatitis, hepatitis, jaundice, palpitation, hallucinations, movement disorders, thrombocytopenia, blood-glucose fluctuations in patients with diabetes, tinnitus, acute renal failure, Stevens-Johnson syndrome, alopecia	Pregnancy: Use with extreme caution; has been associated with fetal abnormalities such as hypospadias, unilateral renal agenesis. Breastfeeding: Use, with caution, in breastfeeding; monitor infant for sedation and lethargy. Caution: Monitor for depression, suicidal ideation, unusual mood, behaviour change; mixed seizure disorder (including absence seizure); renal impairment; encephalopathy Administration: Renal dose adjustment required. Avoid stopping abruptly, which may cause anxiety, insomnia, nausea, pain and sweating. Gradually reduce dose over at least 1 week Counselling: This medicine may cause drowsiness or dizziness. If affected, do not drive or operate heavy machinery.
Ganciclovir Cytomegalovirus (CMV) retinochoroiditis in HIV-infected patients (11.8) CMV oesophagitis, gastritis, neurologic, in HIV-infected patients (11.8)	Injection: 500 mg vial Induction dose: IV: 5 mg/kg twice daily for 3-4 weeks IV: 5 mg/kg twice daily for 3-6 weeks See Section 11.8 for maintenance with valganciclovir, if available.	Common: Nausea, vomiting, diarrhoea, dyspepsia, abdominal pain, constipation, flatulence, dysphagia; hepatic dysfunction; dyspnoea, chest pain, cough, headache, insomnia, convulsions, dizziness, neuropathy, depression, anxiety, confusion, abnormal thinking, fatigue, weight loss, anorexia, infection, fever, night sweats; anaemia, leukopenia, thrombocytopenia, pancytopenia, renal impairment, myalgia, arthralgia, macular oedema, retinal detachment, vitreous floaters, eye pain; ear pain, taste disturbance; dermatitis, pruritus; Injection-site reactions Infrequent or rare: Chest pain, chills, mouth ulceration, cough, dry mouth, drowsiness, arthralgia, pancreatitis, arrhythmias, hypotension, anaphylactic reactions, psychosis, tremor, male infertility, haematuria, disturbances in hearing and vision, alopecia	Pregnancy/breastfeeding: Avoid during pregnancy or breastfeeding unless need justifies the risk. To avoid pregnancy, advise use of an effective contraceptive. Contraindications: Neutropenia (ANC <500) or thrombocytopenia (platelets <25 000); concurrent use with zidovudine Use with caution: Ganciclovir is toxic. Personnel should be adequately protected during handling and administration. If solution comes into contact with skin or mucosa, wash off immediately with soap and water. In patients with renal impairment, renal dose adjustment required. Affects spermatogenesis and fertility. Administration: To avoid phlebitis at the injection site (related to high pH of solution), administer in veins with good flow. Maintain adequate hydration. Monitor blood counts 2-3 times per week.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Gentamicin General dosing for infections, due to susceptible organisms, such as upper UTI (11.44); septic shock (3.1.5); severe pneumonia (3.2.3) Initial empirical antibiotics for emergency management (QC p. 19)	Injection: 10 mg, 40 mg (as sulfate)/ml in 2 ml vial IM/IV (slow): 4–6 mg/kg daily in divided doses every 8 hours for 2 weeks AND ampicillin IV: 240 mg in a single dose AND ampicillin 2 g IV/IM	Common: Nephrotoxicity, ototoxicity Infrequent or rare: Neuromuscular blockade, renal electrolyte wasting (Mg, K, Ca), antibiotic-associated colitis, nausea/vomiting, hypersensitivity reactions	Pregnancy: Avoid unless essential for serious infections; monitor levels to minimize potential for ototoxicity and nephrotoxicity. Breastfeeding: May be used at recommended doses (monitor infant for thrush, diarrhoea) Contraindications: Hypersensitivity to aminoglycoside group of antibiotics Use with caution: In renal impairment, pre-existing tinnitus/hearing loss, conditions with muscular weakness, obesity, the elderly (dosage adjustment required). Monitor renal, auditory, vestibular function, serum-gentamicin concentration. Avoid prolonged use. One-hour (peak) concentration not to exceed 5–10 mg/l (3–5 mg/l for endocarditis) and pre-dose (<i>trough</i>) concentration less than 2 mg/l (less than 1 mg/l for endocarditis). Avoid use with neuromuscular blocking agents (additive toxicity) and other renal/ototoxic agents. Avoid monotherapy with gentamicin especially for severe infections of unclear etiology. Administration: Renal dose adjustment required. Can be dosed daily (4.5 mg/kg every 24 hours) or 1.5 mg/kg every 8 hours. The empirical antibiotic dose for emergency management of 240 mg (QC p. 19) represents the full daily dose for a 60 kg person.
Endocarditis from viridans streptococci- non-complicated cases (as part of combination therapy) (11.10) PID (10.15.5)	IV/IM: 3 mg/kg per 24 hours in 1 dose or in 2 or 3 equally divided doses for 2 weeks AND benzylpenicillin or ceftriaxone or vancomycin IV: 1.5 mg/kg every 8 hours AND clindamycin AND metronidazole		
Cholangitis and peritonitis (10.7a.2) Brucellosis (11.3)	IV: 1.5 mg/kg every 8 hours AND ampicillin AND metronidazole IV: 5 mg/kg daily in divided doses every 8 hours AND doxycycline for 15 days		

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Gentamicin eye drops Bacterial conjunctivitis (10.12.2); corneal ulcer or infective keratitis (10.12.2)	Solution (eye drops): 0.3% 1 drop every 2 hours, reducing frequency as infection is controlled THEN continue for 48 hours after healing is complete.	Common: Burning, stinging, itching, dermatitis	Use with caution: Prolonged use may lead to sensitization and emergence of resistant organisms including fungi; discontinue if there is purulent discharge, inflammation, exacerbation of pain.
Griseofulvin Scalp, skin, groin infections (10.2.7); foot and nail infections (10.2.7)	Tablets: 125 mg, 250 mg Capsules: 250 mg Oral: 500 mg to 1 g daily but not less than 10 mg/kg Duration of treatment depends on the infection and thickness of keratin at site of infection: at least 4 weeks for skin and hair, at least 6 weeks for scalp ringworm and, in severe infection, up to 3 months; 6 months for fingernails, 12 months or more for toe nails.	Common: Nausea, vomiting, diarrhoea, anorexia, headache Infrequent or rare: Leukopenia; hepatotoxicity; sleep disturbances; photosensitivity; systemic lupus erythematosus; rash, toxic epidermal necrolysis, erythema multiforme; peripheral neuropathy, confusion, impaired coordination	Pregnancy: Avoid pregnancy during and for 1 month after treatment; men should not father children within 6 months of treatment. Consider alternative treatments. Breastfeeding: Not recommended; use only if benefit is greater than risk; consider alternatives. Contraindications: Severe liver disease, porphyria, systemic lupus erythematosus Use with caution: In pre-existing hepatic insufficiency (closely monitor hepatic function throughout treatment), blood disorders (monitor blood count weekly during first month of treatment), penicillin allergy (cross-sensitivity may occur) Counselling: Take with milk or food. May impair ability to perform skilled tasks such as operating machinery or driving. Avoid sun exposure. During treatment and for 4 weeks after, drinking alcohol may cause increased heart rate and skin flushing.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Haloperidol Psychoses (including schizophrenia) (10.11) Short-term adjunctive management of agitation, violent behaviour, severe anxiety (QC p. 29) or severe symptoms of acute mania with agitation (10.11.5) Alcohol withdrawal delirium that persists after the stage of tremor and sweating has subsided (3.7) Antiemetic when other agents not available (10.7c)	Tablet: 2 mg; 5 mg Injection: 5 mg in 1 ml ampoule Oral: Initially 1.5-3 mg once daily (typical effective dose is 3-20 mg daily) In elderly or medically ill patients: Oral: Initially 0.5-1 mg once daily (use the lowest effective dose) IM/oral: 2 mg every hour up to 5 doses (maximum dose 10 mg). In elderly and patients with complicating medical illness: IM/oral: 0.5-1 mg every hour up to 3 doses maximum dose 3 mg) See dosing in Section 3.7.	See chlorpromazine. With the exception of tardive dyskinesia, haloperidol has more prominent extrapyramidal side-effects but fewer autonomic side-effects than chlorpromazine.	Pregnancy: Use with caution and at lowest effective dose; reversible respiratory depression, extrapyramidal effects, difficulty feeding have been reported in newborn. Breastfeeding: Caution in breastfeeding (monitor infant for drowsiness) Contraindications: See chlorpromazine. Use with caution: See chlorpromazine. Elderly/debilitated (including HIV stage 3 or 4): See chlorpromazine. Administration: Monitor blood pressure and maintain supine position for 30 minutes after intramuscular injection. Counselling: This medication may impair your ability to perform skilled tasks such as operating machinery or driving.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Hydralazine IV Acute pulmonary oedema with severe hypertension if isosorbide dinitrate not available (3.2.5); severe hypertension in pre-eclampsia and eclampsia (3.2.5) - see IMPAC MCPC (Other indications: Heart failure; hypertension (oral); hypertensive crisis)	Powder for injection: 20 mg (hydrochloride) in ampoule IV (slow): 5 mg diluted with 10 ml sodium chloride 0.9%; may repeat after 30 minutes	Common: Flushing, headache, dizziness, tachycardia, palpitation, oedema Infrequent or rare: Nausea/vomiting, ischaemia, postural hypotension, abnormal liver function, systemic lupus erythematosus-like syndrome, blood disorders (haemolytic anaemia, leukopenia, thrombocytopenia)	Pregnancy: Use only in acute treatment of hypertensive emergencies; avoid large boluses, as fetal distress and arrhythmias have been reported. Breastfeeding: Use, with caution, if benefits are greater than risks; monitor infant for effects such as hypotension, bradycardia, fatigue. Contraindications: Idiopathic systemic lupus erythematosus, severe tachycardia, high output heart failure, myocardial insufficiency due to mechanical obstruction, cor pulmonale, dissecting aortic aneurysm, porphyria Use with caution: In hepatic impairment, renal impairment, coronary artery disease, cerebrovascular disease. May provoke angina (avoid after myocardial infarction until stabilized). Administration: Renal dose adjustment required. Counselling: This medicine may cause dizziness, especially at the start of treatment. If affected, do not drive or operate heavy machinery.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Hydrochlorothiazide Oedema (10.4.3); mild hyperkalaemia (5.2.2) HIV-associated nephropathy (11.31.5) (Other indications: Hypertension; heart failure)	Tablet: 25 mg 25 mg once daily (12.5 mg in elderly). Increase to 50 mg as needed. See Section 11.31.5 for addition hydro-chlorthiazide to enalapril	Common: Dizziness; dry mouth; weakness; muscle cramps; polyuria; orthostatic hypotension; hypokalaemia; hyponatraemia; hypochloreaemic alkalosis; hypomagnesaemia hyperuricaemia Infrequent or rare: Nausea/vomiting, weakness, lethargy, drowsiness; seizures, headache, oliguria, arrhythmias, hypercalcaemia, hyperglycaemia, rash, photosensitivity, altered plasma lipid concentration; rarely, impotence (reversible); blood disorders (including neutropenia, thrombocytopenia); pancreatitis, intrahepatic cholestasis; acute renal failure, hypersensitivity reactions (including pneumonitis, severe skin reactions)	Pregnancy: Avoid use; may cause electrolyte disturbances or neonatal thrombocytopenia. Reduction in maternal blood volume may diminish uteroplacental perfusion. Breastfeeding: Use with caution but unlikely to suppress lactation. Contraindications: Severe renal or severe hepatic impairment; hyponatraemia, hypercalcaemia, refractory hypokalaemia, symptomatic hyperuricaemia; Addison's disease Use with caution: In elderly; electrolytes may need to be monitored with high doses; may aggravate diabetes mellitus and gout; may exacerbate systemic lupus erythematosus; porphyria Counselling: Take the medicine once daily in the morning. While taking this medicine, you may feel dizzy on standing. Get up gradually from sitting or lying to minimize this effect. Sit or lie down if you become dizzy. Pregnancy: May be used at the recommended doses; caution in first trimester due to possibility of oral cleft; limited fetal exposure due to inactivation by placenta Breastfeeding: May be used at recommended doses; caution with high parenteral/oral doses Contraindications: Not relevant to emergency use. For contraindications related to long-term use, see prednisolone.
Hydrocortisone Also see Section 8.2 steroid equivalents table. Anaphylaxis (3.1.3); moderate or severe bronchospasm, if suspect asthma or COPD or unable to take oral medication (3.2.4) Addison's syndrome (acute adrenal insufficiency) (3.4.3); urticaria (10.2.9)	Injection: 100 mg (as sodium succinate) in vial IV (slow): 100 mg in a single dose IV: 100 mg initially AND repeat every 8 hours. Convert to oral dose once patient stable.	Associated with long-term treatment, which is not recommended with IV hydrocortisone. See prednisolone.	

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Hydrocortisone cream Severe inflammatory skin conditions including papular lesions; eosinophilic folliculitis; pityrosporum folliculitis; papular urticaria; eczema; contact atopic dermatitis; numular eczema; seborrhoeic dermatitis; psoriasis (10.2)	Cream: 1% Topical: Apply a small quantity to the affected area 1–2 times daily until improvement occurs.	Infrequent or rare: Exacerbation of local infection, contact dermatitis, perioral dermatitis	Pregnancy: Topical preparations unlikely to cause any adverse effects in pregnancy or breastfeeding, as systemic absorption is expected to be minimal. Breastfeeding: Wipe excess cream from nipple area before feeding. Contraindications: Untreated skin infections, broken skin, rosacea, acne, perioral dermatitis. Occlusive dressings increase penetration into keratinized lesions. Treat secondary infection with an appropriate antimicrobial.
Hydroxypropyl methylcellulose (tears naturale) Chronic soreness of the eyes associated with reduced or abnormal tear secretion; acute viral conjunctivitis (10.12)	Drops: 0.5% Instill frequently (e.g. hourly) for adequate relief.	Common: Eye irritation	
Ibuprofen Mild to moderate pain (20.2, 20.4); musculoskeletal disorders; arthritis (10.13); dysmenorrhoea (10.15); fever (10.1); erythema nodosum (10.2)	Tablet: 200 mg, 400 mg 200–400 mg 3–4 times daily. Maximum 2.4 g daily	Common: Dyspepsia, nausea, diarrhoea; GI ulceration and haemorrhage, raised liver enzymes, headache, dizziness, salt and fluid retention, hypertension Infrequent or rare: Heart failure; hypersensitivity reactions; bronchospasm; renal failure; rarely, hepatic failure; erythema multiforme, toxic dermal necrolysis; oesophageal ulceration, hyperkalaemia	Pregnancy/breastfeeding: Avoid unless potential benefit greater than risk; consider alternatives such as paracetamol or opioids; regular use in third trimester may cause closure of fetal ductus arteriosus <i>in utero</i> , possibly persistent pulmonary hypertension of the newborn, delayed onset and increased duration of labour. Use with caution: /In renal and hepatic disorders, hypersensitivity, and in the elderly. Ibuprofen can reduce the antiplatelet activity of low-dose aspirin and potentially reduce or negate the cardioprotective effect. Counselling: Take with or after food.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Insulin (soluble) Diabetic ketoacidosis (3.4.1) Hyperkalaemia (5.2.2) Overdose of calcium-channel blockers and beta-blockers, given in combination with dextrose (3.8.1) (Other indications: Diabetes mellitus type I; type II after failing oral therapy)	Injection: 40 IU/ml in 10 ml vial; 100 IU/ml in 10 ml vial See Section 3.4.1 IV: 10–15 units in 50 ml D50 (50% dextrose water) infused over 2 hours THEN dextrose infusion + regular blood glucose monitoring See Table in 3.8.1	Common: Hypoglycaemia, weight gain; hypersensitivity reactions; hypokalaemia, transient oedema; local reactions including erythema, itching, lipodystrophy, lipoatrophy	Pregnancy: Generally accepted as safe (insulin requirements should be assessed in each trimester) Breastfeeding: May be used at recommended doses Use with caution: In renal impairment, hepatic impairment, hypokalaemia
Ipratropium bromide Acute wheezing (QC p. 17); COPD, moderate (10.6)	Inhalation (aerosol): 20 mcg / metered dose 20 to 40 mcg 4 times daily (2 puffs) AND inhaled salbutamol	Common: Dry mouth, throat irritation Infrequent or rare: Constipation, tachycardia, atrial fibrillation, urinary retention, acute angle-closure glaucoma	Pregnancy/breastfeeding: Safe in pregnancy. Use with caution: In prostatic hypertrophy. Medical supervision with first dose due to risk of paradoxical bronchospasm. Counselling: Do not use for immediate relief of symptoms.
Isoniazid (INH) (H) TB prevention for PLHIV (13.3)	Tablet: 300 mg Oral: INH 300 mg daily for at least 6 months is recommended, and up to 36 months in HIV-prevalent settings with a high prevalence and transmission of TB AND pyridoxine	Common: Burning, numbness or tingling sensation in hands and feet; drowsiness; anorexia, nausea, abdominal pain; jaundice (other causes excluded), hepatitis; skin rash with or without itching (rarely, but can be serious, e.g. Stevens-Johnson syndrome)	Administration: Desirable to give also pyridoxine 10 mg daily to prevent peripheral neuropathy in PLHIV. Defer IPT in the presence of active hepatitis (acute or chronic), regular and heavy alcohol consumption, and symptoms of peripheral neuropathy. Stop INH if jaundice or skin rash with or without itching. Stop IPT, and start treatment regimen if active TB develops.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Isosorbide dinitrate Pulmonary oedema with severe hypertension (3.2.5)	Tablet (sublingual): 5 mg SL: 5 mg sublingual, repeat in 10–15 minutes, not to exceed 10 mg every 2–3 hours	Common: Throbbing headache, flushing, dizziness, fainting, postural hypotension, tachycardia Infrequent or rare: Paradoxical bradycardia	Pregnancy: Use with caution. Consider alternatives where possible; use minimum effective dose if required in acute situation. Breastfeeding: Use, with caution, if benefit is greater than risk; monitor infant for side-effects. Contraindications: Hypersensitivity to nitrates, hypotension, hypovolaemia, hypertrophic obstructive cardiomyopathy, aortic stenosis, cardiac tamponade, constrictive pericarditis, mitral stenosis; marked anaemia, head trauma, cerebral haemorrhage, angle-closure glaucoma Use with caution: In severe hepatic or renal impairment, hypothyroidism, malnutrition, hypothermia, elderly, or recent MI. Prone to multiple significant drug interaction through CYP3A4 enzyme.
Itraconazole Candidal oesophagitis when fluconazole not available (11.4) Histoplasmosis (11.16) Penicilliosis, mild disease (11.29) Severe disseminated penicilliosis (11.29) Pityriasis versicolor, recurrence (10.2.7)	Capsule: 100 mg; Oral solution 10 mg/ml Oral: 100–200 mg twice daily for 10–14 days (may be increased to a maximum of 400 mg daily) 200 mg 3 times daily for 3 days. THEN 200 mg twice daily for 6–12 months 200 mg twice daily for 8 weeks THEN, in PLHIV, 200 mg daily until 6 months after CD4> 100 Amphotericin B for 14 days THEN itraconazole 200 mg daily for 10 weeks, continuing in PLHIV until 6 months after CD4>100 Pulsed monthly treatment for 3 months	Common: Dyspepsia, anorexia, fatigue, itch Infrequent or rare: Nausea, vomiting, abdominal pain, diarrhoea, constipation, jaundice, hepatitis; heart failure, pulmonary oedema, headache, dizziness, peripheral neuropathy (discontinue treatment); menstrual disorders; hypokalaemia, rash, pruritus, Stevens-Johnson syndrome, alopecia Potentially life-threatening hepatotoxicity reported very rarely; discontinue if signs of hepatitis develop.	Pregnancy: Contraindicated in first trimester; use in second or third trimesters only if drug of choice and no alternatives. Breastfeeding: Not recommended – effects unknown. Use fluconazole if available and indicated. Use with caution: in patients with heart failure or risk factors for heart failure, pre-existing hearing loss, hypersensitivity to other azoles. Monitor liver enzymes in patients with liver disease. Administration: Prone to multiple significant drug interactions (many leading to severe cardiovascular compromise). (An alternative for candida oesophagitis is the itraconazole solution). Counselling: Take capsule with food for best absorption (oral solution should be taken on empty stomach). Do not take antacids within 2 hours of taking this medicine. Tell your doctor if you feel unusually tired or have loss of appetite, nausea, vomiting, abdominal pain, or dark urine.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Ivermectin Strongyloidiasis (11.36)	Tablet: 3 mg, 6 mg 200 mcg/kg as a single dose OR 200 mcg/kg daily for 2 days and then maintenance therapy 6 mg monthly	Common: Strongyloidiasis: Diarrhoea, dizziness, nausea, mild ocular irritation, somnolence Onchocerciasis: Mild Mazzotti reaction within 3 days of treatment (fever, headache, cough, pruritus, conjunctivitis, arthralgia, lymphadenopathy, diarrhoea) Infrequent or rare: Cutaneous or systemic reactions	Pregnancy/breastfeeding: Delay treatment until after delivery and infant is 1 week old. Administration/counselling: Avoid food or alcohol for at least 2 hours before and after a dose. Contraindications: Do not give ivermectin for onchocerciasis in loiasis co-endemic areas.
Filariasis (11.12)	200–400 mcg/kg as a single dose AND albendazole 400 mg twice daily for 2 weeks		
Onchocerciasis (in non Loa Loa endemic areas) (11.28)	150 mcg/kg as a single dose every 6 or 12 months		
Scabies (10.2.4)	200 mcg/kg as a single dose and repeat in 2 weeks.		
Norwegian (crusted) scabies (10.2.4)	Combine ivermectin with topical scabicide (benzyl benzoate or permethrin)		

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Ketamine Sedation during short procedures; induction for intubation (QC p. 28 and 31)	<i>Injection: 50 mg, 100 mg (as hydrochloride) /ml in 10 ml vial</i> IV: 1–2 mg/kg IV over 2 minutes May repeat 0.5 mg/kg IV every 10 minutes as needed OR IM: 4 mg/kg	Common: Raised BP and pulse rate, increased muscle tone (sometimes tonic-clonic and resembling seizures), lacrimation, nausea, vomiting, nystagmus, raised intracranial pressure, diplopia, emergence reactions (may occur after recovery and up to 24 hours), which vary in severity between pleasant dream-like states to vivid imagery, hallucinations, nightmares and emergence delirium (often consisting of dissociative or floating sensations), confusion, excitement, irrational behaviour Infrequent or rare: Raised intraocular pressure, arrhythmias, hypotension, bradycardia, laryngospasm; anxiety, insomnia; increased salivation, apnoea, rashes, injection-site reactions, anaphylaxis	Pregnancy: Safe to use Breastfeeding: Limited data: avoid use. Contraindications: Where elevation of blood pressure would constitute a serious hazard, including eclampsia or pre-eclampsia, severe coronary or myocardial disease, cerebrovascular accident or cerebral trauma Use with caution: In increased cerebrospinal fluid pressure, predisposition to hallucinations or nightmares. Ketamine has abuse potential and can itself cause dependence. Administration: For intravenous injection, dilute 100 mg/ml strength to a concentration of not more than 50 mg/ml with dextrose 5% or sodium chloride 0.9% or water. Give IV slowly; rapid administration may result in respiratory depression and enhanced hypertensive response. Emergence reactions can be eliminated by co-administering benzodiazepine such as diazepam or midazolam and by minimizing stimulation during the recovery period.
Lactulose Hepatic encephalopathy in cirrhosis (10.9.2)	<i>Lactulose solution: 3.1–3.7 g/5 ml</i> 20–30 g (30–45 ml) 3–4 times daily; adjust dose every 1–2 days to produce 2–3 soft stools daily	Common: Flatulence, abdominal discomfort, cramps Infrequent or rare: Diarrhoea, dehydration, hyponatraemia, hypokalaemia	Pregnancy: No evidence of harm Breastfeeding: Use caution. Use with caution: In patients with electrolyte imbalance, diabetes mellitus; solution contains galactose and lactose. Administration: May mix with fruit juice, water, or milk. Onset of action is 1–3 days.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Levonorgestrel For emergency contraception (ECP) (14.5.2)	Tablet: .75 mg; 1.5 mg 1.5 mg taken as a single dose within 120 hours of unprotected sex, sexual assault OR 0.75 mg followed by another dose of 0.75 mg 12 hours later	Common: Nausea/vomiting, breast tenderness, headache, dizziness, abdominal pain Irregular vaginal bleeding for 1–2 days; next menstrual bleeding starts earlier or later than expected	Pregnancy: No harm to fetus if pregnancy should occur Breastfeeding: Unknown safety in breastfeeding but duration of use of ECPs is brief. Administration: The duration of use of ECP is brief, thus less clinical impact is expected in severe cardiovascular complications, angina, migraine, severe liver disease. Counselling: If vomiting occurs within 2 hours after taking levonorgestrel, a replacement dose should be taken. An antiemetic can be taken one-half to one hour before the replacement dose. No protection against STI/HIV
Lidocaine (with or without epinephrine) Local anaesthesia (7.1.2); nerve block (20.3)	Injection: 1%, 2% in vial Topical: 2–4% Local infiltration and nerve block, using 0.5% solution, up to 250 mg (50 ml) in adults Local infiltration and nerve block, using 1% solution, up to 250 mg (25 ml) in adults With epinephrine: 0.5% solution up to 400 mg (80 ml) or 1% solution up to 400 mg (40 ml) in adults	Common: Dizziness, paraesthesia, drowsiness, confusion; apnoea, respiratory depression; coma, seizures; hypotension, arrhythmias, heart block, bradycardia (may lead to cardiac arrest); nystagmus (early sign of overdose)	Pregnancy: Avoid in third trimester Breastfeeding: Amount too small to be harmful Contraindications: Adjacent skin infection, inflamed skin, severe anaemia, heart disease Use with caution: In CHF (lower dosage) and following cardiac surgery, bradycardia, hepatic impairment, severe respiratory depression, and in the elderly
(Other indications: Arrhythmias; spinal and other anaesthesia)			

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Magnesium sulfate Prevention of seizures in severe pre-eclampsia; eclampsia convulsions and prevention of recurrence (QC p. 28) Severe bronchospasm not responding to salbutamol (3.2.4); torsade de pointes/ VF/ pulseless VT from toxicity due to quinine (3.8.1) Aluminium/zinc phosphide toxicity (3.8.1) Tetanus (11.39)	<u>Injection: 500 mg/ml in 2 ml ampoule; 500 mg/ml in 10 ml ampoule (50%)</u> See instructions in QC p. 28. IV: 2 g over 20 minutes IV: 1 g every 6 hours IM: 5 g OR IV: 75 mg/kg loading dose THEN 2–3 g hourly as needed until spasms controlled (with or without diazepam)	Common: Nausea, vomiting, thirst, flushing of skin, hypotension, arrhythmias, coma, respiratory depression, drowsiness, confusion, loss of tendon reflexes, muscle weakness	Pregnancy: Safe for short-term use in third trimester (neonatal respiratory depression if excessive dose) Breastfeeding: Mother treated with parenteral magnesium for pre-eclampsia can breastfeed. Contraindications: Bradycardia or AV block, pre-existing hypomagnesaemia, renal insufficiency/failure Use with caution: In myasthenia gravis, liver or renal impairment Administration: For intravenous injection, dilute 1 part of magnesium sulfate injection 50% with at least 1.5 parts of water for injection. For intramuscular injection, mix magnesium sulfate injection 50% with 1 ml lidocaine injection 2%. Monitor urine output. Before giving next dose, ensure that: • knee jerk is present • urine output >100 ml/4 hours • respiratory rate >16/minute. Otherwise, do not give magnesium and consider calcium gluconate for toxicity. Note: Magnesium sulfate 1 g is approximately equivalent to Mg 4 mmol.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Malathion Head lice (pediculosis capitis) (10.2.8)	0.5% in an aqueous basis Rub 0.5% preparation into dry hair and scalp, allow to dry naturally; THEN remove by washing after 12 hours. THEN Repeat application after 7 days.	Infrequent or rare: Skin irritation, contact dermatitis, allergy	Pregnancy: Avoid use; permethrin preferred. Breastfeeding: Safe to use Counselling: Avoid contact with eyes. Do not use on broken or secondarily infected skin. Use lotion no more than once a week for 3 consecutive weeks.
Crab lice (pediculosis pubis) (10.2.8)	Apply 0.5% aqueous preparation over whole body, allow to dry naturally, THEN wash off after 12 hours or overnight. THEN repeat application after 7 days		
Mebendazole Hookworm (10.18); ascaris (10.7) Ascaris, hookworm prophylaxis every 6 months in adolescent girls and women of childbearing age (19.1) Persistent diarrhoea in immunocompromised patients (10.7d.2) (Other indications: Echinococcus infections prior to surgery or not amenable to surgery, nematode infections including enterobiasis, trichuriasis, capillariasis)	Tablet: 500 mg, 100 mg 500 mg orally in a single dose OR 100 mg twice daily for 3 days (Repeat after 3–4 weeks if eggs persist in stool) 500 mg as a single dose every 6 months as prophylaxis 500 mg twice daily for 5 days	Infrequent or rare: Nausea, vomiting, diarrhoea, abdominal pain or cramps; headache, dizziness; hypersensitivity reaction; with high doses; increased liver enzymes; alopecia; bone marrow depression	Pregnancy: Contraindicated in first trimester, consider alternatives before using in second or third trimester. Breastfeeding: May be used; 2–10% of oral dose absorbed, and some excretion into breast milk expected. Counselling: Take dose between meals.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Meglumine antimoniate	<i>Injection: 30%, equivalent to approximately 8.1% antimony in 5 ml ampoule</i>	Common: Nausea, vomiting, abdominal pain, anorexia; ECG changes (possibly requiring dose reduction or withdrawal); cough; arthralgia, myalgia; elevated liver enzymes, jaundice, renal function impairment; lethargy	Pregnancy: Uncertain safety Breastfeeding: Contraindicated Contraindications: Severe cardiac, liver and kidney disorders. Use local therapy if lesion is close to the eyes, multiple (≥ 3), large (>3 cm diameter), sporotrichoid forms, on the joint, super-infected, or produced by <i>L. braziliensis</i> , <i>L. guyanensis</i> , or <i>L. tropica</i> . Use with caution: The risk of serious, even fatal, toxicity is increased in patients who concomitantly present with: cardiac disease (particularly arrhythmia), renal failure, liver disease, severe malnutrition, severely impaired general condition, advanced HIV infection; pregnancy. Administration: If any of the above cautions is present, provide protein-rich diet throughout treatment and correct iron and other nutritional deficiencies. Monitor cardiac, renal and hepatic function. Successful treatment of mucocutaneous leishmaniasis may induce severe inflammation around lesions (may be life-threatening if pharyngeal or tracheal involvement); may require corticosteroids.
Visceral leishmaniasis (11.20.2)	See table in Section 11.20.2.		
Secondary prophylaxis of leishmaniasis in HIV-infected patient (11.20.3)	IM/IV: 20 mg/kg every 3 or 4 weeks		
Cutaneous leishmaniasis, local treatment (11.20.1)	Intralesional injection: 1 to 5 intralesional injections, every few days or weekly, with or without cryotherapy		
Cutaneous leishmaniasis, systemic treatment (11.20.1)	IV/IM: 20 mg/kg for 21 days		
Mucocutaneous leishmaniasis (<i>L. braziliensis</i>) (11.20.1)	IM: 20 mg/kg daily until slit-skin smears are negative and for at least 4 weeks thereafter THEN, if inadequate response, 10–15 mg/kg every 12 hours for same period if inadequate response. Repeat for at least twice as long if relapse. (If unresponsive to this treatment, treat with pentamidine or amphotericin B.)		

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Melarsoprol Trypanosomiasis, meningoencephalitic stage (11.41)	<u>Injection: 3.6% solution, 5 ml ampoule (180 mg active compound)</u> 2.2 mg/kg slow IV injection per day for 7–10 days	Common: Jarisch-Herxheimer-like reaction (chills; fever; general feeling of illness or discomfort); headache; rigidity; sweating, peripheral neuropathy, reactive encephalopathy in 5–10% patients. Infrequent or rare: myocardial damage, hypertension, hypersensitivity, blood dyscrasias such as agranulocytosis; hepatic and renal dysfunction.	Pregnancy/breastfeeding: Contraindicated in pregnancy Use with caution: Hospitalization and close medical supervision (intensive care) required during treatment. Suspend treatment if reactive encephalopathy. Treat intercurrent infections such as pneumonia and malaria before treatment with melarsoprol. Use with caution in malnutrition (if possible, correct with a protein-rich diet); G6PD deficiency; leprosy (may precipitate erythema nodosum). Administration: Avoid extravasation—injection is very irritating. Patients should be supine and fast for at least 5 hours after injection.
Methadone Acute opioid withdrawal (3.6.2) Opioid substitution therapy (OST) (17.4)	<u>Concentrate for oral liquid: 5 mg/ml, 10 mg/ml (hydrochloride)</u> <u>Oral liquid: 5 mg/5ml, 10 mg/5ml</u> Oral: Initially 15–20 mg. Gradually increase to 40 mg/day. Taper off over 3–28 days. (See Section 3.6.2.) Oral: Initially 20 mg, with an additional 10 mg after 4 hours. If tolerance is low or uncertain, start with a dose of 10 mg (see Section 17.4). (Optimal dosing range is 60–120 mg.)	Common: Drowsiness, dizziness, respiratory depression, QT interval prolongation, dysmenorrhoea, hyperprolactinaemia, dry eyes, dry mouth Rare: torsade de pointes, hypothermia, restlessness, raised intracranial pressure, agitation/confusion (especially in the elderly), urinary retention with high doses (especially in the elderly)	Pregnancy: May be used; not associated with birth defects; caution in third trimester as chronic use is associated with neonatal opioid withdrawal symptoms. Breastfeeding: Monitor adverse effects such as sedation (no adverse effects reported with 20 mg/day or less); infant withdrawal reported with sudden cessation. Contraindications: Acute bronchial asthma or hypercarbia, respiratory depression in absence of appropriate airway equipment, paralytic ileus Do not give to patients showing signs of intoxication from alcohol or depressant drugs (such as diazepam). Use with caution: In patients with renal or hepatic impairment, hypothyroidism, convulsive disorders, decreased respiratory reserve as in asthma, hypotension, elderly, prostatic hyperplasia, adrenal insufficiency, head trauma. Prone to multiple drug interactions. Dose adjustment recommended if severe renal and/or hepatic impairment. Counselling: This medication may impair your ability to perform skilled tasks such as operating machinery and driving.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Methylthioninium chloride (methylene blue) Acute methaemoglobinemia (propanil poisoning) (3.8.1)	Injection: 10 mg/ml in 10 ml ampoule IV (loading dose): 2 mg/kg over 5 minutes THEN further dose of 1 mg/kg if no improvement. See Section 3.8.1 for monitoring and further dosing.	Common: Nausea/vomiting, abdominal pain, chest pain, headache, dizziness, confusion, profuse sweating, hypertension, hypotension, haemolytic anaemia (in G6PD deficiency); methaemoglobinemia with high dosage; bluish skin discoloration; blue saliva, urine, and faeces	Pregnancy/breastfeeding: Uncertain safety in pregnancy and breastfeeding. Use with caution. Contraindications: Severe renal impairment, methaemoglobinemia due to chlorate or induced by sodium nitrite in treatment of cyanide poisoning, G6PD deficiency (may cause haemolytic anaemia) Use with caution: Monitor blood methaemoglobin throughout treatment
Metoclopramide Nausea, vomiting (10.7.3, 14.1.11)	Tablet: 10 mg Injection: 5 mg (hydrochloride)/ml in 2 ml ampoule Oral/IV: 10 mg every 8 hours 15–19 years (<60 kg): 5 mg 3 times daily	Common: Drowsiness, headache restlessness, dizziness Infrequent or rare: Extrapyramidal symptoms (especially children/young adults), hyperprolactinemia, depression, diarrhoea, hypotension, hypertension, neuroleptic malignant syndrome (rare), rash, cardiac conduction abnormalities following IV administration (rare)	Pregnancy/breastfeeding: Considered safe in both pregnancy and breastfeeding (monitor infant for side-effects) Contraindications: Gastrointestinal obstruction (often used to empty the stomach of blood prior to EGD); within 3–4 days after gastrointestinal surgery, convulsive disorders, pheochromocytoma Use with caution: In elderly, children, and young adults; hepatic or renal impairment; Parkinson's disease, epilepsy, depression, porphyria. May mask underlying disorders such as cerebral irritation. Renal dosage adjustment required.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Metronidazole	Tablet: 200 mg, 500 mg Suppositories: 0.5 g, 1 g Gel Injection: 500 mg in 100 ml vial	Common: Nausea/vomiting, diarrhoea, unpleasant metallic taste, dizziness, headache Infrequent or rare: Furred tongue, glossitis stomatitis paraesthesia, increased liver enzymes, blood disorders, myalgia/arthritis, peripheral neuropathy, epileptiform seizures, disulfiram-like reaction, bone marrow depression, alopecia	Pregnancy: May be used if drug of choice. Use has not been associated with increased risk of adverse outcomes. Breastfeeding: May be used at usual doses, but avoid high single-dose therapy or else withhold feeds for 12–24 hours; monitor infant for side-effects; may cause temporary changes to milk taste. Contraindications: Chronic alcohol dependence – disulfiram-like reaction with alcohol occurs. Precautions: Hepatic disease. Prone to multiple drug interactions through CYP enzyme system. Check for interactions with current and new medications. Administration: Tablets should be swallowed whole with water, with or after food. IV: Give over 15–30 minutes. Counselling: Take tablets with food to reduce stomach upset. This medicine may make you feel dizzy or confused. Avoid driving if you are affected. Avoid alcohol during treatment, and for 24 hours after stopping the drug, to prevent nausea, vomiting, flushing, headache, and palpitation. Stop the medicine and inform your doctor if you have any numbness, tingling, pain, or weakness in hands or feet
Bacterial vaginosis (10.15.4)	500 mg twice daily for 5–7 days		
Pelvic inflammatory disease (10.15.5)	Oral: 400–500 mg twice daily for 14 days AND ceftriaxone + doxycycline or tetracycline		
Persistent or chronic diarrhoea in immunocompromised patients (empirical) (10.7d.3)	Oral: 500 mg 3 times daily for 7 days AND cotrimoxazole		
Leg ulcers/pressure sores (10.2.10)	Oral: 400 mg every 8 hours for 7 days		
Acute necrotizing ulcerative gingivitis or periodontitis (10.17.6)	Oral: 200 mg 3 times daily for 7–10 days		
Dental abscess (10.17.5)	Oral: 500 mg 3 times daily AND phenoxymethyl penicillin or amoxicillin		
Peritonsillar abscess (10.17.9)	Oral: 500 mg 3 times daily AND amoxicillin		
Antibiotic-associated colitis; <i>Clostridium difficile</i> colitis (10.7d.2)	Oral: 500 mg 3 times daily for 10–14 days; IV form effective if patient cannot take oral pills (same dose)		

Urogenital trichomoniasis (10.15.4)	Oral: 2 g as a single dose OR 400–500 mg twice daily for 7 days (also treat sexual partners)		
<i>Helicobacter pylori</i> eradication (if allergic to amoxicillin) (10.7a.2)	Oral: 400 mg twice daily AND clarithromycin + omeprazole		
Tetanus (11.39)	Oral/IV: 500 mg 4 times daily for 10 days		
Intestinal amoebiasis (11.11.1) or empirical treatment in dysentery after no clinical improvement from 2 courses antibiotics locally effective for <i>Shigella</i> (10.7d.2)	Oral: 750 mg 3 times daily for 5–10 days THEN diloxanide or iodoquinol or paramomycin		
Amoebic liver abscess (11.11.2)	Oral/IV: 750 mg 3 times daily for 5–10 days		
Severe anaerobic infections- peritonitis or cholangitis (10.7a.2); septic abortion (10.15.6)	Oral/IV: 500 mg 3 times daily AND ceftriaxone or ampicillin + gentamicin		
(Other indications: Dracunculiasis; brain abscess; surgical prophylaxis; animal bites; giardiasis)			

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Miconazole Candida skin infection (11.4) Vulvovaginal candidiasis (10.15.4, 11.4) Oral candidiasis (11.4)	Cream: 2% (nitrate) Suppository: 200 mg Oral gel Gum patch Topical: Apply cream twice daily to clean, dry lesions for 5–7 days; continue at least 10 days after the condition clears. Vaginal suppository: 200 mg inserted daily for 3 days Oral: gel 60 mg 4 times daily for 7 days OR gum patch once daily for 7 days	Infrequent or rare: Local irritation, contact dermatitis (discontinue if sensitization occurs)	Pregnancy/breastfeeding: May be used at recommended doses in pregnancy and breastfeeding. Breastfeeding: Remove excess cream from nipple areas before feeding. Use with caution: Contact with eyes and mucous membranes should be avoided. Counselling: Gum patch: Place rounded side of tablet on upper gum above an incisor tooth and hold upper lip firmly over the gum for 30 seconds using a finger. If tablet detaches within 6 hours, replace with a new tablet. With each dose, alternate sides of the gum.
Midazolam Pretreatment to prevent emergence reaction with ketamine (QC p. 28) Sedation for intubation if not comatose (QC p. 31) Sedation after intubation (QC p. 34)	Injection: (as hydrochloride) 1 mg/ml IV: 0.05 mg/kg over 2 minutes just prior to giving ketamine IV: 0.2 mg/kg Infusion: 0.02–0.1 mg/kg/hour	Common: hypotension, hiccup, cough Infrequent or rare: GI disturbances, heart rate changes, laryngospasm, bronchospasm, respiratory depression and respiratory arrest (particularly with high doses or on rapid injection), drowsiness, confusion, ataxia, amnesia, headache, euphoria, hallucinations, paradoxical excitement and aggression (in elderly), skin reactions, injection-site reactions, anaphylaxis	Pregnancy: Avoid use if possible. High doses during late pregnancy or labour may cause neonatal hypothermia, hypotonia, respiratory depression. Breastfeeding: Present in milk; manufacturer advises avoiding breastfeeding for 24 hours after administration. Contraindications: Marked neuromuscular respiratory weakness including unstable myasthenia gravis; severe respiratory depression; acute pulmonary insufficiency Use with caution: in cardiac disease; respiratory disease; myasthenia gravis. Midazolam is associated with profound sedation when high doses are given IV or when used with certain other drugs.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Miltefosine Visceral leishmaniasis (11.20.2) Secondary prophylaxis in HIV-infected patients with leishmaniasis (11.20.3)	Capsule: 10 mg, 50 mg See table on treatment options in Section 11.20.2. Repeat 28-day courses	Common: Nausea, vomiting, anorexia, diarrhoea – usually brief and resolve as treatment continues. Occasionally severe, requiring treatment interruption. Infrequent or rare: Skin allergy, elevated hepatic transaminases, renal insufficiency	Pregnancy/breastfeeding: Do not use in pregnancy or during breastfeeding and assure adequate contraception for women of childbearing age during treatment and for 3 months afterwards.
Misoprostol Incomplete abortion (10.15.2) Postpartum haemorrhage not responding to oxytocin plus ergometrine (QC p. 26)	Oral tablet: 200 mcg Oral: Single dose of 400 mcg sublingually or 600 mcg by mouth SL: 800 mcg	Common: Diarrhoea (may occasionally be severe and require withdrawal: reduced by giving single doses not exceeding 200 mcg and by avoiding magnesium-containing antacids); abdominal pain, dyspepsia, flatulence, nausea, vomiting, abnormal vaginal bleeding (including intermenstrual bleeding, menorrhagia, postmenopausal bleeding); rash; shivering and fever. Infrequent or rare: Uterine rupture	Use with caution: In conditions where hypotension might precipitate severe complications (cardiovascular or cerebral disease). See other sources for contraindications to use for induction of labour (such as placenta praevia, cephalopelvic distortion, history caesarean section or major uterine surgery, etc.)

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Morphine	Injection: 10 mg (morphine hydrochloride or morphine sulfate) in 1 ml ampoule Oral liquid: 10 mg (morphine hydrochloride or morphine sulfate) in 5 ml Tablet: 10 mg (morphine sulfate) Tablet (prolonged-release): 10 mg, 30 mg, 60 mg (morphine sulfate)	Common: Nausea/vomiting (particularly in initial stages), constipation, dry mouth, drowsiness, anorexia, spasm of urinary and biliary tract, bradycardia, tachycardia, palpitation, euphoria, hallucinations, confusion, hypersensitivity reaction, postural hypotension, dependence, miosis. Larger doses produce respiratory depression, hypotension, muscle rigidity.	Pregnancy: May be used if drug of choice. Not associated with birth defects; high doses or prolonged use near term can cause neonatal respiratory depression and withdrawal symptoms. Breastfeeding: Use caution; monitor adverse effects such as sedation; therapeutic concentrations in breastfeeding infant may be reached with repeated dosing or long-term use. Contraindications: Acute respiratory depression, acute alcoholism, risk of paralytic ileus, raised intracranial pressure or head injury (affects pupillary responses vital for neurological assessment), injection in phaeochromocytoma Use with caution: In renal and hepatic impairment, dependence (severe withdrawal symptoms if withdrawn abruptly), hypothyroidism, convulsive disorders, decreased respiratory reserve and acute asthma, hypotension, prostatic hypertrophy Reduce dose or avoid in elderly and debilitated. Prone to multiple drug interactions through CYP enzymes; check for interactions with new or current medications. Administration: SC dosing not suitable for oedematous patients. Sustained-release tablets should be taken at regular intervals and not on an as-needed basis for episodic or breakthrough pain. Tablets can be crushed
Severe acute pain (20.4)	See dosing and precautions in Section 20.4.		
Chronic pain (20.2)	See dosing in Section 20.2.		
Myocardial infarction (QC p. 8)	IV (slow, 2 mg/min): 10 mg THEN 5–10 mg if necessary		
Difficult breathing in terminal illness (20.5)	2.5 mg every 4–6 hours if not on morphine for pain. If on morphine for pain, increase dose by 25%.		
Mupirocin	Cream (as mupirocin calcium): 2% Ointment: 2% Apply up to 3 times daily for up to 10 days	Infrequent or rare: Local reactions including urticaria, pruritus, burning sensation, rash	Pregnancy/breastfeeding: Safe in pregnancy and breastfeeding Use with caution: Avoid contact with eyes and mouth.
Impetigo (10.2.2); papular urticaria with secondary bacterial infection (10.2.3)			

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Naloxone Opioid overdose (QC p 18)	Injection: 400 mcg/ml (hydrochloride) in 1 ml ampoule IV: 100 mcg in a single dose OR IM: 400 mcg in a single dose OR SC: 800 mcg in a single dose. May repeat every 5 minutes up to 3 times (maximum 10 mg) If response (i.e. respiratory rate >10/minute), start IV infusion: 0.4 mg/hour for 12 hours	Common: Hypotension, hypertension, ventricular tachycardia and fibrillation, cardiac arrest; hyperventilation, dyspnoea, pulmonary oedema Infrequent or rare: Agitation, excitement, paraesthesia	Pregnancy: Use only if potential benefit is greater than risk; may precipitate withdrawal in fetus of an opioid dependant mother Breastfeeding: Unknown safety; currently not recommended. Use with caution: In physical dependence on opioids, other situations where acute withdrawal syndrome may be precipitated; cardiovascular disease Administration: Naloxone effects last only 40 minutes.
Naltrexone Alcohol dependence (16.5)	Tablet: 50 mg Start with 50 mg daily after withdrawal from alcohol or whilst still drinking some alcohol. Maintenance dose: 50–100 mg daily. Important that the patient has not taken any opioid drugs for previous 5 days	Common: Nausea/vomiting, abdominal pain, diarrhoea, constipation, reduced appetite, increased thirst; chest pain; anxiety, sleep disorders, headache, reduced or increased energy, irritability, emotional lability, dizziness; chills; urinary retention; delayed ejaculation; decreased potency; arthralgia, myalgia; increased lacrimation; rash, increased sweating Infrequent or rare: Hepatic dysfunction; suicidal ideation, speech disorders, hallucinations, tremor; idiopathic thrombocytopenia	Pregnancy/breastfeeding: Not recommended during pregnancy or breastfeeding Use with caution: In individuals with hepatic and renal impairment. If feasible, liver function tests should be routinely carried out. No ingestion of other opioid drugs for previous 5 days. Will block effects of other opioid drugs (if analgesia required)
Neostigmine Snake bite neurotoxicity (3.9) (Other indications: Myasthenia gravis, to reverse non-depolarizing muscle relaxants; post-operative non-obstructive urinary retention)	Injection: 500 mcg in 1 ml ampoule Tablet: 15 mg See Section 3.9	Common: Nausea, vomiting, increased salivation, diarrhoea, abdominal cramps. Signs of overdose: Bronchoconstriction, increased bronchial secretions, lacrimation, excessive sweating, involuntary defaecation and micturition, miosis, nystagmus, bradycardia, heart block, arrhythmias, hypotension, agitation, excessive dreaming, weakness eventually leading to fasciculation and paralysis	Pregnancy: Use with caution; no reports of malformation, but neonatal myasthenia gravis and bradycardia are possible in newborns exposed during pregnancy. Breastfeeding: May be used in breastfeeding (monitor infant for adverse effects) Use with caution: In asthma, urinary and intestinal surgery, infection

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Nifurtimox Chagas disease (American trypanosomiasis) (11.42)	Tablet: 30 mg, 120 mg, 250 mg Oral: 8–10 mg/kg daily divided in 2 or 3 doses; for 60 consecutive days	Common: Nausea/vomiting, diarrhoea, abdominal pain, anorexia, central nervous system alterations (sleep disturbances, excitatory states, seizures, psychotic behaviour), tremors, muscle weakness, paraesthesia and polyneuritis	Pregnancy: Uncertain safety in pregnancy, no human data; animal studies have not shown evidence of birth defects. Breastfeeding: Unknown safety in breastfeeding; not currently recommended; use only if potential benefit is greater than risk. Contraindications: Psychiatric or neurological disorders Counselling: Take after meals.
Nitrofurantoin Acute lower urinary tract infection (11.44)	Tablet: 100 mg Oral: 100 mg twice daily for 5 days	Common: Nausea, vomiting, anorexia, diarrhoea, abdominal pain; allergic skin reactions; headache Infrequent or rare: Hepatitis, jaundice; erythema multiforme; pancreatitis; blood disorders; with long-term use, pulmonary fibrosis; possible association with lupus erythematosus-like syndrome; peripheral neuropathy	Pregnancy: May be used at recommended doses. Due to risk of fetal haemolysis, avoid use at or near term in patients with G6PD deficiency. Breastfeeding: May be used at recommended dose except with neonates and infants who are G6PD-deficient; monitor infants for adverse effects. Contraindications: Impaired renal function, G6PD deficiency, porphyria Use with caution: In pulmonary disorders or hepatic impairment, neurological or allergic disorders, anaemia, diabetes mellitus, elderly and debilitated, vitamin B and folate deficiency Counselling: Take with food or milk to reduce nausea and improve absorption. This drug can make your urine a brownish colour.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Omeprazole Gastritis/peptic ulcer disease (PUD) (10.7a.2) <i>Helicobacter pylori</i> gastritis (10.7a.2)	Tablet: 20 mg Oral: 20 mg daily for 4–8 weeks Oral: 20 mg twice daily AND clarithromycin + amoxicillin	Common: Nausea, vomiting, abdominal pain, flatulence, diarrhoea, constipation; headache. Infrequent or rare: Dry mouth, peripheral oedema, dizziness, sleep disturbances, fatigue, paraesthesia, arthralgia, myalgia, rash, pruritus, taste disturbance, stomatitis, hepatitis, jaundice, agitation, impotence, fever, depression, hallucinations, confusion, gynaecomastia, interstitial nephritis, hyponatraemia, blood disorders (including leukopenia, leucocytosis, pancytopenia, thrombocytopenia), visual disturbances, sweating, photosensitivity, alopecia, Stevens–Johnson syndrome, toxic epidermal necrolysis. By decreasing gastric acidity, proton pump inhibitors may increase the risk of gastrointestinal infections (including <i>C. difficile</i> infection).	Pregnancy/breastfeeding: Caution in pregnancy and breastfeeding. Use only when treatments with antacids and H ₂ antagonists have failed. Use with caution: In renal impairment, hepatic impairment, and the elderly. Prone to multiple drug interactions. Do not administer in combination with clopidogrel. Counselling: Swallow the tablet whole. Do not crush or chew it.
Ondansetron Hyperemesis gravidarum (14.1.1.1); moderate nausea or vomiting or after chemotherapy for 1–2 days (10.7c.3.2.3) Severe vomiting (Other indications: Prior to chemotherapy causing severe vomiting)	Tablet: 4 mg, 8 mg Injection: 2 mg/ml in 2ml ampoule Liquid: 4 mg in 5 ml Oral: 4 mg every 12 hours: increase to 8 mg if this dose not effective. IV: 8 mg over 15 minutes every 12 hours OR 1 mg/hour infused continuously for up to 24 hours. Oral: Up to 24 mg daily oral or IV	Common: Constipation, headache, flushing; injection-site reactions; transient rise in hepatic enzymes Infrequent or rare: Hiccups, hypotension, bradycardia, chest pain, arrhythmias, movement disorders, seizures; on IV administration, rarely, dizziness, transient visual disturbances (very rarely, transient blindness)	Pregnancy/breastfeeding: No increased risks found in pregnancy or breastfeeding Administration: Give IV doses ≤8 mg over at least 5 minutes and doses >8 mg over at least 15 minutes. No dose adjustment in the elderly or if renal impairment. In severe liver impairment, do not exceed maximum dose of 8 mg.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Oseltamivir Pandemic H1N1: patients with severe influenza-like illness or at risk for severe ILI (QC p 20, 11.17)	Capsule: 30 mg, 45 mg, 75 mg Suspension: 12 mg/ml >40 kg: 75 mg twice daily for 10 days (Note: In severe illness may use 150 mg twice daily)	Common: Nausea, vomiting, abdominal pain, diarrhoea; headache; conjunctivitis Infrequent or rare: Rash, hepatitis, arrhythmias, neuropsychiatric disorders (in children and adolescents), visual disturbances, Stevens-Johnson syndrome, toxic epidermal necrolysis	Pregnancy: Unknown safety in pregnancy Breastfeeding: Not recommended until more known; use only if potential benefit is greater than risk.
Oxytocin Treatment of postpartum and post-abortion haemorrhage (QC pp. 25–26) (Other indications: prevention of postpartum haemorrhage - when the anterior shoulder is delivered or immediately after delivery)	Injection: 10 IU in 1 ml ampoule IM: 10 IU AND Start IV fluids with 20 IU oxytocin at 60 drops/minute See QC p. 25. Continue oxytocin at 20 IU at 20 drops/minute for at least 1 hour after bleeding stops.	Common: Nausea/vomiting, arrhythmia, headache Infrequent or rare: Disseminated intravascular coagulation, rash; anaphylactoid reactions (with dyspnoea, hypotension, or shock); uterine spasm (may occur at low doses); uterine hyperstimulation (usually with excessive doses; may cause fetal distress, asphyxia, and death, or may lead to hypertonicity, tetanic contractions, soft-tissue damage or uterine rupture); water intoxication and hyponatraemia associated with high doses with large infusion volumes of electrolyte-free fluid; in overdose, placental abruption, amniotic fluid embolism	Pregnancy/breastfeeding: Not known to be harmful in breastfeeding, as rapidly inactivated in GI tract Contraindications: Hypertonic uterine contractions, mechanical obstruction to delivery, fetal distress; any condition where spontaneous labour or vaginal delivery inadvisable. Avoid prolonged administration in oxytocin-resistant uterine inertia, severe pre-eclamptic toxemia, severe cardiovascular disease. Use with caution: Monitor fetal heart rate and uterine motility (discontinue immediately if uterine hyperactivity/fetal distress). To avoid water intoxication with hyponatraemia: (1) use electrolyte-containing diluent (not glucose); (2) increase oxytocin concentration to reduce fluid; (3) restrict fluid intake by mouth; (4) monitor fluid and electrolytes.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Paracetamol (acetaminophen) Mild to moderate pain (20.2, 20.4); fever (10.1) (Other indications: Acute migraine attacks; tension headache)	Tablet: 250 mg, 500 mg Dispersible tablets: 500 mg Suppositories: 250 mg, 500 mg Oral: 0.5–1 g every 4–6 hours, (max 4 g daily; max 2 g daily if hepatic impairment, cirrhosis) Rectally: 0.5–1 g every 4–6 hours	Common: Increased transaminases Infrequent or rare: Urticarial or erythematous rash, blood disorders, liver damage following overdosage	Pregnancy/breastfeeding: Considered safe in both pregnancy and breastfeeding (monitor infant for side-effects) Use with caution: In hepatic impairment, renal impairment, alcohol dependence Counselling: Do not exceed 4 grams in 24 hours.
Paromomycin Local treatment of cutaneous leishmaniasis (11.20.1) Visceral leishmaniasis caused by certain species	Injection: 750 mg base (11 mg base = 15 mg paromomycin sulfate) Ointment: 15% paromomycin + 12% methyl benzethonium chloride Ointment twice daily for up to 20 days See table in Section 11.20.2	Common: Injection site reactions, elevated liver enzymes Infrequent or rare: Ototoxicity (reversible at recommended dosage), nephrotoxicity, neurotoxicity (numbness, skin tingling, muscle twitching, convulsions; neuromuscular blockage, respiratory paralysis reported following high doses)	Pregnancy: Unknown safety in pregnancy; no human data. Use only if potential benefit is greater than risk. Breastfeeding: Unknown safety in breastfeeding, but poorly absorbed from GI tract so excretion into breast milk likely to be minimal. Contraindications: Hypersensitivity to aminoglycosides, course of paromomycin treatment in preceding 3 months, concurrent administration of nephrotoxic or ototoxic drugs including aminoglycosides, renal impairment

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Pentamidine Cutaneous leishmaniasis (due to severe side-effects, recommended only if no other treatment available) (11.20.1) Leishmaniasis, secondary prophylaxis in HIV-infected patients (11.20.3) Severe <i>Pneumocystis jirovecii</i> (PCP) pneumonia, treatment if not able to tolerate or unresponsive to cotrimoxazole (10.6.3) Human African trypanosomiasis (<i>T. b. gambiense</i>)- haemolympathic stage (11.41)	Powder for injection: 200 mg, 300 mg (isetonate) in vial IM/IV: 2–3 mg/kg once daily or every second day for 4–7 doses 4 mg/kg (300 mg) every 3–4 weeks Slow IV/deep IM: 4 mg/kg daily for 5 days; then reduce dose to 2 mg/kg daily to complete 21 days IM (deep): 4 mg/kg daily for 7 consecutive days	Common: Nausea, vomiting, diarrhoea, taste disturbances; severe reactions, sometimes fatal (hypotension, hypoglycaemia, pancreatitis, arrhythmias); leukopenia, thrombocytopenia; acute renal failure, hypocalcaemia Infrequent or rare: Azotaemia, abnormal liver-function tests, anaemia, hyperkalaemia, dizziness, syncope, flushing, hyperglycaemia, rash, Stevens-Johnson syndrome; on inhalation, bronchoconstriction (may be prevented by prior use of bronchodilators), cough, shortness of breath; discomfort, pain, induration, abscess formation, muscle necrosis at injection site	Pregnancy: Potentially fatal visceral leishmaniasis and PCP pneumonia (cotrimoxazole is preferred) should be treated in pregnancy. Breastfeeding: Manufacturer advises avoiding unless essential; not known to be harmful, as rapidly inactivated in GI tract. Contraindications: In severe renal impairment Administration: Risk of severe hypotension following administration. Establish baseline blood pressure and administer with patient lying down; monitor blood pressure closely during administration and at regular intervals until treatment concluded. Avoid direct intravenous injection whenever possible and never give rapidly. Reconstitute with 3–5 ml of water for injection; dilute further to 50–250 ml with glucose 5% or normal saline; give over at least 60 minutes. IM injections should be deep and preferably given into the buttock. Pentamidine is toxic – protect health workers during handling and administration.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Permethrin Scabies (10.2.3); pediculosis (body lice) (10.2.8)	Cream: 5% Topical: Apply cream over whole body AND Wash off after 8-12 hours (if hands washed with soap within 8 hours after application, treat again) THEN Repeat after 7 days as necessary. Apply to damp hair and leave for 10 minutes before washing off.	Common: Temporary increase in itch, redness, swelling that usually accompany lice or mite infestation Infrequent or rare: Local irritation, rash, oedema	Pregnancy: May be used; systemic absorption expected to be minimal. Breastfeeding: May be used; systemic absorption expected to be minimal, but avoid application to nipple areas (or withhold breastfeeding during treatment). Contraindications: Do not use on inflamed or broken skin. Counselling: Avoid contact with eyes, mouth, and inside the nose. Itch may persist for 2-3 weeks after scabies treatment or 7-10 days after lice treatment. This may not indicate ongoing infection. Scabies: Remember to apply also between fingers and toes, under nails, in skin folds, navel, between the buttocks, and in groin area. If you wash your hands or any other treated parts of the body during the treatment period, you should reapply the lotion to the washed areas.
Pediculosis capitis (head lice) (10.2.8)			

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Phenobarbital Status epilepticus (3.5) Epilepsy (10.10c.2)	Injection: 200 mg/ml (phenobarbital sodium) Tablet: 15–100 mg (phenobarbital) Loading dose: IV: 5–15 mg/kg bolus over 1 hour Oral: Initiate at 60 mg daily; then maintain at 60–180 mg daily.	Common: Drowsiness, incoordination, restlessness and confusion (in elderly), impaired memory and cognition, hyperactivity (particularly in the elderly), allergic skin reactions, paradoxical excitement, sleep disorders. With IV: hypotension, respiratory depression, laryngospasm Infrequent or rare: Hepatitis; cholestasis; behavioural disturbances, nystagmus, irritability, lethargy, depression, ataxia, hallucinations; osteomalacia; megaloblastic anaemia (may be treated with folic acid); agranulocytosis, thrombocytopenia; allergic skin reactions; very rarely Stevens-Johnson syndrome and toxic epidermal necrolysis; status epilepticus (on treatment withdrawal); Dupuytren's contracture; lymphadenopathy	Pregnancy: Not recommended; adverse effects on neurobehavioural development have been reported. Breastfeeding: Use with caution in breastfeeding; avoid large doses and monitor for adverse effects; may accumulate in breast milk. Contraindications: Porphyrria, absense seizures Use with caution: In the elderly, impaired renal or hepatic function, respiratory depression Avoid sudden withdrawal. Prone to multiple drug interactions through CYP enzymes; check for interactions with all new and current medications. Counselling: Take once daily at bedtime. May cause drowsiness and affect your ability to drive or operate machinery. Avoid these activities at least until you know how this medicine affects you. Do not stop taking this medicine suddenly without your clinician's advice. Avoid drinking alcohol, as the medicine may increase the effects of alcohol.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Phenoxymethyl-penicillin (penicillin V) Streptococcal pharyngitis (10.17.9, 11.32) Cutaneous anthrax, non-severe, if known antibiotic sensitivity (10.2.10) Dental abscess (10.17.5) Prevention of recurrent rheumatic fever (11.32) Erysipelas (10.2.2) (Other indications: Otitis media, post-splenectomy prophylaxis)	Tablet: 500 mg Oral: 500 mg twice daily for 10 days Oral: 500 mg every 6 hours for 7–10 days Oral: 250 mg every 6 hours for 5 days Oral: 500 mg twice daily Oral: 500 mg every 6 hours for 5–10 days	See benzylpenicillin.	Pregnancy/breastfeeding: Considered safe in both pregnancy and breastfeeding (monitor infant for side-effects) Contraindications: Hypersensitivity to penicillins Use with caution: In renal impairment. Oral penicillin should not be used for the treatment of severe infections. Counselling: Take 1 hour before meals or on an empty stomach.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Phenytoin Generalized tonic-clonic seizures; partial seizures (10.10c.2) Status epilepticus (3.5)	Capsule: 25 mg, 50 mg, 100 mg Injection: 50 mg/ml in, 5 ml vial Tablet: 25 mg, 50 mg, 100 mg Tablet (Chewable): 50 mg Oral: Start at 150–200 mg daily, increase by increments of 25–30 mg to reach maintenance at 200–400 mg daily Loading dose: IV (slowly over 60): 15 mg/kg in normal saline, at a rate of not more than 50 mg/minute (monitor BP and ECG). Then oral maintenance dose as above.	Common: Nausea, vomiting, constipation; insomnia, transient nervousness, tremor, paraesthesia, dizziness, headache, anorexia; gingival hypertrophy and tenderness; rash (discontinue; if mild, re-introduce cautiously but discontinue immediately if recurrence); acne, hirsutism, coarse facies Infrequent or rare: Hepatotoxicity, peripheral neuropathy, dyskinesia, lymphadenopathy, osteomalacia; blood disorders (including megaloblastic anaemia (may be treated with folic acid)), leukopenia (if severe, progressive, or clinically apparent leukopenia develops, withdraw drug, replacing with suitable alternative), thrombocytopenia, aplastic anaemia, polyarteritis nodosa, lupus erythematosus, Stevens-Johnson syndrome, toxic epidermal necrolysis; pneumonitis, interstitial nephritis; with excessive dosage, nystagmus, diplopia, slurred speech, ataxia, confusion, hyperglycaemia	Pregnancy: Risk of teratogenicity; use only if benefit is greater than risk (provide adequate folic acid supplementation to mother); monitor for neonatal bleeding if vitamin K not given at birth. Breastfeeding: May be used in breastfeeding, monitor for sedation and decreased sucking; consider infant serum level monitoring. Contraindications: Porphyria. Avoid parenteral use in sinus bradycardia, sinoatrial block, second- and third-degree heart block, Adams-Stokes syndrome. Use with caution: In hepatic impairment, diabetes mellitus, hypotension, heart failure. Resuscitation facilities must be available for intravenous administration. Administration: Administer phenytoin IV in normal saline and not in same line as diazepam. IV line should be running well, as the drug is caustic and will cause local damage if it extravasates. Monitor blood counts. When decision is made to withdraw treatment, do so preferably over 6 months at a rate not greater than 25 mg each week or 100 mg each month. Counselling: Seek immediate medical attention if you have symptoms such as sore throat, rash, mouth ulcers, bruising, or bleeding. This medication may impair ability to perform skilled tasks such as operating machinery or driving; avoid these activities at least until you know how this medicine affects you. Take with or after food. Good dental hygiene can help to prevent gum enlargement. Do not stop this medicine suddenly without your clinician's advice.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Pilocarpine eye drops Chronic open-angle glaucoma (10.12.4) Emergency treatment of acute angle-closure glaucoma (before surgery) (10.12.2) (Other indications: Ocular hypertension; to antagonize effects of mydriasis and cycloplegia following surgery or ophthalmoscopic examination)	Drops: 2%, 4% (hydrochloride or nitrate) 1 drop (2% or 4% solution) up to 4 times daily 1 drop (2% solution) every 10 minutes for 30–60 minutes; then 1 drop every 1–3 hours until intraocular pressure subsides	Common: Eye pain, blurred vision, ciliary spasm; ciliary spasm leads to headache and brow ache, which may be most severe in the initial 2–4 weeks of treatment. Infrequent or rare: Lacrimation, myopia, conjunctival vascular congestion, superficial keratitis, vitreous haemorrhage, increased pupillary block; lens opacities (following prolonged use); rarely, systemic effects including hypertension, tachycardia, bronchial spasm, pulmonary oedema, salivation, sweating, nausea, vomiting, diarrhoea	Contraindications: Acute iritis, acute uveitis, anterior uveitis, some forms of secondary glaucoma; acute inflammation of anterior segment. Use not advisable after angle-closure surgery (risk of posterior synechiae). Use with caution: In retinal disease, conjunctival or corneal damage (monitor intraocular pressure in chronic open-angle glaucoma and in long-term treatment); cardiac disease, hypertension, asthma, peptic ulceration, urinary tract obstruction, Parkinson's disease. Withdraw treatment if symptoms of systemic toxicity develop. Administration: A darkly pigmented iris may require higher concentration of the miotic or more frequent administration; care should be taken to avoid overdosage. Counselling: If you are using more than one type of eye drop, put in pilocarpine drops last. Causes difficulty with adapting to the dark and may cause accommodation spasm. Avoid skilled tasks, for example, operating machinery or driving, until vision is clear.
Podophyllum resin External anogenital warts; plantar warts (10.2.3)	Solution: 10–25% Apply carefully, avoiding contact with normal tissue; rinse off after 1–6 hours. May be repeated at weekly intervals but no more than 4 times in all. Only a few warts should be treated at any one time.	Common: Irritation, staining of the skin Infrequent or rare: Systemic effects resulting from cutaneous absorption include nausea, vomiting, abdominal pain, diarrhoea; also, transient leukopenia and thrombocytopenia; renal failure; delayed neurotoxicity including visual and auditory hallucinations, delusions, disorientation, confusion, delirium following excessive application	Pregnancy/breastfeeding: Contraindicated in pregnancy and breastfeeding Use with caution: Avoid use on large areas; very irritant to eyes (keep away from face); avoid contact with normal skin, mucus membranes, open wounds. Administration: Must be applied by a trained health worker Counselling: Avoid contact with face and other sensitive areas.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Povidone iodine (povidone-iodine) Skin disinfection (10.2) Antiseptic for minor wounds and burns (4)	Solution: 10% Apply undiluted solution to the skin area. Apply undiluted solution to the affected area twice daily.	Infrequent or rare: Irritation of skin and mucous membranes; may interfere with thyroid function tests	Contraindications: Avoid regular or prolonged use in patients with thyroid disorders or those taking lithium. Administration: Do not use on large open wounds; may produce systemic adverse effects such as metabolic acidosis, hypernatraemia, impairment of renal function.
Polyethylene glycol electrolyte solution (osmotically balanced mixture) Acute iron poisoning and overdose with highly toxic sustained-release preparations, e.g. calcium channel blockers (3.8.1)	Solution Bowel irrigation: 2 litres per hour for adults if tablets are present beyond the stomach	Common: Nausea, vomiting, abdominal cramps, bloating Infrequent or rare: Aspiration pneumonia Note: Risk of fluid and electrolyte imbalance if solution is not correctly formulated	Pregnancy: Unknown safety in pregnancy. Breastfeeding: Not recommended until more known; use only if potential benefit is greater than risk. Contraindications: Ileus, GI haemorrhage, haemodynamic instability, uncontrollable vomiting, bowel obstruction or perforation, decreased consciousness with unprotected airway.
Potassium permanganate Wet dressings to assist healing of suppurating superficial wounds, tropical ulcers; pemphigus; tinea pedis; infected eczema (10.2)	Aqueous solution: 1:10 000 (0.01%) Apply dressings soaked in a 1:10 000 solution to affected area until superficial crusts can be gently separated. Change dressings 2 or 3 times daily. Bathe severe weeping lesions in a 1:10 000 solution every 8 hours.	Common: Irritant to mucous membranes; skin and fabrics can be stained brown.	Use with caution: Avoid occlusive dressings. Administration: In exudative eczematous areas, treatment should be stopped when the skin becomes dry. Note: Potassium permanganate is sometimes supplied as an aqueous stock solution of 1:1000 (0.1%) for dilution before use. To be diluted 1 in 10 to provide a 0.01% (1 in 10 000) solution.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Potassium chloride (KCl) Hypokalaemia, mild to moderate (5.2.2) Hypokalaemia, severe (5.2.2)	Solution: 11.2% in 20 ml ampoule (equiv to K⁺ 1.5 mmol/ml, Cl⁻ 1.5 mmol/ml); Oral preparations Oral: 20–50 mmol daily after meals Slow IV infusion: 20–40 mmol/l in normal saline (not to exceed 10–20 mmol/hour)	Common: Nausea/vomiting (severe symptoms may indicate obstruction of oesophagus or small bowel), hyperkalaemia (especially in renal insufficiency), rapid infusion toxic to heart	Pregnancy/breastfeeding: May be used for supplementation in pregnancy and breastfeeding; monitor electrolytes to keep maternal serum levels within normal range. Contraindications: Severe renal impairment and plasma potassium concentration above 5 mmol/l Use with caution: In elderly, mild/moderate renal impairment, history of peptic ulcer Administration: Monitor potassium levels. Potassium salts cause nausea and vomiting; therefore, poor compliance is a major limitation to their effectiveness; where appropriate, potassium-sparing diuretics are preferable. Counselling: If you have severe nausea and vomiting, stop the medicine and inform your clinician.
Praziquantel Schistosomiasis (11.34)	Tablet: 600 mg Oral: 40 mg/kg as a single dose	Usually mild and transient with short course. Many adverse effects result from death of the parasite and are more severe with high parasite burden. Common: Dizziness (dose dependent), headache, malaise, drowsiness; nausea, vomiting, abdominal pain, diarrhoea, anorexia, colic; reversible rises in hepatic transaminases; rectal bleeding infrequent or rare; arrhythmia	Pregnancy: Use not recommended in first trimester; consider alternatives. Caution in second and third trimesters; use only if treatment of choice. Breastfeeding: May be used for single-day treatment during breastfeeding Contraindications: Ocular cysticercosis Counselling: Take with food. Swallow with plenty of water to prevent vomiting due to bitter taste. Tablet may be cut into halves or quarters, but do not chew. The medicine may make you feel drowsy or dizzy; if you are affected, do not drive or operate machinery until 24 hours after finishing your course.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Prednisolone Also see Section 8.2 steroid equivalents table.	Tablet: 5 mg, 25 mg	Overdosage or prolonged use can exaggerate some of the normal physiological actions of corticosteroids, leading to mineralocorticoid and glucocorticoid side-effects.	Pregnancy: May be used at the recommended doses; caution in first trimester due to possibility of oral cleft; limited transplacental transfer. Monitor blood glucose, especially in diabetics.
Moderate to life-threatening wheezing (3.2.4)	Oral: 40–60 mg	<i>Mineralocorticoid side-effects:</i> Hypertension, sodium and water retention, potassium and calcium loss	Breastfeeding: May be used at recommended doses; amount in milk low at doses up to 80 mg (monitor infant's adrenal function if dose is higher)
<i>Pneumocystis jirovecii</i> pneumonia (PCP pneumonia) (10.6.3)	Oral: 40 mg twice daily for 5 days THEN 40 mg daily for 5 days THEN 20 mg daily for 11 days to complete 21 days of treatment	Common: Dyspepsia, increased susceptibility to infection (oral, vaginal, intertriginous candidiasis), masking of signs of infected acne, oedema, hypertension, hypokalaemia, hyperglycaemia, weight gain, osteoporosis, spontaneous fractures, increased appetite, delayed wound healing, skin atrophy, growth retardation in children, myopathy, muscle weakness, wasting (particularly symptomatic on drug withdrawal) fat redistribution (producing cushingoid appearance), amenorrhoea, psychosis, euphoria, depression, adrenal suppression, bruising	Contraindications: Systemic infection (unless life-threatening or specific antimicrobial therapy given). Avoid live virus vaccines in those receiving immunosuppressive doses.
Management of severe persistent asthma (10.6.4)	Oral: 0.5 mg/kg daily (reassess weekly; taper when patient stable for 1 week)		Use with caution: In infections, hypertension, recent myocardial infarction, congestive heart failure, renal impairment, hepatic impairment, diabetes mellitus including family history, osteoporosis, glaucoma including family history, corneal perforation, severe affective disorder, epilepsy, psoriasis, peptic ulcer, hypothyroidism, history of steroid myopathy, and in the elderly; can activate or exacerbate TB, amoebiasis, strongyloidiasis.
Type 2 lepra reaction (11.21)	See Section 11.21 for dosing.		Prolonged treatment leads to adrenal suppression, which persists for years after stopping treatment. Cushingoid features are increasingly likely with doses above 7.5 mg daily. Risk of chickenpox, measles, and activation of tuberculosis increased.

Other indications: With antineoplastic drugs for acute and chronic leukaemias; lymphoma; suppression of inflammatory and allergic reactions; inflammation of the eye; myasthenia gravis)	Note on steroid dosing: Patients on long term steroid therapy are at risk for a blunted stress response in conditions causing physiologic stress (e.g., severe infection, trauma, during surgery). As a result, standard steroid doses may need to be supplemented. Patients on steroid replacement for primary dysfunction of the hypothalamus-pituitary-adrenal axis HPA axis (for example, Addison disease, hypopituitarism) SHOULD receive higher dose of steroids in conditions causing physiologic stress. A common IV regimen is hydrocortisone 50-100 mg every 8 hours for 2 days. Consult a specialist for assistance on how to provide this supplemental steroid treatment.	Infrequent or rare: Acute pancreatitis, peptic and oesophageal ulceration, vertebral compression fracture, aseptic necrosis of the talus or femoral and humoral heads; facial erythema, suppression of skin test reactions, hyperhidrosis, skin bruising, telangiectasia, myocardial rupture following recent myocardial infarction, congestive heart failure, leucocytosis, hypersensitivity reactions (including anaphylaxis), thromboembolism, malaise, hiccups, headache, vertigo. <i>Glucocorticoid side-effects:</i> Diabetes and osteoporosis can result in osteoporotic fractures, particularly in the elderly. CNS side-effects: Psychological dependence, insomnia, aggravation of schizophrenia, aggravation of epilepsy. Ophthalmologic side-effects: Glaucoma, papilloedema, posterior subcapsular cataracts, corneal or scleral thinning, exacerbation of opthalmic viral or fungal disease, increased intra-ocular pressure, exophthalmos	Administration: Monitor weight, blood pressure, fluid and electrolyte balance, and blood glucose levels throughout prolonged treatment. The suppressive action of a corticosteroid on cortisol secretion is least when it is given as a single dose in the morning. During corticosteroid withdrawal the dose may be reduced rapidly down to physiological doses (equivalent to prednisolone 7.5 mg daily) and then reduced more slowly. Assessment of the disease may be needed during withdrawal to ensure that relapse does not occur. Counselling: Take with food to reduce stomach upset. Tell your doctor immediately if you have any signs of infection. If you have been on this medicine for more than 3 weeks, don't stop the treatment suddenly. Tell your doctor, dentist, or pharmacist that you are on steroids before undergoing any new treatment.
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Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Primaquine Radical treatment of <i>P. vivax</i> and <i>P. ovale</i> malaria (after standard chloroquine therapy) (11.25.3)	Tablet: 7.5 mg, 15 mg Oral: 250 mcg/kg daily for 14 days OR 30 mg daily for 14 days	Common: Nausea, vomiting, anorexia, abdominal cramps; dizziness, headache Infrequent or rare: Haemolytic anaemia (frequently in G6PD deficiency; withdraw treatment); methaemoglobinemia (withdraw treatment); haemoglobinuria; agranulocytosis, granulocytopenia, leukopenia	Pregnancy: Uncertain safety in pregnancy, use not recommended. Breastfeeding: Use with caution; limited data. Monitor infants for adverse effects (e.g. haemolysis, jaundice). Avoid in neonates and infants who are G6PD-deficient. Contraindications: Conditions that predispose to granulocytopenia. Monitor blood count. Exclude G6PD deficiency before radical treatment for <i>P. vivax</i> and <i>P. ovale</i> but not before single-dose gametocytocidal treatment. Counselling: Give with food if severe nausea/vomiting or abdominal cramps occur.
Procaine benzylpenicillin G Neurosyphilis (11.37) Tropical ulcer (10.2.10) Borreliosis (louse-borne relapsing fever) if not able to take orally (10.1) (Other indications: Diphtheria; animal bites)	Powder for Injection: 1 g vial (1 million IU), 3 g vial (3 million IU) IM: 2.4 g (2.4 million IU) daily AND oral <i>probenecid</i> 500 mg 4 times daily for 10–14 days 0.5 g (600 000 IU) daily for 2–4 weeks IM: 0.6 to 0.8 g (600 000 to 800 000 IU) as a single dose	See benzylpenicillin. Pain and inflammation at injection site; Jarisch-Herxheimer reaction (rigors, fever, and hypotension) usually within several hours after treatment of syphilis or borreliosis; probably due to release of endotoxin. Accidental intravascular administration may result in anxiety, agitation, fear of death, hallucinations. These usually resolve in 15–30 minutes and rarely last beyond 24 hours.	Pregnancy/breastfeeding: Considered safe in both pregnancy and breastfeeding (monitor infant for side-effects). Contraindications: Hypersensitivity to penicillins or if IV administration needed Use with caution: In renal failure. Observe patient with syphilis or borreliosis for several hours after treatment. Administration: Give by deep IM injection. (Do not give IV.) Can be used for daily outpatient treatment. Probenecid is added to increase serum penicillin levels in neurosyphilis.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Propranolol Primary prevention of variceal bleeding in patients with documented varices (10.9) (Other indications: migraine prophylaxis, essential tremor, stable angina, hypertension, some arrhythmias)	Tablet. 20 mg; 40 mg (hydrochloride) Titrate to achieve a 25% reduction in the heart rate	Common: Nausea/vomiting, diarrhoea, bradycardia, bronchospasm, cold extremities, fatigue, sleep disturbances including nightmares; heart failure, hypotension, conduction disorders, peripheral vasoconstriction, exacerbation of intermittent claudication, Raynaud's phenomenon; alteration of glucose and lipid metabolism Infrequent or rare: Rash, dry eyes (reversible), sexual dysfunction, exacerbation of psoriasis, purpura, thrombocytopenia	Pregnancy: Use with caution. May cause intrauterine growth restriction, neonatal hypoglycaemia, and bradycardia; risk greater in severe hypertension. Breastfeeding: Present in milk; safe in usual dosage; monitor infant. Contraindications: Asthma or history of obstructive airway disease, uncontrolled heart failure, Prinzmetal's angina, marked bradycardia, hypotension, sick sinus syndrome, second- or third-degree atrioventricular block, cardiogenic shock, metabolic acidosis, severe peripheral arterial disease; phaeochromocytoma Use with caution: In first-degree atrioventricular block; renal impairment; liver disease; portal hypertension; diabetes mellitus; myasthenia gravis; history of hypersensitivity Administration: When stopping treatment, reduce dosage gradually over at least 2 weeks. Counselling: Do not stop taking this medicine suddenly without your clinician's advice.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Pyridoxine (vitamin B6) Ethylene glycol poisoning (3.8.1) Peripheral neuropathy (10.10a.6); neuropathic signs on INH (13.3, 15.4.2) Prevention of peripheral neuropathy with INH for prophylaxis or treatment for TB (15.4.2) Antiemetic in pregnancy (14.1.1.1)	Tablet: 25 mg Oral: 50 mg every 6 hours for 6 doses Oral: 50–75 mg daily Oral: 10 mg daily Oral: 25 mg, up to 3–4 times daily	Infrequent or rare: Sensory neuropathy reported with high doses given for extended periods	Pregnancy/breastfeeding: May be used in pregnancy and breastfeeding; doses >200 mg/day may suppress lactation.
Pyrimethamine Toxoplasmosis in immunodeficiency (11.40) Chorioretinitis (10.12.6)	Tablet: 25 mg, 50 mg Oral: 100–200 mg as a single dose THEN 50 mg daily for at least 6 weeks AND sulfadiazine + folic acid for 6 weeks Oral: 75 mg daily for 3 days THEN 25 mg daily for 4 weeks AND sulfadiazine + folic acid for same duration (In unresponsive patients, 50 mg daily for a further 4 weeks)	Common: Nausea/vomiting, diarrhoea, depression of haematopoiesis with high doses, megaloblastic anaemia, rashes, insomnia, CNS toxicity (at high doses)	Pregnancy: May be used during pregnancy if it is the drug of choice; avoid in first trimester when possible. Breastfeeding: Use, with caution, if it is the drug of choice. Contraindications: Megaloblastic anaemia Use with caution: In hepatic and renal impairment For treatment of toxoplasmosis, pyrimethamine must always be taken with sulfadiazine and should be administered with folic acid when available. Administration: Give with food if GI disturbances occur. Monitor blood counts in prolonged treatment and give folate supplements throughout treatment.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Quinine Severe <i>P. falciparum</i> malaria if parenteral artesunate not available (QC p. 20, 11.25.5)	Tablet: 300 mg Infusion: 300 mg/ml in 2 ml ampoule IV/IM (in anterior thigh): 20 mg/kg over 4 hours; THEN 10 mg/kg every 8 hours until oral medication is possible Oral: 600 mg quinine sulfate 3 times daily for 7 days AND clindamycin or doxycycline	Common: Nausea, vomiting, diarrhoea, CNS disturbances, reversible hearing loss, cinchonism, tinnitus, headache, hot and flushed skin, nausea, abdominal pain, rashes, visual disturbances (including temporary blindness), confusion, fever, rash, hypoglycaemia (after parenteral administration), thrombocytopenia, ECG changes Infrequent or rare: Angioedema, intravascular haemolysis, acute renal failure, prolonged QT interval	Pregnancy: Use only if benefit is greater than risk: avoid in first trimester when possible, as high doses (>1 g total) may cause fetal deafness. Breastfeeding: Caution; monitor for adverse effects. Avoid in G6PD-deficient infants. Contraindications: Haemoglobinuria, optic neuritis, tinnitus Use with caution: In atrial fibrillation, conduction defects, heart block, renal impairment, G6PD deficiency, myasthenia gravis. With IV use, monitor signs of cardiac toxicity and blood glucose levels. Administration: Oral: If part of all of a dose is vomited within 1 hour, the same amount must be re-administered immediately. IV: Do not give as intravenous bolus injection. Infuse IV quinine over 4 hours, preferably in glucose 5% to reduce risk of hypoglycaemia.
Quinine + clindamycin Non-severe <i>P. falciparum</i> malaria in first-trimester pregnancy (11.25.7); <i>P. falciparum</i> malaria in travellers (11.25.3); second-line treatment for malaria (11.25.4)	Separate blisters with tablets of 150 or 300 mg quinine and tablets of 100 mg doxycycline Oral: 600 mg of quinine salt given 3 times a day (every 8 hours) for 7 days and 600 mg of clindamycin base twice daily for 7 days	As for quinine and clindamycin	See doxycycline and quinine. Additionally: Use with caution: Rifampicin reduces the plasma concentration of quinine, leading to increased treatment failures. Avoid antiarrhythmics, such as flecainide and amiodarone. Antihistamines, such as terfenadine, and antipsychotic drugs, such as pimoziide and thioridazine, can increase risk of arrhythmias. Cimetidine can increase quinine levels.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Rifampicin	Capsule or tablet: 150 mg, 300 mg	Common: Nausea, vomiting, diarrhoea, anorexia, headache, drowsiness; arthralgia, myalgia (in the first weeks of treatment). Those occurring mainly on intermittent therapy include influenza-like symptoms (chills, fever, dizziness, bone pain); urine, saliva, other body secretions coloured orange-red. Infrequent or rare: Respiratory symptoms (including shortness of breath); collapse and shock; haemolytic anaemia, thrombocytopenic purpura, disseminated intravascular coagulation, leukopenia, eosinophilia; acute renal failure; alterations of liver function, jaundice; flushing, urticaria, rashes, exfoliative dermatitis, toxic epidermal necrolysis, Stevens-Johnson syndrome, pemphigoid reactions; oedema; psychoses; adrenal insufficiency; muscular weakness, myopathy; menstrual disturbances	Pregnancy: Considered safe Breastfeeding: May be used, with caution, at recommended doses Counselling: Urine, tears, saliva, and sputum may become coloured orange-red- do not worry about it as the colour change is harmless.
Buruli ulcer (in combination therapy with streptomycin) (10.2.10)	See Section 10.2 for dosing.		
Leprosy- paucibacillary (11.21)	Oral: 600 mg rifampicin once monthly for 6 months AND daily dapsone		
Leprosy- multibacillary (11.21)	600 mg rifampicin once monthly for 12 months AND dapsone + clofazimine. See Section 11.21 for details of dosing.		

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Rifampicin + isoniazid + pyrazinamide + ethambutol hydrochloride (R + H+ Z+ E) Tuberculosis (13.10 HIV/TB co-management, 15 TB)	Tablet: fixed dose combination rifampicin 150 mg, isoniazid 75 mg, pyrazinamide 400 mg, ethambutol 275 mg 2HRZE/4HR Oral: Rifampicin 10 mg/kg daily (max 300 mg) AND isoniazid 5 mg/kg (max 600 mg) AND pyrazinamide 25 mg/kg AND ethambutol 15 mg/kg once daily OR rifampicin 10 mg/kg daily (maximum 600 mg) AND isoniazid 5 mg/kg (maximum 900 mg) AND pyrazinamide 25 mg/kg AND ethambutol 15 mg/kg 3 times weekly (given as DOT; 3 times weekly regimen NOT recommended for HIV-positive patients and those from high HIV prevalent settings)	See rifampicin, isoniazid, ethambutol, pyrazinamide. Common: Hyperuricaemia, polyarthritits, nausea Infrequent or rare: Hepatotoxicity (including fever, anorexia, hepatomegaly, splenomegaly, jaundice, liver failure), flushing, dysuria, sideroblastic anaemia, photosensitivity	Pregnancy/breastfeeding: Considered safe in pregnancy; Benefits of treating TB in pregnant and breastfeeding women outweigh risks of drug side-effects to either mother or infant. Monitor infant for signs of pyridoxine deficiency or jaundice; consider maternal or fetal supplementation. Contraindications: Hypersensitivity to rifampicin, hepatic disease, porphyria, optic neuritis, severe renal impairment Use with caution: In hepatic or renal impairment, diabetes mellitus, gout, chronic alcohol dependence, elderly, epilepsy, history of psychosis. Prophylactic pyridoxine 10 mg daily indicated, particularly in HIV-positive patients. Administration: Ocular examination recommended before and during treatment. Three times per week is acceptable alternative provided that patient is receiving directly observed therapy, and is not living with HIV or living in an HIV-prevalent setting. See Section 15. Counselling: Discontinue treatment and seek immediate medical attention if you develop persistent nausea, vomiting, malaise, yellow discoloration of the white of your eye or urine. Tell your health care provider immediately about any changes in your vision. If you are using a combined oral contraceptive (the Pill), patch, vaginal ring, or progestin-only pills, use additional contraception, such as condoms.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Salbutamol Acute bronchospasm (QC p. 17, 3.2.4) Asthma and COPD (10.6) Hyperkalemia (5.2.2)	Inhalation respirator solution for use in nebulizers: 5 mg (as sulfate)/ml Aerosol: 100 mcg (as sulfate) per dose See QC p.17 and 3.2.4 for dosing. See Sections 10.6.4, 10.6.5. By nebulizer: 10–20 mg OR IV: 0.5 mg (500 mcg). Administration should be slow, over 15–20 minutes. If neither of these are available, give salbutamol 1200 mcg by metered-dose inhaler with spacer (12 puffs).	Common: Palpitations, fine tremor (usually hands), headache. Infrequent or rare: With inhaled dosage forms, hyperglycaemia and hypokalaemia after high doses; muscle cramps; arrhythmias, tachycardia, insomnia, paradoxical bronchospasm, urticaria/angioedema	Pregnancy: May be used at recommended doses; asthma management for pregnant and non-pregnant women should be the same. Breastfeeding: May be used at recommended doses (monitor infant) Use with caution: In hyperthyroidism, myocardial insufficiency, arrhythmias, susceptibility to QT interval prolongation, hypertension, diabetes mellitus
Salicylic acid Hyperkeratotic conditions, including warts; adjunct in treatment of psoriasis, ringworm, seborrhoeic dermatitis, ichthyosis (10.2)	Solution: 5% Apply directly to affected area once daily, starting with lower strength preparations; gradually increase strength until satisfactory response obtained.	Infrequent or rare: Local irritation, dermatitis, toxicity with excessive application or treatment of large areas	Contraindications: Broken or inflamed skin. Use with caution: In significant peripheral neuropathy; in diabetics at risk of neuropathic ulcers Administration: Avoid application to large areas. Counselling: Avoid contact with eyes, lips, and inside of your nose. Protect surrounding skin; rub warts gently with file or pumice stone once weekly.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Selenium sulfide Pityriasis versicolor (10.2.7)	Detergent-based suspension 2%, lotion Apply lotion with small amount of water to entire affected area; rinse off after 10 minutes. Repeat daily for 7–14 days. OR Apply to affected area at bedtime; rinse off in morning; repeat 1–6 times over 2 weeks; repeat course if necessary. Detergent-based suspension/shampoo: Massage 5–10 ml into wet hair and leave for 2–3 minutes before rinsing thoroughly; repeat twice weekly for 2 weeks; then once weekly for 2 weeks; thereafter only when needed.	Common: Local irritation, hair discolouration or loss. Absorption may result in systemic toxicity including tremors, weakness, lethargy, pain in lower abdomen, occasional vomiting (symptoms usually resolve within 10 days)	Pregnancy: May be used on the scalp; it should not be used on the body to treat skin infections, as it may be absorbed through the skin. Breastfeeding: Uncertain safety in breastfeeding; use with caution. Use with caution: Do not apply to damaged skin (risk of systemic toxicity); avoid contact with eyes; do not use within 48 hours of applying preparations for hair colouring, straightening, or permanent wave. Administration: To minimize absorption, rinse hair thoroughly after use and remove all traces from skin (including nails). Note: Selenium sulfide is widely used in proprietary shampoos.
Senna Constipation (10.7d.4, 20.2)	Tablet: 7.5 mg (sennosides) Oral: 2–4 tablets, usually at night. Initial dose should be low; then gradually increased to 30 mg.	Common: Abdominal discomfort, cramps Infrequent or rare: Hypokalaemia (with prolonged use or overdosage)	Pregnancy: If dietary and lifestyle changes fail to control constipation in pregnancy, moderate doses of poorly absorbed laxatives may be used. A bulk-forming laxative should be tried first. An osmotic laxative, such as lactulose, can also be used. Bisacodyl or senna may be suitable, if a stimulant effect is necessary. Breastfeeding: Not known to be harmful Use with caution: Avoid prolonged use unless indicated to prevent faecal impaction.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Sodium bicarbonate Treatment of cardiotoxicity from drug overdose (e.g. tricyclic antidepressant, carbamazepine); alkalization of urine to enhance excretion of salicylate and chlorphenox pesticides; correction of metabolic acidosis (3.8.1)	Injection: IV: 4.2%, 8.4% (intravenous concentrations 1 ampoule = 50 meq = 4.2 grams = 100 mmol) IV: Emergency dosing: Initially, 1–2 mmol/kg over 1–2 minutes. Additional dose: 0.5 mmol/kg every 10 minutes. Maintenance: 100–150 mmol sodium bicarbonate in 1 litre 5% dextrose at 250 ml/hour.	No common side-effects. Infrequent or rare: severe allergic reactions (rash; hives; difficulty breathing; tightness in chest; swelling of mouth, face, lips, or tongue); irritability; muscle spasms or twitching; pain, redness, or swelling at the injection site Overly aggressive therapy with sodium bicarbonate injection can result in metabolic alkalosis (associated with muscle twitching, irritability, tetany) and hypernatraemia. Therefore, blood pH and electrolytes should be monitored.	Pregnancy/breastfeeding: Unknown safety in pregnancy and breastfeeding; use with caution, if benefit is greater than risk.
Sodium cromoglycate Allergic conjunctivitis; seasonal keratoconjunctivitis (10.1.2)	Ophthalmic drops: 2% Apply 4 times daily.	Common: Burning, stinging	Administration: Start treatment with cromoglycate 1 month before the onset of the hay fever season. Counselling: It can take 3–4 weeks to reach full effect.
Sodium nitrite Cyanide poisoning (3.8.2)	Injection: 30 mg/ml in 10 ml ampoule (3% solution) Slow IV infusion: 10 ml of 3% solution over 2–4 minutes THEN sodium thiosulfate	Common: Nausea/vomiting, abdominal pain, vasodilatation (resulting in syncope, hypotension, tachycardia, flushing), headache, methaemo-globinaemia, cyanosis, dyspnoea/tachypnea	Pregnancy/breastfeeding: Unknown safety in pregnancy and breastfeeding; use with caution; cyanide poisoning likely to be the more significant risk. Use with caution: In severe cardiovascular or cerebrovascular disease. Monitor plasma methaemoglobin levels.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Sodium nitroprusside MAO-I toxicity (3.8.1)	Powder for infusion: 50 mg in ampoule IV: Initially 0.3–0.5 mcg/kg/minute increase gradually to 0.5–6 mcg/kg/minute for the desired hemodynamic effect or the appearance of headache or nausea (maximum 8 mcg/kg/minute). Stop infusion if response satisfactory after 10 minutes.	Common: Severe hypotension, headache, dizziness, retching, abdominal pain, perspiration, palpitation, arrhythmias, apprehension, retrosternal discomfort	Pregnancy: Not recommended in hypertensive crisis in pregnancy. Breastfeeding: Use, with caution, if benefits are greater than risks; monitor infant for effects such as hypotension, bradycardia, fatigue. Contraindications: Severe hepatic impairment, compensatory hypertension, severe vitamin B12 deficiency
(Other indications: Hypertensive crisis) Sodium stibogluconate (pentavalent antimony compound) Cutaneous leishmaniasis (11.20.1) Secondary prophylaxis of visceral leishmaniasis (11.20.3) Post kala-azar dermal leishmaniasis, alternative treatment (11.20.2)	Injection: 100 mg/ml vial IV/IM: 20 mg/kg daily for 21 days IV/IM: 20 mg/kg per month See table in Section 11.20.2	Common: Nausea/vomiting, anorexia, abdominal pain, diarrhoea, ECG changes; coughing (see Cautions); headache, lethargy; arthralgia, myalgia Infrequent or rare: Jaundice, flushing, bleeding from nose or gum, substernal pain, vertigo, fever, sweating, rash; also reported, pancreatitis and anaphylaxis; pain and thrombosis on intravenous administration; intramuscular injection also painful	Pregnancy: Uncertain safety in pregnancy; use, with caution, if benefit is greater than risk. Breastfeeding: Limited information suggests that doses up to 1.4 g daily produce low levels in milk, not expected to cause any adverse effects, especially if the infant is older than 2 months; if withholding nursing during therapy is preferred, breastfeeding can be resumed 24–48 hours after last dose. Contraindications: Cardiac, liver and kidney disorders. Use with caution: Successful treatment of mucocutaneous leishmaniasis may induce severe inflammation around the lesions (may be life-threatening if pharyngeal or tracheal involvement); may require corticosteroid. Administration: IV injections must be given slowly over 5 minutes (to reduce risk of local thrombosis) and stopped if coughing or substernal pain.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Sodium thiosulfate Cyanide poisoning (together with sodium nitrite) (3.8.1) Pityriasis versicolor (10.2.7)	Injection: 250 mg/ml in 50 ml ampoule (25% solution) Topical solution: 15% IV: 50 ml of 25% solution (12.5 g) over 10 minutes Topical: Apply twice daily for 4 weeks.	Sodium thiosulfate has low toxicity. Osmotic disturbances may occur but at the recommended doses are usually mild. Infrequent or rare: When used in cyanide poisoning, symptoms of thiocyanate toxicity may occur: pain in the joints, blurred vision, hyperreflexia, muscle cramps, nausea and vomiting, agitation, delusions, hallucinations, tinnitus	Pregnancy/breastfeeding: Unknown safety in pregnancy and breastfeeding; use with caution. Cyanide poisoning likely to be the more significant risk.
Spectinomycin Alternative treatment for gonorrhoea without dissemination (11.13); gonococcal conjunctivitis (10.12.2) Disseminated gonococcal infection (11.1.3)	Powder for injection: 2 g (as hydrochloride) in vial IM: 2 g as a single dose IM: 2 g twice daily for 7 days (some data suggest that 3 days is adequate)	Infrequent or rare: Dizziness; nausea; urticaria; chills; fever; headache, insomnia, mild to moderate pain after injection, anaphylaxis	Pregnancy/breastfeeding: Use only if indicated. No controlled studies or reports teratogenicity. Use with caution: In renal impairment
Spironolactone Oedema (10.4.3); ascites (10.9) (Other indications: Nephritic syndrome; primary hyperaldosteronism; moderate to severe heart failure in patients taking an ACE inhibitor and a beta-blocker)	Tablet: 25 mg Oral: 100–200 mg daily. Increase if necessary to 400 mg daily in resistant oedema (usual maintenance dose 25–200 mg daily) AND furosemide	Common: Hyperkalaemia, hyponatraemia, hyperchloraemic acidosis, weakness, headache, nausea, vomiting, diarrhoea, breast tenderness Infrequent or rare: GI cramps, drowsiness, menstrual irregularities, transient increase in blood urea nitrogen, gynaecomastia, impotence, rash, ataxia, hepatotoxicity	Pregnancy: Not recommended; consider alternatives where possible. Breastfeeding: Use, with caution, if benefit is greater than risk; amount to infant appears very small, unlikely to cause adverse effects; theoretically may suppress lactation due to diuresis Contraindications: Hyperkalaemia, hyponatraemia, severe renal impairment, Addison's disease. Monitor blood urea nitrogen and plasma electrolytes; discontinue if hyperkalaemia. Use with caution: In elderly; in patients with diabetes mellitus, renal impairment, hepatic impairment, porphyria. Avoid concurrent administration of potassium supplements.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Streptomycin Tuberculosis (initial phase of combination therapy in TB patients returning after defaulting or relapsing from their first treatment course) (15) Buruli ulcer (10.2.10)	Injection: 1 g vial IM: 15 mg/kg daily. Indicated in combination as 2HRZES/1HRZE/5HRE. IM: 15 mg/kg daily for 8 weeks (in combination therapy with rifampicin 10 mg/kg daily)	See gentamicin. Also, hypersensitivity reactions, paraesthesia of mouth	Pregnancy: Avoid streptomycin during pregnancy; auditory or vestibular nerve damage possible Breastfeeding: May be used at recommended doses (monitor infant for thrush or diarrhoea) Contraindications: Hearing disorders, myasthenia gravis Use with caution: In patients with renal impairment; elderly patients. Patients over 60 years or those weighing <50 kg may not tolerate doses above 500–750 mg daily. Administration: Monitor auditory, and vestibular function. If poor renal function, adjust dose.
Sulfadiazine Toxoplasmosis in immunodeficiency (11.40); chorioretinitis (10.12) Toxoplasmosis (in second and third trimesters of pregnancy if fetal infection has been documented) Rheumatic fever, secondary prophylaxis (11.32)	Tablet: 500 mg Oral: 4–6 g daily in 4 divided doses for at least 6 weeks AND <i>folic acid</i> + pyrimethamine for 6 weeks Oral: 4 g daily in 4 divided doses AND <i>folic acid</i> + pyrimethamine Oral: 1 g daily	Common: Nausea/vomiting, diarrhoea Infrequent or rare: Hypersensitivity reactions (Stevens-Johnson syndrome and toxic epidermal necrolysis; discontinue if rash develops), systemic lupus erythematosus, myocarditis, serum sickness, crystalluria resulting in haematuria, blood disorders (discontinue if develops), liver damage, cough/shortness of breath, pancreatitis, CNS problems (convulsions, ataxia, hallucinations), electrolyte disturbances	Pregnancy: Uncertain safety in pregnancy. Use, with caution, only if potential benefit is greater than risk; avoid in first trimester due to bone marrow toxicity; avoid in third trimester due to risk of neonatal jaundice. Breastfeeding: Limited data; use with caution. Avoid in G6PD-deficient neonates and infants; monitor infants for adverse effects (e.g. haemolysis, jaundice). Contraindications: Hypersensitivity to sulphonamides, porphyria Use with caution: In hepatic or renal impairment. Maintain adequate fluid intake to avoid crystalluria. Avoid in blood disorders (except with specialist supervision); monitor blood counts; predisposition to folate deficiency, elderly, asthma, G6PD deficiency Administration: For the treatment of toxoplasmosis, pyrimethamine must always be taken with sulfadiazine.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Sulfadoxine with pyrimethamine (SP) Intermittent preventive therapy in pregnancy (IPTp) where stable transmission of <i>P. falciparum</i> malaria (11.25.8)	Tablet: sulfadoxine 500 mg + pyrimethamine 25 mg Oral: sulfadoxine 1.5 g with pyrimethamine 75 mg (3 tablets) as a single dose under direct observation- give twice during pregnancy, in second and third trimester, 4 weeks apart; if HIV-positive, give 3 doses (see Section 11.25.8.).	Common: Nausea, vomiting, diarrhoea, feeling of fullness; rash, itch Infrequent or rare: Hypersensitivity reaction (Stevens-Johnson syndrome and toxic epidermal necrolysis); hepatitis; cough, dyspnoea; blood disorders (leukopenia, thrombocytopenia, megaloblastic anaemia, purpura)	Pregnancy: Use during pregnancy if treatment of choice; consider folic acid supplementation 5 mg/day. Breastfeeding: Unknown safety; not recommended Contraindications: Already taking cotrimoxazole, hypersensitive to sulfonamides or pyrimethamine, severe hepatic or renal impairment. Avoid in blood disorders (except with specialist supervision). Withdraw treatment if blood disorder: rash, sore throat, mouth ulcers, shortness of breath, or cough occurs. Use with caution: In G6PD deficiency, predisposition to folate deficiency
Suramin Trypanosomiasis (first stage, <i>T.b. rhodesiense</i> infection) (11.41)	Powder for injection: 1 g in vial Slow IV injection: 3.3 mg/kg as single dose (after test dose) followed by weekly increments of 6.7 mg/kg, then 10 mg/kg, 13.3 mg/kg, 16.7 mg/kg, 16.7 mg/kg in weeks 2 through 6, respectively	Infrequent or rare: Immediate and potentially fatal shock and unconsciousness; abdominal pain, diarrhoea, stomal ulceration, dermatitis, abscess, painful joints	Pregnancy: Uncertain safety in pregnancy; use, with caution, only if potential benefit is greater than risk (e.g. in trypanosomiasis). Breastfeeding: Unknown safety in breastfeeding; avoid if possible. Contraindications: Hypersensitivity; totally blind patients with onchocerciasis (unless they require relief from the intensely itchy lesions not relieved by safer alternative) Use with caution: In anaphylaxis, renal/hepatic impairment, total blindness, elderly

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Terbinafine Dermatophytosis (ringworm) (10.2.7) Cutaneous candidiasis (10.2.9, 11.4); pityriasis versicolor (10.2.7)	Cream: 1% Ointment: 1% (hydrochloride) Apply thinly 1–2 times daily for up to 1 week (tinea pedis), 1–2 weeks (tinea corporis and tinea cruris) Apply thinly 1–2 times daily for up to 2 weeks	Infrequent: Occasional local irritation and hypersensitivity reactions include mild burning sensation, erythema, itching. If severe, treatment should be discontinued.	Use with caution: Avoid contact with eyes and mucous membranes.
Tetracaine (amethocaine) eye drops Short-acting local anaesthesia of the cornea and conjunctiva	Drops: 0.5% (hydrochloride) Instill 1 drop	Common: Burning, stinging, redness Infrequent or rare, allergic reactions	Contraindications: Hypersensitivity to ester-type local anaesthetics; eye inflammation or infection. Use with caution: Avoid prolonged use (risk of severe keratitis, permanent corneal opacification, scarring, delayed corneal healing). Protect eye from dust and bacterial contamination until sensation is fully restored. Never give patients anaesthetic drops to take home.
Tetracycline Cholera (10.7d.2) Acne (10.2.3) Late latent syphilis, syphilis of undetermined duration, and late syphilis (11.3.7) Syphilis with penicillin allergy in non-pregnant patient (11.3.7)	Tablet: 500 mg Oral: 500 mg 4 times daily for 3 days Oral: 250–500 mg twice daily Oral: 500 mg 4 times daily for 30 days Oral: 500 mg 4 times daily for 15 days	Common: Nausea/vomiting, diarrhoea, increased BUN, phototoxicity, rash, increased intracranial pressure, discoloration of teeth/enamel hypoplasia (young children), antibiotic-associated colitis, hypersensitivity reaction Infrequent or rare: Hepatotoxicity, burning or stinging	Pregnancy: Contraindicated after week 8 due to effects on fetal bone growth and dental discolouration; short courses can be used if alternative not appropriate: implicated in causing maternal hepatotoxicity, especially in third trimester (dose-related) Breastfeeding: Uncertain safety; avoid if possible. Contraindications: Tetracycline hypersensitivity. Use with caution: In renal or hepatic impairment

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Tetracycline eye ointment Cobra spit (3.9); bacterial conjunctivitis; corneal erosion (10.12.2) Trachoma, continuous intensive treatment (10.12.5)	Ointment: 1% (hydrochloride) Apply directly to the eye, 1 application 3–4 times daily Apply directly to the eye, 1 application of ointment in each eye twice daily for at least 6 weeks.	Infrequent or rare: Rash; stinging, burning, No reports of tooth discoloration at usual topical doses	Contraindications: Hypersensitivity to tetracycline group of antibiotics Use with caution: Prolonged use may lead to overgrowth of non-susceptible organisms.
Thiamine Chronic thiamine deficiency (as may occur in alcohol abuse and dependence) (3.5) Ethylene glycol poisoning (3.8.1) Persistent vomiting in pregnancy >2 weeks (14.1.11)	Tablet: 50 mg Injection (vitamin B1) (hydrochloride) IM/IV: 100 mg daily for 5 days THEN switch to oral 100 mg daily IM/IV: 100 mg every 8 hours for 6 doses Oral: 100 mg daily- until persistent vomiting stops	Toxic effects are unlikely since any excess thiamine is excreted. Infrequent or rare: Anaphylaxis	Pregnancy/breastfeeding: Severely thiamine-deficient mothers should avoid breastfeeding. Administration: If IV, infuse over 30 minutes.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Tranexamic acid (TXA) Trauma patients with ongoing significant haemorrhage or at risk of significant haemorrhage, within 3-4 hours (4.2)	<i>Injection, solution: 100 mg/ml (10 ml)</i> IV: loading dose of 1 g over 10 minutes THEN infusion of 1 g over 8 hours.	Common: Nausea/vomiting, diarrhoea, allergic skin reactions. Rapid intravenous injection may cause dizziness and/or hypotension. To avoid this response, the solution should not be injected more rapidly than 1 ml per minute. Infrequent or rare: Thromboembolic events, disturbances in colour vision (discontinue)	Pregnancy: TXA has been used in pregnancy and no harmful effects have been reported. In the CRASH-2 trial, pregnancy was not an exclusion criteria. Weigh the potential risks and benefits for each woman. Breastfeeding: Very small amounts pass into breast milk; an antifibrinolytic effect in the infant is unlikely. Contraindications: History of thromboembolic disease Administration: As early as possible, within 3-4 hours of injury. Reduce injection dose in patients with renal insufficiency. TXA solution for injection should not be mixed with blood for transfusion or with infusion solutions containing penicillin or mannitol.
Tretinoin Severe acne (10.2.3)	<i>Gel: tretinoin 0.01% (tretinoin is the acid form of vitamin A)</i> Apply thinly 1-2 times daily. Several months of treatment may be needed to achieve optimal response. Treatment should continue until no new lesions develop in 2 weeks.	Common: Local reactions include burning, erythema, stinging, pruritus, dry or peeling skin (discontinue if severe). Increased sensitivity to UVB light or sunlight; temporary changes of skin pigmentation reported Infrequent or rare: Eye irritation and oedema, blistering or crusting of skin	Pregnancy: Topical retinoids contraindicated in pregnancy; women of child-bearing age must use effective contraception (oral progestogen-only contraceptives not considered sufficiently effective). Use with caution: Topical retinoids should be avoided in severe acne involving large areas. Avoid contact with eyes, nostrils, mouth, mucous membranes; eczematous, broken, or sunburned skin. Avoid exposure to UV light (including sunlight, solariums). If sun exposure is unavoidable, an appropriate sunscreen or protective clothing should be used. Avoid use of retinoids with abrasive cleaners, comedogenic or astringent cosmetics. Allow peeling (e.g. resulting from use of benzoyl peroxide) to subside before using a topical retinoid. Counselling: Some redness and skin peeling may occur initially but settles with time.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Triclabendazole Fascioliasis (11.11)	Tablet: 250 mg Oral: 10 mg/kg in a single dose In treatment failure re-administer 10 mg/kg. THEN follow by another dose 12–24 hours later (giving a total dose of 20 mg/kg)	Common: Nausea, vomiting, diarrhoea, headache, biliary colic due to obstructing worms	Use with caution: In severe fascioliasis, biliary colic can occur due to obstruction by dying worms.
Urea Hydrating agent and keratolytic for dry, scaling, and itching skin conditions, including mild psoriasis, xeroderma (10.2)	Cream or ointment: 5%, 10% Apply directly to affected area twice daily, preferably to damp skin.	Common: Transient stinging and local irritation	Use with caution: Avoid application to face or broken skin; avoid contact with eyes.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Valproic acid (sodium valproate) Epilepsy (10.10c) Mood stabilization in bipolar disorder, acute mania (10.11.5)	Tablet: 200 mg, 500 mg Liquid: 200 mg/ml Oral: Initiate at 400 mg daily; increase by 200 mg daily to maximum of 2 g daily in divided doses. Oral: Initiate at 500 mg at night; gradually increase by 200 mg every 7 days until response; typical dose: 1–2 g daily In elderly or medically ill patients including HIV stage 3 or 4: initiate at 200 mg in morning and at night. Increase dose 200 mg every 7 days until clinical response. See 10.11.5	Common: Nausea, gastric irritation, diarrhoea, increased appetite; weight gain; ataxia, tremor; paraesthesia, drowsiness; elevated liver transaminases, hyperammonaemia Infrequent or rare: Hepatotoxicity (can be fatal), pancreatitis, hyponatraemia from drinking excess fluid; blood dyscrasias (anaemia, leukopenia, pancytopenia, thrombocytopenia); severe allergic reaction (including toxic epidermal necrolysis, Stevens-Johnson syndrome). Transient hair loss (regrowth may be curly); increased alertness, aggression, hyperactivity, behavioural disturbances, vasculitis; lethargy, drowsiness, confusion, stupor, hallucinations, menstrual disturbances, hearing loss, rash, peripheral oedema, increase in bleeding time, extrapyramidal symptoms, dementia, encephalopathy, coma, gynaecomastia, Fanconi's syndrome, hirsutism, acne, enuresis, hyponatraemia Withdraw treatment immediately if persistent vomiting and abdominal pain, anorexia, jaundice, oedema, malaise, drowsiness, loss of seizure control.	Pregnancy: Risk of teratogenicity; use only if benefit is greater than risk; consider folic acid supplementation 5 mg/day and vitamin K supplementation. Breastfeeding: May be used in breastfeeding. Small amounts excreted in breast milk; use minimum effective dose and monitor infant for adverse effects (e.g. jaundice). Use with caution: Monitor coagulation studies and liver function tests regularly during therapy. Prone to multiple drug interactions through CYP enzymes; check for interactions with all new and current medications. Elderly: See cautions in Section 10.11.5. Administration: Do not use for alcohol withdrawal. Counselling: Take with food to reduce stomach upset. Your appetite may increase when taking this medicines; pay attention to your diet to avoid weight gain. This medication may impair your ability to perform hazardous activities requiring mental alertness. Tell your clinician immediately if fever, rash, abdominal pain, vomiting, yellow eyes or urine, bruising, or bleeding develops. Do not stop taking this medicine suddenly unless advised by your doctor.

Drug Indication	Formulations Dosage	Adverse effects	Special groups/comments
Vitamin B12 (hydroxocobalamin) Pernicious anaemia and other macrocytic anaemia without neurological involvement Pernicious anaemia and other macrocytic anaemias with neurological involvement Prophylaxis and treatment of other macrocytic anaemias due to vitamin B12 deficiency	Intramuscular anhydrous hydroxo-cobalamin 1 mg/ml IM: 1 mg 2–4 times weekly for 2 weeks; then 1 mg every 3 months. IM: 1 mg on alternate days until no further improvement; then 1 mg every 2 months IM: 1 mg IM every 2–3 months	Infrequent or rare: Nausea, headache, dizziness; fever, hypersensitivity reactions including rash and pruritus, anaphylaxis; injection-site pain; hypokalaemia during initial treatment	Pregnancy: Safe, but megaloblastic anaemia of pregnancy is usually due to folate deficiency and should be treated with folate plus vitamin B12. Breastfeeding: Safe to use. Contraindications: Sensitivity to B12 (hydroxocobalamin). Use with caution: Establish which deficiency is present – vitamin B12 or folate – with a marrow examination and treat the underlying cause. Always give B12 with folic acid in pernicious anaemia; folic acid given alone can precipitate neuropathy if there is underlying unrecognized vitamin B12 deficiency. Cardiac arrhythmias secondary to hypokalaemia have been reported during initial therapy; therefore, potassium should be monitored during this period. Administration: Do not give by IV injection. Store below 25 °C; protect from light.
Vitamin K (phytonadione) Rodenticide poisoning (3.8.1) with no bleeding but prolonged INR Warfarin therapy (3.8.1) with INR > 9.0 but no bleeding Rodenticide poisoning or warfarin therapy (3.8.1) with severe haemorrhage	Injection: 10 mg/ml in 5 ml ampoule Tablet: 10 mg Oral: 10–20 mg; may need to continue for weeks if long-acting anticoagulant rodenticide Oral: 2.5–5 mg IV: 10 mg	Common: Pain, tenderness, erythema at IM site Infrequent or rare: Hypersensitivity reactions	Pregnancy: Use only if benefit is greater than risk. Breastfeeding: May be used short-term; caution if chronic dosing required Use with caution: In elderly and hepatic impairment Administration: IV vitamin K should be given slowly over 20 minutes. Note: Vitamin K is not an antidote to heparin.